

Endpoint convergence for the planar Schrödinger equation

OpenAI

September 24, 2026

Abstract

We resolve the planar Sobolev endpoint of Carleson’s convergence problem: for initial data in $H^{1/3}(\mathbb{R}^2)$, the Schrödinger evolution converges almost everywhere to the initial data as $t \downarrow 0$. The evolution is defined by taking the Gaussian regularization limit first; on one full-measure spatial set, this limit exists for every $0 < t < 1$, and the resulting evolution is continuous at $t = 0$.

Contents

1	Introduction	2
1.1	History and significance	3
1.2	Three ingredients of the proof	3
1.3	Organization and dependencies	4
2	Packets, localization, and fractal estimates	4
2.1	Conventions and the compact atom class	4
2.2	Exact packets and spatial localization	6
2.3	The refined fractal estimate and unit-cube suprema	8
2.4	Weights and cap rescaling	9
3	A power saving for transverse inputs	11
3.1	Uniform multilinear estimates and the tube model	12
3.2	Information at spatial scales	15
3.3	Direction costs and equality of the profiles	16
3.4	Projection growth on a block	19
3.5	Selecting angular scales and differentiating	22
3.6	Hills and finitely many jumps	23
3.7	The partitioning contradiction	24
4	A sparse estimate for packet arrays	27
4.1	A local norm and an exact-field reduction	27
4.2	The iteration and broad saving	30
4.3	Narrow decoupling and the child measure	32
4.4	Terminal estimate and summation	34
5	A bilinear frame estimate without a scale loss	36
5.1	Reduction to small, exactly localized frequency pieces	37
5.2	A positive frame iteration and smooth deletion masks	38
5.3	Energy flux and the quartic potential	42
5.4	Stopping and sampling the two separated groups	46

6	Flat frequency arrays	49
6.1	A one-dimensional distribution estimate	49
6.2	Projection of a thin frequency strip	52
7	Rich tubes lie in a short list of plates	54
8	The dyadic distribution estimate	60
8.1	Parameters and recursive estimates	61
8.2	Peeling to a common rich measure	63
8.3	Plates and coarse frequency intervals	64
8.4	Flattening all intermediate scales	66
8.5	Two bands and the bilinear frame estimate	68
8.6	Summing the bands and closing the induction	70
9	Endpoint square summation and the order of limits	72
9.1	From packet distributions to a short-time maximal estimate	72
9.2	Two dual estimates on a short time interval	74
9.3	The common time tree and the Sobolev square norm	76
9.4	Recovering the Gaussian-first, all-real-time limit	79
	References	82

1 Introduction

For $f \in L^2(\mathbb{R}^2)$ we use the Fourier transform

$$\widehat{f}(\xi) = \int_{\mathbb{R}^2} e^{-ix \cdot \xi} f(x) dx, \quad \|f\|_{H^{1/3}}^2 = (2\pi)^{-2} \int_{\mathbb{R}^2} (1 + |\xi|^2)^{1/3} |\widehat{f}(\xi)|^2 d\xi.$$

The first formula is initially defined on integrable functions and then extended by continuity to L^2 . For $a > 0$ put

$$U_a f(x, t) = (2\pi)^{-2} \int_{\mathbb{R}^2} e^{ix \cdot \xi - it|\xi|^2 - a|\xi|^2} \widehat{f}(\xi) d\xi. \quad (1.1)$$

This integral is absolutely convergent by Cauchy–Schwarz. At a Lebesgue point of f , write

$$f^*(x) = \lim_{r \downarrow 0} \frac{1}{\pi r^2} \int_{|y-x| < r} f(y) dy.$$

Our main result includes the order of limits in the definition of the evolution.

Theorem 1.1 (Planar endpoint convergence). *For every $f \in H^{1/3}(\mathbb{R}^2)$ there is a set $E_f \subset \mathbb{R}^2$ of full Lebesgue measure such that, for each $x \in E_f$, the finite limit*

$$u_f(x, t) = \lim_{a \downarrow 0} U_a f(x, t)$$

exists for every real $0 < t < 1$, and

$$\lim_{t \downarrow 0} u_f(x, t) = f^*(x).$$

In fact $u_f(x, \cdot)$ extends continuously to $[0, 1)$ with value $f^(x)$ at zero, and the Gaussian limit is uniform over $0 < t < 1$.*

1.1 History and significance

Carleson [7] introduced the pointwise convergence problem and proved its one-dimensional positive result at regularity $s = 1/4$. Dahlberg and Kenig [9] showed that this threshold is necessary. Sjölin [23] and Vega [26] independently obtained convergence for $s > 1/2$ in higher dimensions. In the plane, Bourgain and Moyua–Vargas–Vega [2, 20] developed the connection with Fourier restriction. The bilinear approach of Tao and Vargas [25] gave $s > 15/32$; Tao’s sharp bilinear restriction theorem [24] improved this to $s > 2/5$, and Lee [18] obtained $s > 3/8$. In dimensions $n \geq 3$, Bourgain [3] obtained the sufficient range $s > 1/2 - 1/(4n)$.

Bourgain’s later counterexample [4, Proposition 1] forces $s \geq n/[2(n+1)]$ when $n \geq 2$, but does not rule out equality. Du, Guth and Li [11, Theorems 1.1–1.2] proved planar convergence and a local L^3 maximal estimate for $s > 1/3$. Their proof combined polynomial partitioning with wave packets and refined Strichartz estimates, using ℓ^2 decoupling to control packets tangent to the partitioning surface. Du, Guth, Li and Zhang [12, Theorems 1.1 and 4.2] then obtained higher-dimensional convergence for $s > (n+1)/[2(n+2)]$ and developed multilinear refined Strichartz estimates. Du and Zhang [13, Theorem 1.1] established convergence for $s > n/[2(n+1)]$ when $n \geq 3$. A recent Bessel-potential extension still states its positive $p = 2$ range with the strict inequality $s > n/[2(n+1)]$ [21, Theorems 1.3–1.4].

The choice of pointwise representative also has a history: Sjögren and Sjölin [22, Theorem 2] constructed, for $s > 1/2$, representatives continuous on the entire time line for almost every spatial point, using spatial or space-time averages.

Theorem 1.1 resolves positively the planar Sobolev endpoint of Carleson’s convergence problem: it supplies the equality case $s = 1/3$, with the order of limits and the all-real-times representative specified above. Its maximal estimate has local L^1 output. The stronger question of an endpoint local L^3 maximal estimate is not asserted here.

1.2 Three ingredients of the proof

The main analytic conclusion is a local maximal estimate at the exact Sobolev exponent. Write $S(t)f = e^{it\Delta}f$ for the L^2 evolution with Fourier multiplier $e^{-it|\xi|^2}$. For bounded-frequency data, the pointwise maximal estimate takes the form

$$\int_{B(x_0,1)} \sup_{0 \leq t \leq 1} |S(t)f(x)| \, dx \leq C \|f\|_{H^{1/3}}, \quad (1.2)$$

with a constant independent of the center x_0 and the frequency bound. Theorem 9.6 proves this estimate. The final section then constructs the representative in Theorem 1.1; it does not infer an all-times assertion by intersecting time-dependent exceptional sets.

We work first at unit frequency, with packets of spatial width L and propagation time L^2 . A measure satisfying a two-dimensional ball bound models the spatial area on a measurable time graph. The proof has three main components.

A transverse power saving. Multilinear restriction [1] and the refined fractal restriction estimate of Du and Zhang [13, Theorem 1.6] reduce a hypothetical saturated transverse configuration to weighted incidences between points and tubes. Conditional information across scales has additive profiles whose extremal relations come from three transverse directions. A robust projection theorem [17, Theorem 1] restricts the derivatives of these profiles to zero and one. A further argument rules out arbitrarily many alternations; polynomial partitioning [15, Theorem 4.1] then excludes the remaining jump. This gives a fixed power saving, which is propagated to the sparse packet estimate in Proposition 4.1.

A bilinear estimate for smooth packet arrays. Theorem 5.1 treats packets with general smooth profiles and separated frequency groups. Positive masks remove large pieces of a tight frequency frame, charging them to the lost energy. A quartic potential, together with the curvature identity for the parabola, controls changes in the spatial smoothing scale. The resulting estimate is uniform in the number of scales. This uniformity is needed for the long, nearly flat frequency configurations that arise in the proof of Theorem 8.1.

Square summation at the endpoint. Theorem 8.1 yields a short-time weak estimate with some fixed output exponent $p > 2$, with no growing frequency loss. To combine frequencies, we linearize all of them on the same time graph. A binary time tree separates heavier children from their lighter siblings. Spatial localization supplies decay in the relative depth of this tree, while $p > 2$ supplies a positive convexity deficit that sums powers of the lighter-child masses across frequency scales. Together these estimates retain the Sobolev square norm and prove (1.2).

The packet and measure formulations allow arbitrary bounded phase-space counts and sufficiently smooth compact profiles. In particular, the bilinear estimate applies to the composite packets created by localization; no exact evolution equation is imposed on those profiles.

1.3 Organization and dependencies

Section 2 fixes the packet conventions and the restriction estimates used with small losses. The transverse and sparse arguments follow in Sections 3 and 4. The independent frame estimate is proved before its applications, in Section 5. Sections 6 and 7 provide the flat-frequency and geometric reductions, and Section 8 proves the uniform dyadic distribution estimate. Section 9 performs the endpoint summation and proves Theorem 1.1. Figure 1 records the logical order. The distinction between a fixed power saving and an estimate uniform in the packet scale is essential throughout.

2 Packets, localization, and fractal estimates

We first fix the packet conventions and the analytic estimates used in the transverse argument. There are two packet classes: compactly supported smooth atoms, which are convenient for the later geometric arguments, and exact Schrödinger packets, which permit the use of restriction estimates. The latter have rapidly decreasing spatial tails. Keeping these two classes distinct will also make the localization errors explicit.

2.1 Conventions and the compact atom class

For frequency data $v \in L^2(\mathbb{R}^2)$ supported in a fixed bounded set, put

$$Ev(x, t) = \int_{\mathbb{R}^2} e^{i(x \cdot \xi - t|\xi|^2)} v(\xi) d\xi. \quad (2.1)$$

Thus extension input norms are frequency-space L^2 norms; the Fourier normalization in Section 1 differs only by the fixed factor $(2\pi)^{-2}$. In particular, with that Fourier convention, $\|f\|_2^2 = (2\pi)^{-2} \|\hat{f}\|_2^2$. All frequency sets below lie in fixed bounded boxes unless another scale is specified. The direction associated with ξ is the line parallel to $(2\xi, 1)$, or its unit representative with positive last coordinate. On a fixed bounded frequency set, distances between frequencies and between these unit representatives are comparable.

Unless stated otherwise, all measures are nonnegative Borel measures on space-time $\mathbb{R}^3$. A ball bound is required for balls with arbitrary centers. Its normalizing parameter may

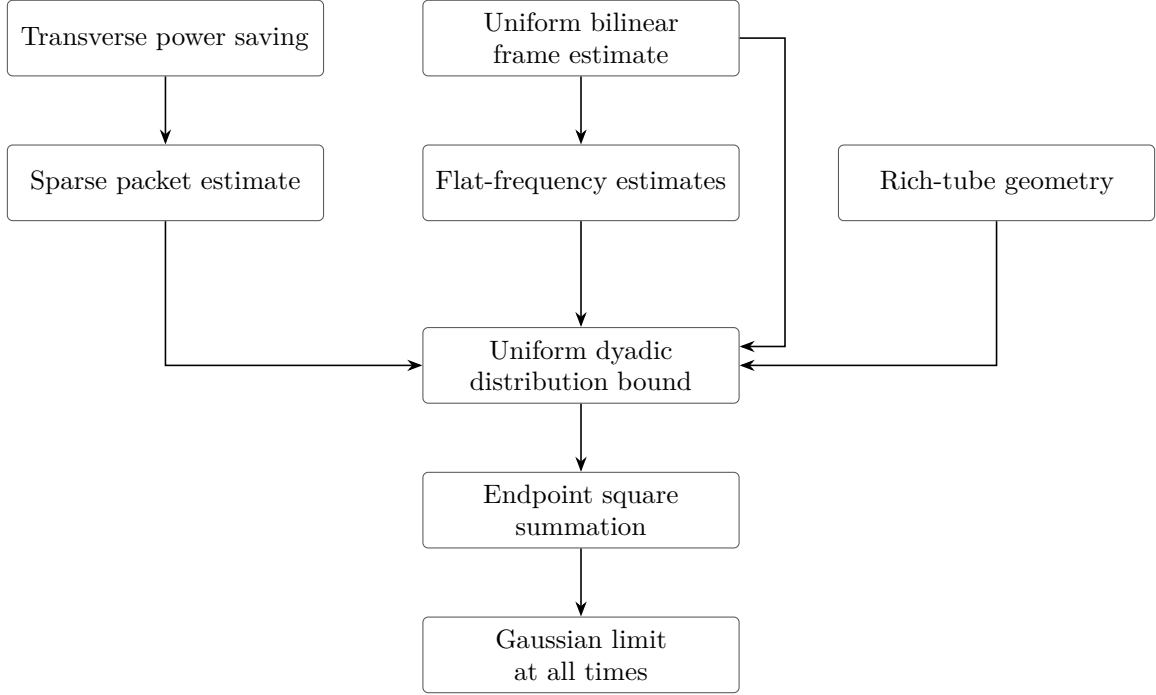


Figure 1: Dependencies of the proof. Small scale losses are allowed in the transverse and sparse arguments; the frame and dyadic distribution estimates must be uniform in the packet scale.

be assumed positive, since the zero case is immediate. Lattice partitions are half-open; a supremum on a cell may be taken on its closure. Fixed enlargements, boundedly overlapping grids, and fixed changes of frequency boxes cost constants independent of the large scales and of the number of packets. All distribution thresholds are positive.

An estimate with a free factor S^ϵ is asserted for each fixed $\epsilon > 0$, with a constant that may depend on ϵ . We can therefore use a smaller exponent to absorb a fixed number of logarithms. On a countersequence, $o(1)$ denotes a quantity tending to zero; simultaneous statements at countably many fixed parameters are obtained by subsequences and diagonal choices. Parameters reused in different arguments have only their local meaning.

Definition 2.1 (Compact atoms). Let $L \geq 1$, and set $R = L^2$. An atom with label α is the field

$$A_\alpha(x, t) = L^{-1} e^{i(x \cdot \omega_\alpha - t |\omega_\alpha|^2)} H_\alpha \left(\frac{x - b_\alpha - 2\omega_\alpha t}{L}, \frac{t}{L^2} \right). \quad (2.2)$$

The frequencies ω_α lie in a fixed square. The labels obey a joint phase-space bound: every unit ball in $\mathbb{R}^2 \times \mathbb{R}^2$ contains at most a fixed number of the pairs

$$\left(\frac{b_\alpha}{L}, L\omega_\alpha \right).$$

The profiles $H_\alpha(y, \tau)$ are supported in a fixed bounded box, and have uniform bounds for sufficiently many derivatives. More precisely, each application requires a finite integer J and fixed bounds $\|\partial^\beta H_\alpha\|_\infty \leq C_\beta$ for $|\beta| \leq J$. The integer J is chosen after the fixed parameters of that application; it does not grow with L or with the array cardinality. A packet field is a finite sum $\sum_\alpha c_\alpha A_\alpha$, and its coefficient energy is $\sum_\alpha |c_\alpha|^2$, without weights.

The joint phase-space condition does not require bounded multiplicity of frequencies alone. In particular, arbitrarily many spatial translates of one frequency are allowed. Constants in

packet estimates may depend on the fixed support, multiplicity, and derivative bounds in Definition 2.1. Whenever a later estimate permits an arbitrary fixed decay order, sufficiently many of these finitely many derivatives are to be fixed first.

Suppose that μ is restricted to $|t| \leq CL^2$ and satisfies

$$\mu(B(z, r)) \leq mr^2 \quad (z \in \mathbb{R}^3, r \geq 1). \quad (2.3)$$

The additional sparsity assumption, with the fixed choice $M = 100$, is

$$\int_{\mathbb{R}^3} \left(1 + \frac{|x - b_\alpha - 2\omega_\alpha t|}{L}\right)^{-M} d\mu(x, t) \leq \frac{mL^3}{D} \quad \text{for every label used.} \quad (2.4)$$

Other sufficiently large fixed values of M would work. The ball bound alone implies (2.4) with $D = 1$ up to a fixed constant. Indeed, the ρ -neighborhood of one trajectory on this time slab is covered by $O(1 + L^2/\rho)$ balls of radius $C\rho$. For $\rho = 2^k L$, its mass is at most

$$Cm(2^k L^3 + 2^{2k} L^2).$$

Summing these bounds against 2^{-Mk} , $k \geq 0$, proves the assertion.

The normalization in (2.2) is chosen to be stable under cap rescaling. For labels in a square of side h^{-1} centered at v , where $1 \leq h \leq L$, set

$$(x', t') = \left(\frac{x - 2vt}{h}, \frac{t}{h^2}\right), \quad L' = \frac{L}{h}, \quad b'_\alpha = \frac{b_\alpha}{h}, \quad \omega'_\alpha = h(\omega_\alpha - v). \quad (2.5)$$

An exact substitution gives

$$A_\alpha(x, t) = h^{-1} e^{i(x \cdot v - t|v|^2)} A'_\alpha(x', t'), \quad (2.6)$$

where A'_α has the same profile and the parameters in (2.5). The coefficients are unchanged, and

$$\left(\frac{b'_\alpha}{L'}, L'\omega'_\alpha\right) = \left(\frac{b_\alpha}{L}, L\omega_\alpha - Lv\right),$$

so the phase-space ball counts are unchanged as well. If μ' is the pushforward of μ , an inverse image of an r -ball is covered by $O(h)$ balls of radius Chr . Hence μ' has ball parameter $m' = Cmh^3$. The integrand in (2.4) is invariant under this change, and $m'(L')^3 = CmL^3$, so its sparsity parameter is preserved up to the allowed fixed constants. On a time slab of length $O((L')^2)$ centered at zero, a compact packet meets only $O(1)$ spatial squares of side $(L')^2$. Thus the energies assigned to these squares sum with bounded overlap.

2.2 Exact packets and spatial localization

The corresponding energy statement for exact fields requires controlling the tails rather than declaring the packets compactly supported.

Lemma 2.2 (Exact packet localization). *Let $s \geq 1$, let v have frequency support in a fixed bounded set, and consider $|t| \leq Cs^2$. There is a decomposition $v = \sum_{\omega, n} a_{\omega, n} v_{\omega, n}$, with $\omega \in s^{-1}\mathbb{Z}^2$ in a fixed bounded enlargement and $b_n = sn$, $n \in \mathbb{Z}^2$, such that*

$$\sum_{\omega, n} |a_{\omega, n}|^2 \leq C\|v\|_2^2, \quad (2.7)$$

$$\left\| \sum_{(\omega, n) \in \mathcal{I}} a_{\omega, n} v_{\omega, n} \right\|_2^2 \leq C \sum_{(\omega, n) \in \mathcal{I}} |a_{\omega, n}|^2 \quad \text{for every subfamily } \mathcal{I}, \quad (2.8)$$

$$|Ev_{\omega, n}(x, t)| \leq C_N s^{-1} \left(1 + \frac{|x - b_n - 2t\omega|}{s}\right)^{-N} \quad (N \geq 1). \quad (2.9)$$

The constants are independent of the number of terms retained. The phase-space label counts are bounded, and every packet has frequency support in an $O(s^{-1})$ neighborhood of ω .

Fix $0 < \eta < 1$. On a set Ω in this time slab, retain the packets whose trajectories come within $s^{1+\eta}$ of Ω at the same time. For every $A > 0$, the sum of the omitted packets is bounded pointwise on Ω by $C_{A,\eta}s^{-A}\|v\|_2$. For a partition into spatial squares of side s^2 , extended over the common time slab, the retained data v_Q can consequently be chosen so that

$$\sum_Q \|v_Q\|_2^2 \leq C\|v\|_2^2, \quad |Ev - Ev_Q| \leq C_{A,\eta}s^{-A}\|v\|_2 \quad \text{on } Q \times [-Cs^2, Cs^2]. \quad (2.10)$$

The same conclusions hold on translated time slabs after modulating the initial datum and writing the trajectories in the recentered time $t - t_0$.

Proof. Take a smooth bounded-overlap partition $\{\chi_\omega\}$ of frequency space at resolution s^{-1} . We can arrange that $\chi_\omega(\omega + \eta/s)$ is supported in $[-1, 1]^2$. Put

$$g_\omega(\eta) = s^{-1}(\chi_\omega v)(\omega + \eta/s).$$

Then $\|g_\omega\|_2 = \|\chi_\omega v\|_2$. On $[-\pi, \pi]^2$, expand g_ω in the orthonormal Fourier basis:

$$g_\omega(\eta) = \sum_{n \in \mathbb{Z}^2} a_{\omega,n} (2\pi)^{-1} e^{-in \cdot \eta}.$$

Choose one smooth cutoff ψ supported in $(-\pi, \pi)^2$ and equal to one on $[-1, 1]^2$. Multiplying the series by ψ leaves g_ω unchanged, and gives the packet data

$$v_{\omega,n}(\xi) = \frac{s}{2\pi} \psi(s(\xi - \omega)) e^{-isn \cdot (\xi - \omega)}.$$

Parseval, followed by the bounded overlap of the frequency partition, gives (2.7). Multiplication by ψ is bounded on L^2 , and its translated frequency supports have bounded overlap. Applying these two facts to any selected Fourier coefficients proves (2.8). This also proves convergence of the decomposition in L^2 . Since all frequency supports remain in a fixed bounded set, the corresponding extensions converge uniformly in (x, t) .

Changing variables $\xi = \omega + \eta/s$ gives

$$Ev_{\omega,n}(x, t) = \frac{s^{-1}}{2\pi} e^{i(x \cdot \omega - t|\omega|^2)} \int \psi(\eta) e^{i((x - sn - 2t\omega)/s) \cdot \eta - i(t/s^2)|\eta|^2} d\eta. \quad (2.11)$$

For $|t|/s^2 \leq C$, all derivatives of $\psi(\eta) e^{-i(t/s^2)|\eta|^2}$ are bounded. Repeated integration by parts proves (2.9). In particular the profile in (2.11) is Schwartz in its first argument, uniformly on the prescribed time interval.

For clarity, the tail bound is uniform even for infinite packet expansions. At a fixed point, there are $O(s^2)$ frequency centers. For each one the normalized spatial labels form a translate of $\mathbb{Z}^2$, and, for $N > 1$,

$$\sum_{n: |x - sn - 2t\omega| > s^{1+\eta}} \left(1 + \frac{|x - sn - 2t\omega|}{s}\right)^{-2N} \leq C_N s^{-\eta(2N-2)}.$$

The factor s^{-2} from the squared packet bound cancels the $O(s^2)$ frequency count. Cauchy-Schwarz and (2.7) therefore bound the discarded field by $C_N s^{-\eta(N-1)}\|v\|_2$. Choose N sufficiently large in terms of A and η .

Finally, a bounded-velocity trajectory traverses distance $O(s^2)$ in the time slab. Since $s^{1+\eta} \leq s^2$, its padded trajectory meets only $O(1)$ squares of side s^2 . Each coefficient is consequently assigned to only $O(1)$ of the subfamilies defining v_Q . Equation (2.8) proves their summed energy bound, and the preceding tail estimate proves the second part of (2.10). To center the time interval at t_0 , replace $v(\xi)$ by $e^{-it_0|\xi|^2}v(\xi)$ before decomposing it. $\square$

In applications involving polynomially many boxes or a measure of polynomial total mass, the exponent A is chosen large enough that all these tail errors remain smaller than any prescribed negative power. If only a subpower padding is desired on a countersequence, first fix a positive padding exponent for each required estimate and then let it decrease through a diagonal choice. This does not assert a uniform tail constant as $\eta \downarrow 0$.

2.3 The refined fractal estimate and unit-cube suprema

We use the following specialization of the refined fractal estimate of Du–Zhang. We include the reductions from its uniform occupancy formulation, since both the weights and the suprema will matter later.

Lemma 2.3 (Refined fractal estimate). *Let $S \geq 1$. Let $\mathcal{Q}$ be a collection of distinct unit lattice cubes in a ball of radius CS in $\mathbb{R}^3$. Suppose that*

$$\#\{Q \in \mathcal{Q} : Q \cap B(z, r) \neq \emptyset\} \leq \gamma r^2 \quad (1 \leq r \leq CS), \quad \#\{Q \in \mathcal{Q} : Q \cap B(z, \sqrt{S}) \neq \emptyset\} \leq \lambda_0 \quad (2.12)$$

for all centers z , with fixed enlargements of the radius range allowed. If v has frequency support in a fixed bounded set, then

$$\int_{\bigcup \mathcal{Q}} |Ev|^2 dx dt \leq C_\epsilon S^\epsilon (\gamma \lambda_0 S)^{1/3} \|v\|_2^2, \quad (2.13)$$

$$\sum_{Q \in \mathcal{Q}} \sup_Q |Ev|^2 \leq C_\epsilon S^\epsilon (\gamma \lambda_0 S)^{1/3} \|v\|_2^2. \quad (2.14)$$

These estimates are uniform under translations of the collection of cubes.

Proof. For unit-ball frequency support, Theorem 1.6 of [13], specialized to spatial dimension 2 and counting exponent 2, gives

$$\|Ev\|_{L^2(X)} \leq C_\epsilon S^\epsilon \gamma^{1/6} \lambda^{1/6} S^{1/6} \|v\|_2$$

when every occupied square-root-scale lattice cube contains a comparable number λ of the unit cubes of X . Its time sign is changed by reflection, and its physical-space input norm is converted to the frequency-space norm by Plancherel. A bounded frequency enlargement is reduced to unit frequency by a fixed parabolic dilation and bounded cube coverings. None of these operations changes a large-scale exponent.

For the asserted upper occupancy version, take an ambient radius comparable to S whose square root is an integer, large enough to contain the given ball. Its square-root grid then contains whole unit lattice cubes. The occupancy of each parent is at most $C\lambda_0$, since such a parent is covered by a fixed number of $\sqrt{S}$ -balls. Sort the parents by their dyadic occupancy λ . Every resulting union retains the counting bound $C\gamma r^2$. Apply the quoted theorem to each union, square it, and sum. The geometric sum of $\lambda^{1/3}$ is at most $C\lambda_0^{1/3}$; using a logarithmic bound would also suffice after decreasing the chosen small exponent. Renaming that exponent proves (2.13). Ball centers can be translated at the cost of replacing $v(\xi)$ by $e^{i(y \cdot \xi - u|\xi|^2)} v(\xi)$, which preserves its norm and support.

To obtain suprema, let $F = Ev$. Its space-time Fourier transform is supported on a fixed compact portion of the paraboloid. Choose a Schwartz function ρ whose Fourier transform is one on that compact set, so that $F = F * \rho$. For every unit cube Q and every $z \in Q$,

$$\sup_{y \in Q} |F(y)| \leq C_N \int_{\mathbb{R}^3} (1 + |w|)^{-N} |F(z + w)| dw. \quad (2.15)$$

Indeed, taking the supremum over $y - z$ in a fixed bounded set in the convolution formula leaves a rapidly decreasing majorant for the kernel. Cauchy–Schwarz with this integrable weight, followed by integration over $z \in Q$ and summation in Q , gives

$$\sum_{Q \in \mathcal{Q}} \sup_Q |F|^2 \leq C_N \int_{\mathbb{R}^3} (1 + |w|)^{-N} \int_{\bigcup \mathcal{Q}} |Ev(z + w)|^2 dz dw.$$

For each fixed $w = (y, u)$ the inner field is the extension of the modulated datum above. Apply (2.13) on the unchanged domain and integrate the weight. This proves (2.14) without enlarging the ambient scale according to the translation. $\square$

The same reproducing argument applies separately to factors in a product. For example, if $F_i = Ev_i$, $i = 1, 2, 3$, then

$$\sum_{Q \in \mathcal{Q}} \prod_{i=1}^3 \sup_Q |F_i| \leq C_N \int_{(\mathbb{R}^3)^3} \prod_{i=1}^3 (1 + |w_i|)^{-N} \int_{\bigcup \mathcal{Q}} \prod_{i=1}^3 |Ev_i(z + w_i)| dz dw_1 dw_2 dw_3. \quad (2.16)$$

To see this, multiply (2.15) for the three fields, integrate over each unit cube, and use Tonelli. For each fixed triple of shifts, the three exact fields are tested at the same integration point on the same domain. Their data have independent unimodular modulations, which preserve supports, norms, and frequency transversality. This is the version used when applying multilinear restriction to products of cube suprema.

2.4 Weights and cap rescaling

Lemma 2.4 (Weighted refined estimate). *Let ν be supported in a ball of radius CS , $S \geq 1$, and suppose*

$$\nu(B(z, r)) \leq mr^2 \quad (r \geq 1), \quad \nu(B(z, \sqrt{S})) \leq mSd \quad (z \in \text{supp } \nu), \quad (2.17)$$

where $m, d > 0$. For bounded frequency support,

$$\int |Ev|^2 d\nu \leq C_\epsilon S^\epsilon m S^{2/3} d^{1/3} \|v\|_2^2. \quad (2.18)$$

The same bound holds for $\sum_Q \nu(Q) \sup_Q |Ev|^2$, summed over unit cubes. The ball bound alone gives this conclusion with $d = 1$, up to a fixed constant.

Proof. Fixed-radius enlargements of the second bound in (2.17) follow by a bounded covering with balls of radius $\sqrt{S}$ centered on the support. This also extends it to arbitrary centers, up to a fixed constant. Every unit cube has mass at most Cm . Split the positive-mass cubes into classes

$$m\chi < \nu(Q) \leq 2m\chi, \quad \chi \in \{C_0 2^{-k} : k \geq 0\},$$

with harmless endpoint adjustments and fixed C_0 . Cubes in one class have counting parameter $C\chi^{-1}$: the cubes meeting an r -ball have disjoint masses inside a fixed enlargement of that ball. Their square-root-scale occupancy is at most $CSd\chi^{-1}$ by the same argument. Lemma 2.3 therefore gives

$$\begin{aligned} \sum_{Q \text{ in the } \chi \text{ class}} \nu(Q) \sup_Q |Ev|^2 &\leq C_\epsilon m\chi S^\epsilon (\chi^{-1} \cdot Sd\chi^{-1} \cdot S)^{1/3} \|v\|_2^2 \\ &= C_\epsilon S^\epsilon m S^{2/3} d^{1/3} \chi^{1/3} \|v\|_2^2. \end{aligned}$$

The series $\sum_\chi \chi^{1/3}$ converges. Summing proves both assertions, including measures with arbitrarily small positive cube masses; no lower mass cutoff is needed here. Finally the first ball bound, evaluated at $r = \sqrt{S}$, supplies $d = 1$. $\square$

Lemma 2.5 (Weighted cap estimate). *Fix $\alpha_0 < 1/2$. Suppose that $S \geq 1$, that ν is supported in a ball of radius CS and satisfies $\nu(B(z, r)) \leq mr^2$ for $r \geq 1$, and that v is supported in a frequency ball $B(v_0, q)$, with v_0 in a fixed bounded set and*

$$S^{-\alpha_0} \leq q \leq 1.$$

Then, for every $\epsilon > 0$,

$$\int |Ev|^2 d\nu \leq C_{\epsilon, \alpha_0} S^\epsilon m S^{2/3} q^{1/3} \|v\|_2^2. \quad (2.19)$$

The estimate also holds with the integral replaced by $\sum_Q \nu(Q) \sup_Q |Ev|^2$ over unit cubes. Bounded translations, and averages of translations with a fixed integrable Schwartz weight, obey the same bounds.

Proof. Translate the space-time support ball to the origin, absorbing the translation in a unimodular factor on v . Write

$$g(\eta) = q v(v_0 + q\eta), \quad (y, \tau) = (q(x - 2v_0 t), q^2 t), \quad S' = Sq^2.$$

Then $\|g\|_2 = \|v\|_2$, $\text{supp } g \subset B(0, 1)$, and

$$Ev(x, t) = q e^{i(x \cdot v_0 - t|v_0|^2)} Eg(y, \tau). \quad (2.20)$$

Let ν' be the pushforward of ν . The inverse map is

$$t = q^{-2}\tau, \quad x = q^{-1}y + 2v_0 q^{-2}\tau.$$

Consequently, the inverse image of a ball of radius $r \geq 1$ is a bounded shear of a box with side lengths $r/q, r/q, r/q^2$. It is covered by $O(q^{-1})$ balls of radius Cr/q . The original ball bound applies at these radii and gives

$$\nu'(B(z, r)) \leq Cmq^{-3}r^2 \quad (r \geq 1). \quad (2.21)$$

There is no Jacobian multiplier on this pushforward measure.

The rescaled time interval has length $O(S')$, but the spatial projection has diameter $O(Sq) = O(S'/q)$. Partition the latter into squares Q of side S' . On the common time slab, use Lemma 2.2 at packet width $s = \sqrt{S'}$, with any fixed padding exponent in $(0, 1)$. Let g_Q denote its localized data. Then

$$\sum_Q \|g_Q\|_2^2 \leq C\|g\|_2^2, \quad |Eg - Eg_Q| \leq C_A(S')^{-A}\|g\|_2 \quad \text{on the corresponding prism,}$$

where A can be made arbitrarily large. Each restricted measure lies in a ball of radius CS' , so Lemma 2.4 with $d = 1$ and (2.21) gives

$$\sum_Q \int_{Q \times [-CS', CS']} |Eg_Q|^2 d\nu' \leq C_\epsilon(S')^\epsilon m q^{-3} (S')^{2/3} \|g\|_2^2.$$

The sum uses the localized energies, rather than the number of spatial squares. This distinction is essential for the cap power.

For completeness, $\nu'(\mathbb{R}^3) = \nu(\mathbb{R}^3) \leq C m S^2$, whereas $S' \geq S^{1-2\alpha_0}$. Thus the integrated squared error, after multiplication by the q^2 in (2.20), is at most

$$C_A m q^2 S^2 (S')^{-2A} \|g\|_2^2.$$

Choose A in terms of α_0 so that this is bounded by the desired right side; bounded S is absorbed in the constant. For the principal term,

$$q^2 m q^{-3} (S q^2)^{2/3} = m S^{2/3} q^{1/3}. \quad (2.22)$$

Since $(S')^\epsilon \leq S^\epsilon$, this proves (2.19).

Finally, square (2.15) by weighted Cauchy–Schwarz, integrate in $z \in Q$ against $d\nu(z)$, and sum the original unit cubes. This bounds the sum of weighted suprema by

$$C_N \int_{\mathbb{R}^3} (1 + |w|)^{-N} \int_{\mathbb{R}^3} |Ev(z + w)|^2 d\nu(z) dw.$$

For every shift the field is an exact extension of modulated data with the same cap support and norm. Apply the already proved cap estimate and integrate the rapidly decreasing weight. This proves the supremum assertion and the asserted translated versions. $\square$

Remark 2.6 (Enlarged cubes in the inductive formulation). The proof of Lemma 2.3 is sufficient for all later applications. For comparison, we record an alternative reduction of (2.13) to the inductive Proposition 3.1 of [13]. Choose $K = S^\delta$ with $0 < \delta < 1/4$ sufficiently small relative to the desired loss, and enlarge occupied unit cubes to occupied K^2 -cubes. Selecting one original unit cube in each occupied large cube shows, by ball covering, that the large cubes have counting parameter $O(\gamma)$ for $r \geq K^2$, and square-root-scale occupancy $O(\lambda_0)$. Normalize $\|v\|_2 = 1$ and discard large cubes with L^6 norm below S^{-A} . There are polynomially many cubes, so their total L^2 contribution is arbitrarily small when A is sufficiently large. The remaining L^6 norms have only $O(\log S)$ dyadic classes, since $|Ev| \leq C$ and each large cube has volume K^6 .

After fixing a norm class, recompute its parent occupancies and sort them dyadically. On each resulting union Y of M_Y large cubes, Proposition 3.1 gives

$$\|Ev\|_{L^6(Y)} \leq C_\epsilon M_Y^{-1/3} (\gamma \lambda_0 S)^{1/6} S^\epsilon \|v\|_2.$$

Hölder contributes $|Y|^{1/3} = M_Y^{1/3} K^2$ on passing to L^2 , cancelling $M_Y^{-1/3}$. After squaring, the additional loss is K^4 ; the finitely many logarithmic losses and this factor are absorbed by the choice of δ and a smaller initial ϵ .

The grids in this deduction can be made compatible as follows. Increase the ambient scale by a bounded factor so that $S^{1/2}/S^{2\delta}$ is an integer. Such scales have successive ratios tending to one. Use nested grids at these two scales and a bounded family of translated parent grids. For sufficiently large S , each original unit cube can be assigned to a parent grid in which it is farther than $2\sqrt{3}K^2$ from the parent boundary; keep only the occupied large cubes within that parent. Split the original collection according to these finitely many assignments. The same original ball counts control each part, while translation is absorbed into the datum as in Lemma 2.3. Bounded scales are covered by a fixed constant. Thus neither boundary cubes nor rounding changes the resulting estimate.

3 A power saving for transverse inputs

The first step is a rigidity statement for almost extremizers of the fractal estimate. A hypothetical extremizing configuration produces a probability distribution on points and incident tubes. Information at successive spatial scales first forces three identical one-dimensional profiles, then forces those profiles to alternate only finitely often between slopes zero and one. Polynomial partitioning rules out the remaining transition.

Proposition 3.1 (Transverse power saving). *Fix a bounded frequency region. There are $c > 0$ and $L_0 < \infty$, depending only on that region, for which the following configuration cannot exist when $L \geq L_0$. Let $\mathcal{Q}$ be unit lattice cubes inside a ball of radius L^2 in $\mathbb{R}^3$, with*

$$\#\mathcal{Q} \geq L^{4-c}, \quad \#\{Q \in \mathcal{Q} : Q \cap B(z, r) \neq \emptyset\} \leq L^c r^2 \quad (r \geq 1),$$

where cube counts may equivalently use their centers, up to fixed constants. Suppose v_1, v_2, v_3 have frequency support in the fixed region, $\|v_i\|_2 \leq 1$, and

$$\sup_Q |Ev_i| \geq L^{-4/3-c} \quad (Q \in \mathcal{Q}, i = 1, 2, 3).$$

Then it is impossible that every triple of unit frequency directions, one from each support, has absolute determinant at least L^{-c} .

We argue by contradiction along $L \rightarrow \infty$ and $c \downarrow 0$. Every $o(1)$ in this section refers to this sequence or a subsequence. A diagonal choice converts estimates valid with every fixed positive power loss into $L^{o(1)}$ estimates; constants are fixed before making that choice. All rounding, support enlargements, and packet padding are included in these subpower errors. The same convention applies when a scale is a fixed positive power of L .

3.1 Uniform multilinear estimates and the tube model

We need the multilinear estimates when transversality itself deteriorates by a subpower. An unspecified transversality-dependent constant in the restriction theorem would not justify this use. The following finite induction supplies the required uniformity.

Lemma 3.2 (Multilinear estimates with subpower transversality). *Suppose three exact extension inputs have bounded frequency supports, and every triple of their unit directions has determinant at least ν , where $0 < \nu \leq 1$. For each $\epsilon > 0$ there are $C_\epsilon, M_\epsilon < \infty$ such that, on every ball of radius $S \geq 2$,*

$$\int_{B_S} \prod_{i=1}^3 |Ev_i(z)| dz \leq C_\epsilon \nu^{-M_\epsilon} S^\epsilon \prod_{i=1}^3 \|v_i\|_2. \quad (3.1)$$

The same conclusion holds for the sum over unit lattice cubes in the ball of $\prod_i \sup_Q |Ev_i|$. In particular, when $\nu = S^{-o(1)}$ both bounds are $S^{o(1)} \prod_i \|v_i\|_2$. For weighted tubes of width Δ in a bounded region, the corresponding multilinear Kakeya bound is

$$\int \prod_{i=1}^3 \left(\sum_T w_i(T) \mathbf{1}_T \right)^{1/2} \leq C_\epsilon \nu^{-M_\epsilon} \Delta^{3-\epsilon} \prod_{i=1}^3 \left(\sum_T w_i(T) \right)^{1/2}. \quad (3.2)$$

Here fixed enlargements of tubes and of the ambient region are allowed; if those enlargements are subpower on a sequence, they cost a subpower.

Proof. Partition each direction support into neighborhoods of radius a sufficiently small fixed power of ν . There are at most $\nu^{-O(1)}$ such neighborhoods. For each triple choose representative directions and send them to the coordinate axes by a linear map. Its norm, inverse norm, and volume distortion are bounded by fixed powers of ν^{-1} . The transformed direction neighborhoods lie in fixed small neighborhoods of the axes. The weighted multilinear Kakeya inequality of [1, Theorem 1.15 and Remark 1.10], applied there and summed over triples, proves (3.2). Weights follow first by repetitions and then by approximation. Widths larger than a fixed constant can instead be treated by the trivial bound, after increasing the inverse-power constant.

For completeness, use this Kakeya inequality to obtain restriction uniformity. Let $A_B(S, \nu)$ be the least normalized product-integral constant for inputs supported in $[-B, B]^2$. Fix padding exponent $\theta > 0$ and Kakeya loss $\sigma > 0$. Cover a ball of radius S by cubes of side comparable to $\sqrt{S}$. The exact packet decomposition restricts each input on a smaller cube to packets incident to its $S^\theta \sqrt{S}$ enlargement. Its retained exact input has squared norm at most a fixed constant times the incident coefficient energy. Its support is within $O_B(S^{-1/2})$ of the original support: only caps intersecting that support occur, and every enlarged cap lies within this distance of such an intersection. Thus determinant continuity preserves $\nu/2$ transversality when $S^{-1/2} \leq c_B \nu$, without any openness or connectedness assumption on the supports.

Apply the smaller-scale product estimate in every cube. Its geometric coefficient is the product of the three square roots of incident energies. After division of coordinates by S , the padded tubes have width $\Delta = S^{-1/2+\theta}$. Enlarge them by a fixed factor, so an incident tube contains the whole smaller grid cube. Equation (3.2), divided by that cube's volume $S^{-3/2}$, gives the bound

$$C_{B,\theta,\sigma} \nu^{-M_\sigma} S^{3/2} (S^{-1/2+\theta})^{3-\sigma} \prod_i \|v_i\|_2 \leq C_{B,\theta,\sigma} \nu^{-M_\sigma} S^\beta \prod_i \|v_i\|_2, \quad \beta = 3\theta + \sigma/2,$$

for the sum of these geometric coefficients. To bound discarded-field errors, normalize each parent input, use rapid tails for a discarded factor and the fixed-frequency L^∞ bound for the others, and expand the three-factor product. The seven error terms together contribute at most $C_{B,\theta,N} S^{3-N} \prod_i \|v_i\|_2$, for every prescribed N . This is relative to the current *parent* norms; no lower bound for a retained local norm is required. We obtain the recurrence

$$A_B(S, \nu) \leq C_{B,\theta,\sigma} \nu^{-M_\sigma} S^\beta A_{B+1}(C_B \sqrt{S}, \nu/2) + C_{B,\theta,N} S^{3-N}, \quad S^{-1/2} \leq c_B \nu. \quad (3.3)$$

Spatial translations only modulate the exact inputs. Taking $N > 3$ and setting $\tilde{A}_B = 1 + A_B$ absorbs the additive error in the same multiplicative recurrence for $\tilde{A}_B$.

Fix the target $\epsilon > 0$. Choose a finite J with $3 \cdot 2^{-J} < \epsilon/2$, then choose θ, σ with $2\beta < \epsilon/2$. The finitely many radii defined by $S_0 = S$, $S_{j+1} = C_{B+j} \sqrt{S_j}$ are comparable to $S^{2^{-j}}$, with constants depending only on B, J . If $S \geq (C_{B,J}/\nu)^{2^J}$ for a sufficiently large constant, all J support-enlargement conditions hold. Iterating (3.3) and using the terminal trivial bound $\tilde{A}_{B+J}(R, \nu') \lesssim_{B,J} R^3$ gives

$$\tilde{A}_B(S, \nu) \leq C_{B,\epsilon} \nu^{-M_{B,\epsilon}} S^\beta \sum_{j=0}^{J-1} 2^{-j+3 \cdot 2^{-j}} \leq C_{B,\epsilon} \nu^{-M_{B,\epsilon}} S^\epsilon.$$

In the complementary range, the trivial bound $A_B(S, \nu) \lesssim_B S^3$ is absorbed by increasing the fixed inverse-transversality exponent to at least $3 \cdot 2^J$. This proves (3.1) for all S, ν . On a subpower-transverse sequence the threshold is eventually satisfied for each fixed J , and every intermediate enlarged support remains subpower transverse. This is a finite version of the restriction bootstrapping in [1, Proposition 2.1]; polynomial dependence is supplied by this argument, not by the statement of their restriction theorem alone.

Finally, fixed-frequency reproducing kernels give, for $z \in Q$,

$$\sup_Q |Ev_i| \lesssim_N \int (1 + |w|)^{-N} |Ev_i(z + w)| dw.$$

Integrate the product over Q and sum. For each triple of shifts the three shifted fields are exact extensions on the same domain, with independently modulated inputs of unchanged norms and supports. Apply (3.1) and integrate the Schwartz weights. $\square$

Lemma 3.3 (Reduction to weighted incident tubes). *A countersequence to Proposition 3.1 yields a set P in a fixed bounded region and three weighted tube families satisfying*

$$|P| = L^{2+o(1)}, \quad |P \cap B(z, r)| \leq L^{2+o(1)} r^2 \quad (L^{-1} \leq r \leq 1). \quad (3.4)$$

The points are centers of distinct cubes of side comparable to L^{-1} . The tubes have width $L^{-1+o(1)}$, their coefficient weights obey $\sum_T w_i(T) \lesssim 1$, and every cross-family direction triple has determinant at least $L^{-o(1)}$. At each $p \in P$ there is a retained incident subfamily of total weight $e_i^(p) = L^{-4/3+o(1)}$ such that its normalized weight law satisfies, for fixed $\kappa > 0$, $u_0 > 0$,*

$$\mathbb{P}(\text{dir } T_i \in B(v, L^{-u}) \mid X = p) \leq L^{-\kappa u + o(1)} \quad (0 < u < u_0, u \in \mathbb{Q}). \quad (3.5)$$

This bound is uniform in p, v for each fixed rational u . The retained subfamily may depend on p .

Proof. Sort occupied L -boxes by their number of selected unit cubes. A dyadic occupancy class retains $L^{4-o(1)}$ cubes. The squared supremum sum for each field on this class is at least $L^{4/3-o(1)}$. By Lemma 2.3 at radius L^2 , a class with occupancy at most $L^{2-\delta}$, for fixed $\delta > 0$, instead has squared sum at most $L^{4/3-\delta/3+o(1)}$. Thus some retained class has $L^{2-o(1)}$ cubes in each occupied box. The upper counting bound gives occupancy at most $L^{2+o(1)}$. Divide coordinates by L^2 and take the centers of these boxes. Counting cubes in an enlarged ball and dividing by the minimum occupancy proves (3.4). Translation of the original ambient ball merely modulates the exact data.

Apply Lemma 2.2 at packet width L on the time interval of length $O(L^2)$. After division of coordinates by L^2 , the resulting tubes have width $L^{-1+o(1)}$. Give each tube the squared coefficient of its packet as weight. On each retained box, packets reaching a padded enlargement give an exact local input and an error $O(L^{-100})$. The padding exponent tends sufficiently slowly to zero; the rapid-tail estimates permit this by diagonalization. By (2.8), the squared norm of this exact local input is bounded by the incident coefficient energy $e_i(p) = \sum_{T \ni p} w_i(T)$ up to a constant. On the at least $L^{2-o(1)}$ selected unit cubes in this box the required squared supremum sum is $L^{-2/3-o(1)}$. The weighted fractal estimate at radius L bounds it above by $L^{2/3+o(1)} e_i(p)$, so

$$e_i(p) \geq L^{-4/3+o(1)}.$$

Lemma 3.2, in its Kakeya form and with $\Delta = L^{-1+o(1)}$, yields

$$\sum_{p \in P} \prod_{i=1}^3 e_i(p)^{1/2} \leq L^{o(1)}.$$

For fixed $\delta > 0$, points at which some $e_i(p) \geq L^{-4/3+\delta}$ therefore number at most $L^{2-\delta/2+o(1)}$. Discarding these points by a diagonal choice leaves (3.4) and makes all three energies $L^{-4/3+o(1)}$ uniformly.

It remains to remove concentration in direction without losing the local lower bound. Choose $0 < \kappa < 1/3$ and $0 < u_0 < 1/4$. For each fixed rational $0 < u < u_0$, put $q = L^{-u}$ and partition directions into bounded-overlap bins of diameter comparable to q . Delete incident bins having weight greater than $q^\kappa e_i(p)$. There are $O(q^{-\kappa})$ such bins. On the original box of radius $O(L)$, Lemma 2.5 applies with $S = L$; the packet frequency enlargement $O(L^{-1})$ is smaller than the bin width q . Cauchy–Schwarz over the deleted bins then bounds the squared supremum sum of their combined field by

$$L^{2/3+o(1)} q^{1/3-\kappa} e_i(p).$$

Here the chosen unit cubes have local density $L^{o(1)}$, and the energies of all deleted bins sum to $O(e_i(p))$. The estimate also holds for arbitrary subfamilies of these bins. For every fixed $u > 0$ this is negligible relative to $L^{-2/3-o(1)}$.

Delete successively for any fixed finite list of rational u 's, assigning a packet deleted more than once to just one bin. The triangle inequality in the square-sum norm shows that the surviving field retains its lower local norm. The unrestricted local estimate then gives surviving weight $e_i^*(p) = L^{-4/3+o(1)}$. Every surviving bin has original weight at most $q^\kappa e_i(p)$; normalizing by $e_i^*(p)$ costs only $L^{o(1)}$. Boundedly many neighboring bins cover a ball, and a diagonal choice proves (3.5) at all fixed rational scales. All estimates were uniform in p , so this point-dependent pruning supplies legitimate conditional laws without requiring a single globally pruned family. $\square$

3.2 Information at spatial scales

From now on only the abstract configuration of Lemma 3.3 is used. Draw X uniformly from P . Given X , draw six tubes T_i, T'_i independently, two from each retained family. A tube label includes its exact direction. For rational $0 < s \leq 1$, let X_s be the cell containing X in nested dyadic grids of side comparable to L^{-s} ; use the trivial partition at $s = 0$. For a discrete variable Y and arbitrary conditioning V , write

$$I_L(Y \mid V) = -\log_L \mathbb{P}(Y = Y(\omega) \mid V)$$

for its information evaluated at the sampled labels. Conditional probabilities may be fixed in any version on null sets. This is the exponent of the reciprocal probability of the realized label, rather than its expectation: a sampled label of conditional probability $L^{-a+o(1)}$ has information $a + o(1)$. We retain these sampled exponents in one joint limiting law so that comparisons at different spatial scales refer to the same configuration.

Lemma 3.4 (Limiting information calculus). *After a subsequence, all the countably many information variables used below have a common limiting joint law. For spatial, coordinate, and direction partitions with at most $L^{C+o(1)}$ possible labels, their conditional informations are uniformly integrable and have bounded limiting values. In this law the chain rule remains exact and*

$$I(Y \mid V) \geq I(Y \mid V, W) \quad \text{almost surely.} \quad (3.6)$$

If, given Z, V , the possible values of Y on an event E_L number at most $L^{\gamma+o(1)}$, include its indicator in the joint limit. On the event where the limiting indicator equals one, $I(Y \mid Z, V) \leq \gamma$. Alternatively, an event specified by a finite system of sampled inequalities can be treated by open-set implications with fixed slack before that slack tends to zero; no identification of arbitrary boundary events is implicit. Consequently variables determining one another up to $L^{o(1)}$ possibilities have identical limiting information, also after additional conditioning. Finite systems of limiting inequalities holding with positive probability pass to finite L with any fixed positive slack.

Proof. A variable with at most $L^{C+o(1)}$ labels satisfies, uniformly in conditioning,

$$\mathbb{P}\{I_L(Y \mid V) > C + \eta\} \leq L^{-\eta+o(1)}.$$

The same counting argument with a varying threshold proves uniform integrability. For the conditioning assertion put $p = \mathbb{P}(Y \mid V)$ and $q = \mathbb{P}(Y \mid V, W)$ at the sampled labels. Conditional on (V, W) , the expectation of p/q is at most one. Thus

$$\mathbb{P}\{I_L(Y \mid V) - I_L(Y \mid V, W) < -\eta\} \leq L^{-\eta}.$$

This proves the almost-sure sign in the limit; no finite-scale pointwise monotonicity is asserted. The chain rule is pointwise already at finite scale. The determination rule follows by summing over the stated small set of possible labels. This argument also works on restricted events, since it bounds the total mass of the bad low-probability labels in the event's permitted set. If the event indicator is included in the joint limit, apply weak convergence to the open set where it is one and the information is strictly above $\gamma + \eta$, and then let $\eta \downarrow 0$. For events defined through other limiting variables, the same argument uses strict inequalities and slack, as detailed in the zero-ratio application below.

Tightness and diagonal weak convergence on a countable product give the joint law. Include in this product all rational block and prefix parameters, all finitely many choices of reference and replacement tubes, their direction bins, coordinate grids, projections, the information differences used below, and the angular quantities to be introduced. Direction grids use deterministic representatives of occupied bins. Realized bases and inverse bases have norms $L^{o(1)}$, so every coordinate partition has at most $L^{C+o(1)}$ labels. Standard open-set inequalities for weak convergence justify the assertion about finite systems after enlarging their bounds by slack. Uniform integrability permits passage of expectations in the cost estimates below. $\square$

Lemma 3.5 (The spatial profiles). *In the limiting law set*

$$F(s) = I(X_s), \quad G_i(s) = I(X_s \mid T_i), \quad G'_i(s) = I(X_s \mid T'_i).$$

These functions extend from rational scales to nondecreasing Lipschitz functions on $[0, 1]$, vanishing at zero. Their Lipschitz constants are respectively 3 and 1. Almost surely,

$$F(s) \geq 2s, \quad F(1) = 2, \quad G_i(1), G'_i(1) \geq \frac{2}{3}. \quad (3.7)$$

Proof. The exact chain increment of G_i between s and $s + t$ is $I(X_{s+t} \mid X_s, T_i)$. Inside a parent, the tube meets at most $L^{t+o(1)}$ child cubes, including its padding. The determination rule bounds this increment between zero and t ; the unrestricted spatial count gives the bounds zero and $3t$ for F . Rational comparisons simultaneously yield the Lipschitz extensions.

The ball count in (3.4) gives every X_s -cell mass at most $L^{-2s+o(1)}$, proving $F(s) \geq 2s$. An X_1 -cell contains boundedly many points and X is uniform, so $F(1) = 2$. Finally the retained weights give

$$\mathbb{P}(X = p, T_i = T) = |P|^{-1} \frac{w_i(T) \mathbf{1}_{\{T \text{ retained at } p\}}}{e_i^*(p)} \leq L^{-2/3+o(1)} w_i(T).$$

The probability of sampling a tube whose marginal is smaller than $L^{-\eta} w_i(T)$ is at most $L^{-\eta} \sum_T w_i(T) = O(L^{-\eta})$. Outside this event, every conditional point mass is at most $L^{-2/3+\eta+o(1)}$. Letting $\eta \downarrow 0$ proves the conditional endpoint bounds, also for all primed tubes. This uses only summability of weights, so exact tube labels need not have bounded cardinality. $\square$

3.3 Direction costs and equality of the profiles

To compare the three tube-conditional profiles, we use coordinates adapted to one reference tube from each family. Specifying those coordinates requires revealing coarse direction bins. This can lower the spatial information; the costs below measure that loss, and their telescoping bound will let the loss tend to zero with the block size.

Fix a rational block $[s, s + b] \subset [0, 1]$ and rational $b_* \geq b$. Let $C = X_s$ and let D record direction bins of precision L^{-b_*} for a fixed sublist of the six tubes, including one reference

tube from each family. Within the parent subtract its center, multiply coordinates by L^s , and use the approximate basis supplied by the three reference bins. Its norms and inverse norms are $L^{o(1)}$. For rational $0 < t \leq b$ let Z_i be its i th coordinate grid index at resolution L^{-t} , and write $Z = (Z_1, Z_2, Z_3)$ and $Z_{\hat{i}} = (Z_j)_{j \neq i}$. These coordinate grids are nested in t . When extra bins are added to D , the basis is still selected using only the three reference bins, so the coordinate variables do not change.

Lemma 3.6 (Block comparison and telescoping costs). *Write $J = F(s+t) - F(s)$ and $J_i = G_i(s+t) - G_i(s)$ for the chosen reference tubes. Define*

$$\ell = I(X_{s+t} | C) - I(X_{s+t} | C, D), \quad \ell_i = I(X_{s+t} | C, T_i) - I(X_{s+t} | C, T_i, D).$$

All these limiting costs are nonnegative, and

$$\sum_{i=1}^3 (J_i - \ell_i) \leq \sum_{i=1}^3 I(Z_i | Z_{\hat{i}}, C, D) \leq I(Z | C, D) = J - \ell. \quad (3.8)$$

For disjoint blocks or disjoint prefixes, with the same D , the sum of expectations of each type of cost is $O(b_)$. The same statement holds with any one fixed other tube in place of T_i . All coordinatewise deficits obtained by using any ordering in the second inequality of (3.8) are nonnegative.*

Proof. The spatial child and the three coordinate labels determine one another up to $L^{o(1)}$ choices. A reference tube determines its two off-axis coordinates to $L^{o(1)}$ choices: both its width and the error in its approximate direction, relative to the parent, are at most $L^{-t+o(1)}$. Conditional information calculus therefore gives

$$J_i - \ell_i = I(Z_i | C, D, T_i) = I(Z_i | Z_{\hat{i}}, C, D, T_i) \leq I(Z_i | Z_{\hat{i}}, C, D).$$

For each ordering of the coordinates, its chain-rule term conditions on a subset of $Z_{\hat{i}}$, hence dominates the last expression. Sum the three terms to obtain (3.8).

At finite scale, the expected cost is the conditional mutual information, divided by $\log L$, between the child and D given the parent (and, when present, the same fixed tube). Along the nested spatial filtration these mutual informations telescope. Inserting the omitted gaps between disjoint blocks only adds nonnegative expected terms. Their sum is at most $H(D)/\log L = O(b_*)$, because there are $L^{O(b_*)}$ direction-bin lists. Fixed tube conditioning is never changed within this telescoping argument. Passage to the limit follows from Lemma 3.4. $\square$

Lemma 3.7 (Additivity and replacement invariance). *Almost surely, simultaneously for every choice of one of the two tubes in each family,*

$$F = G_1 + G_2 + G_3, \quad G'_i = G_i, \quad G_i(1) = \frac{2}{3}.$$

In any block above, including a prefix and with any extra direction bins in D ,

$$\begin{aligned} I(Z_i | C, D) &= J_i + O\left(\sum_k \ell_k\right), \\ I(Z_i, Z_j | C, D) &= J_i + J_j + O\left(\sum_k \ell_k\right). \end{aligned} \quad (3.9)$$

The absolute constants in these error bounds are fixed.

Proof. Partition a fixed rational spatial interval into blocks of length at most $b = b_*$. By (3.8), the positive part of the excess of the summed G_i increment over the F increment has expectation $O(b)$. Let $b \downarrow 0$ to obtain the interval inequality almost surely. At $[0, 1]$, (3.7) forces equality and $G_i(1) = 2/3$. The interval deficits are nonnegative and additive, so their zero full-interval total forces all rational subinterval deficits to vanish. Continuity gives $F = \sum_i G_i$. Repeat for the finitely many replacement triples; subtracting identities gives $G'_i = G_i$.

For any coordinate ordering, its i th chain term dominates $I(Z_i | Z_{\hat{i}}, C, D)$ and then $J_i - \ell_i$. The sum of the surpluses is exactly $\sum_i \ell_i - \ell \leq \sum_i \ell_i$, using $J = \sum_i J_i$. Take a chosen coordinate, or chosen pair, first in the ordering. This gives both bounds in (3.9), with each individual error controlled by the total cost. $\square$

Lemma 3.8 (One common profile). *Almost surely,*

$$G_1 = G_2 = G_3 =: G, \quad F = 3G. \quad (3.10)$$

In particular $0 \leq \dot{G} \leq 1$, $G(1) = 2/3$, and $3G(s) \geq 2s$.

Proof. In the exact basis of T_1, T_2, T_3 , normalize the i th coordinate of the direction of T'_i to one, and call its other two ratios a_j ($j \neq i$). Replacement transversality and Cramer's rule bound the normalizing denominator, its inverse, and all resulting ratios by subpowers. Include in the joint limit the values $\min\{1, \max\{0, -\log_L |a_j|\}\}$, with value one at zero. The two limiting values cannot both be positive: that would put T'_i within a fixed positive-power angular ball about T_i , with probability tending to zero by (3.5) and conditional independence given X .

Let r_j be the limiting clipped value for a_j . We make the restricted-event comparison on $\{r_j = 0\}$ precise. Fix a rational $0 < \eta < t/2$. At finite scale, $r_{j,L} < \eta/2$ implies $|a_j| > L^{-\eta/2}$. The approximate basis fixed by D differs from the exact basis by $L^{-b_*+o(1)}$, including the subpower condition numbers. Its corresponding coefficient thus has magnitude at least $L^{-\eta}$ for large L . Given (T'_i, Z_j, C, D) alone, possible child positions on this event number at most $L^{O(\eta)+o(1)}$: the approximate j th coordinate parametrizes the tube with inverse slope at most $L^{\eta+o(1)}$. This permitted set depends only on those displayed labels. Exact reference directions need not be added to the conditioning.

Apply the restricted determination bound and the information conditioning rule. The probability that $r_{j,L} < \eta/2$ and

$$I_L(X_{s+t} | C, D, T'_i) - I_L(Z_j | C, D) > C\eta + \eta$$

tends to zero, with a fixed absolute C . The joint limit therefore gives the corresponding comparison on the open event $\{r_j < \eta/2\}$ with error at most $(C+1)\eta$; one can first use any strict upper error bound and then decrease it. This is an implication about the support of the joint limiting law and requires no coupling of finite-scale and limiting events. Let rational $\eta \downarrow 0$. On $\{r_j = 0\}$ we obtain

$$G'_i(s+t) - G'_i(s) - \ell'_i \leq I(Z_j | C, D),$$

where D includes the alternative bin and ℓ'_i is its analogous cost. Replacement invariance and (3.9) now compare the i and j profile increments up to the costs.

Take dyadic partitions of the exponent interval, with block size $b = b_*$. On each block distribute its cost uniformly over its length. The integral of this density is the cost; hence its expected integral is $O(b)$ by Lemma 3.6. Summability over dyadic b shows that all normalized costs tend to zero for almost every pair consisting of a path and an exponent. At simultaneous differentiability points it follows that $\dot{G}_i \leq \dot{G}_j$ whenever the corresponding

limiting ratio information is zero. The restricted-event determination just used is applied before differentiation, so no uniform positive lower bound for the coefficient is required.

For almost every path each vertex i therefore has an edge $i \rightarrow j$ for which $\hat{G}_i \leq \hat{G}_j$ almost everywhere. Since both integrals are $2/3$, this forces $G_i = G_j$. Every equivalence class of equal profiles has at least two members. With three families there is only one such class, proving (3.10). $\square$

3.4 Projection growth on a block

The next input is the robust form of the discretized projection theorem: one direction improves the projection of every sufficiently large subset. That quantifier is essential because the subset we construct will depend on the direction.

Lemma 3.9 (Robust planar discretized projection). *Fix $0 < d < 1$ and $\rho_0 > 0$. There are $\varepsilon_{\text{pr}} > 0$ and $\eta_{\text{pr}} > 0$ with the following property for all sufficiently small Δ . Suppose $A \subset B(0, 1) \subset \mathbb{R}^2$, with $\mathcal{N}_\Delta$ denoting covering number by radius- Δ balls, satisfies*

$$\Delta^{-2d+\eta_{\text{pr}}} \leq \mathcal{N}_\Delta(A) \leq \Delta^{-2d-\eta_{\text{pr}}}, \quad \mathcal{N}_\Delta(A \cap B(z, r)) \leq \Delta^{-\eta_{\text{pr}}} r^{\rho_0} \mathcal{N}_\Delta(A)$$

for $\Delta \leq r \leq 1$. Let σ be a probability on projection lines obeying $\sigma(B(V, r)) \leq \Delta^{-\eta_{\text{pr}}} r^{\rho_0}$ on the same range. Some $V \in \text{supp } \sigma$ then satisfies

$$\mathcal{N}_\Delta(\pi_V A') \geq \Delta^{-d-\varepsilon_{\text{pr}}} \quad \text{for every } A' \subset A \text{ with } \mathcal{N}_\Delta(A') \geq \Delta^{\eta_{\text{pr}}} \mathcal{N}_\Delta(A).$$

One may decrease both the tolerance and the gain.

Proof. This is the $n = 2$, $m = 1$, $\alpha = 2d$ case of [17, Theorem 1], a general form of the planar Bourgain discretized projection theorem. In dimension two its Grassmannian nonconcentration condition is the displayed angular-ball condition. That theorem gives the conclusion simultaneously for every large subset, for a direction set of probability at least one minus a positive power of Δ . Decreasing the tolerance relative to its exponent gives the stated formulation. $\square$

Take T_1, T_2, T_3 as reference tubes and write the direction of T'_1 in their exact basis as $(1, a_2, a_3)$. Given X and these three exact references, redraw T'_1 from its original conditional law. Let $h_j(u)$, for rational $0 < u < u_0$, be the limiting value of the negative base- L logarithm of the resulting marginal mass in the ball of radius L^{-u} centered at the sampled a_j .

Lemma 3.10 (Angular marginal information). *The functions h_2, h_3 are nonnegative and nondecreasing on their rational domains, and almost surely*

$$h_2(u) + h_3(u) \geq \kappa u \quad (0 < u < u_0, u \in \mathbb{Q}). \quad (3.11)$$

Their values agree with those of marginal coordinate-cell information at comparable resolution; the analogous joint assertion also holds.

Proof. The exact coordinate map and its inverse have subpower norm on the realized support. A joint ball in (a_2, a_3) therefore pulls back to at most two direction balls of radius $L^{-u+o(1)}$. The alternative direction law, conditional on X and the exact references, is still its original law given X . Equation (3.5), enlarged by a subpower using covering, bounds this joint ball mass by $L^{-\kappa u+o(1)}$.

Here ball and grid informations have the same limit. Conditional on the exact point and references, take grids at comparable sizes. If p_z is the mass of a sampled cell and q_z the total mass of its boundedly many neighbors, then

$$\sum_{z: q_z > L^\eta p_z} p_z \leq L^{-\eta} \sum_z q_z \lesssim L^{-\eta}.$$

A boundedly finer grid and the determination rule handle balls not contained in one grid cell. The argument is identical for joint and marginal grids. The chain rule and (3.6) bound joint cell information by the sum of marginal informations. This proves (3.11); monotonicity follows directly from nesting of centered balls. $\square$

Lemma 3.11 (The projection block test). *Fix $0 < d < 1$, $d_0 > 0$, and $\kappa_0 > 0$. There is $\zeta > 0$, depending only on these parameters, for which the following conditions cannot hold on an event of positive limiting probability. Fix rational $[s, s + b] \subset [0, 1]$, with $b < u_0$, rational $b_* \geq b$, and $j \in \{2, 3\}$. At direction precision L^{-b_*} let D_0 be the reference triple of bins and let D_1 add the alternative bin. Require*

- (i) $|G(s + b) - G(s) - db| \leq \zeta b$, and $G(s + u) - G(s) \geq d_0 u - \zeta b$;
- (ii) all costs in Lemma 3.6, for both D_0 and D_1 and each reference tube or T'_1 , are at most ζb ;
- (iii) $h_j(u) \geq \kappa_0 u - \zeta b$.

The prefix, cost, and angular requirements need be imposed only on a fixed finite rational grid of $0 < u \leq b$ containing b , with mesh at most a sufficiently small fixed multiple of ζb . Decreasing ζ preserves the conclusion. All power-error constants are independent of s, b, b_ ; the meshes can be fixed in relative coordinates u/b , and their finite cardinalities only affect fixed constants in the later cost sums.*

Proof. Put $K = L^b$ and use the same reference coordinates for D_0, D_1 . Let $Y = (Z_1, Z_j)$ at the endpoint or a tested prefix. Equations (3.9)–(3.10) imply, with either conditioning C, D_0 or C, D_1 ,

$$I(Y_b \mid C, D_a) = 2db + O(\zeta b), \quad I(Y_u \mid C, D_a) \geq 2d_0 u - O(\zeta b), \quad a = 0, 1. \quad (3.12)$$

Let $\tilde{a}_j$ be the approximate direction ratio fixed by D_1 , and let $\Pi(Y_b)$ record the interval of length K^{-1} containing $z_j - \tilde{a}_j z_1$ at the pair-cell center. The exact and approximate ratios differ by at most $L^{-b_*+o(1)}$. Given T'_1, C, D_1 , this projection ranges over $L^{o(1)}$ cells. Indeed, the approximate basis is already fixed by D_0 , and the exact central line of T'_1 is part of its label. Its slope differs from the representative slope by $L^{-b_*+o(1)}$, while its width in parent coordinates is at most $L^{s-1+o(1)} \leq L^{-b+o(1)}$. Along a segment of length $L^{o(1)}$ both errors cause only $L^{-b+o(1)}$ projection variation. Hidden exact reference directions do not enter this coordinate system. Also Y_b and X_{s+b} determine each other to $L^{o(1)}$ choices with these labels: coordinate one parametrizes the alternative tube with subpower distortion. Consequently

$$\begin{aligned} I(Y_b \mid \Pi, C, D_1) &\geq I(Y_b \mid \Pi, C, D_1, T'_1) \\ &= I(X_{s+b} \mid C, D_1, T'_1) \geq G(s + b) - G(s) - \zeta b. \end{aligned}$$

Since Π is determined by Y_b, D_1 , its chain rule together with (3.12) gives

$$I(\Pi \mid C, D_1) \leq db + O(\zeta b). \quad (3.13)$$

All projection information variables are among those included in the joint limit.

We detail the passage from these sampled inequalities to one common set and one direction measure. Enlarge the inequalities by fixed slack and pass to finite L . There is a success event of probability at least $p_* > 0$ along a subsequence. Fix parent and reference-bin labels (c, d_0^{bin}) whose conditional success probability p is at least p_* . No lower bound on the probability of this individual fiber is needed. Denote its conditional law by μ . Absorb all fixed slack into

$\alpha = C\zeta + o(1)$, enlarging the absolute constant C as necessary. On success,

$$\begin{aligned} K^{-2d-\alpha} &\leq \mu(Y_b = y) \leq K^{-2d+\alpha}, \\ K^{-2d-\alpha} &\leq \mu(Y_b = y \mid A_{\text{bin}} = a) \leq K^{-2d+\alpha}, \\ \mu(Y_u = y_u) &\leq K^\alpha L^{-2d_0 u}, \\ \mu(\Pi = \pi \mid A_{\text{bin}} = a) &\geq K^{-d-\alpha}, \end{aligned} \tag{3.14}$$

where A_{bin} is the additional direction bin. These are statements about the indicated labels whenever they have any successful realization.

Let A consist of the centers of all endpoint pair cells with a successful realization in this fixed parent/reference fiber. The first line of (3.14) and total success mass at least p give

$$pK^{2d-\alpha} \leq |A| \leq K^{2d+\alpha}. \tag{3.15}$$

A prefix cell containing a successful endpoint has mass at most the third line of (3.14). The nested grids put all mass of that endpoint cell in its prefix. Dividing by the minimum endpoint mass bounds the number of good endpoints in that prefix. A ball of radius r meets boundedly many cells at a comparable coarser tested prefix size; the finite mesh costs $K^{O(\zeta)}$. Thus, for $K^{-1} \leq r \leq 1$,

$$\frac{\mathcal{N}_{K^{-1}}(A \cap B(z, r))}{\mathcal{N}_{K^{-1}}(A)} \lesssim p^{-1} K^{O(\zeta)+o(1)} r^{d_0}. \tag{3.16}$$

In fact the preceding argument gives r^{2d_0} before weakening the exponent and absorbing rounding. For the untested shortest prefixes the allowed power factor suffices.

Retain bins a whose conditional success probability is at least $p/2$. The successful mass in these bins is at least $p/2$. For each retained bin, let $A_a \subset A$ be its successful endpoint cells. The conditional upper pair-cell mass gives

$$|A_a| \geq (p/2)K^{2d-\alpha}, \quad |A_a|/|A| \geq (p/2)K^{-2\alpha}.$$

The lower projection-cell mass in (3.14) gives

$$\mathcal{N}_{K^{-1}}(\pi_{\tilde{a}_j(a)} A_a) \leq K^{d+O(\zeta)+o(1)}, \tag{3.17}$$

where initially $\pi_{\tilde{a}_j}(z_1, z_j) = z_j - \tilde{a}_j z_1$. Passing to a unit vector for this projection only shrinks interval lengths.

It remains to put a nonconcentrating probability on these directions. Restrict the actual samples to success and the retained bins, normalize, and push forward to the line spanned by $(-\tilde{a}_j, 1)$. This costs at most $2/p$ in mass. To estimate a slope ball, first condition on the exact point X and exact reference tubes, before any restriction by the additional bin. These labels determine (C, D_0) , so fixing that fiber does not change the original alternative law once these exact labels are given. If that ball contains any successful alternative, the centered marginal bound at this alternative, supplied by (iii), bounds the *original* alternative probability of the entire ball after a fixed radius enlargement. Choosing a coarser tested radius costs $K^{O(\zeta)+o(1)}$. Therefore its restricted mass is at most

$$K^{O(\zeta)+o(1)} r^{\kappa_0} \quad (K^{-1} \leq r \leq 1).$$

Integrate this bound over the exact points and references under the fixed fiber, and then normalize. This operation introduces no reciprocal probability of an individual additional bin; conditional independence supplies the original alternative law at the stage where the angular estimate is used.

All slopes have magnitude $L^{o(1)}$. The sine-angle formula shows that the slopes in a projective ball of radius r occupy an interval of length $rL^{o(1)}$; exact-to-approximate slope errors are $K^{-1}L^{o(1)}$. Covering and radius rounding thus give, for the pushforward probability σ ,

$$\sigma(B(V, r)) \leq K^{O(\zeta)+o(1)} r^{\kappa_0}, \quad K^{-1} \leq r \leq 1. \quad (3.18)$$

The coordinate range of A is at most $L^{o(1)}$. Rescale it into the unit ball. At resolution K^{-1} this merges or covers at most $L^{o(1)}$ grid cells per ball and changes every displayed power error by $o(1)$ only. Use covering numbers after this rescaling.

Choose $0 < \rho_0 < \min\{d_0, \kappa_0\}$, apply Lemma 3.9, and then take ζ so small that every $O(\zeta)$ error is below its tolerance and its projection gain. Equations (3.15), (3.16), and (3.18) meet its hypotheses. For every direction in the finite support of σ , however, (3.17) gives a subset of the *same* A of relative covering number $K^{-O(\zeta)-o(1)}$ with too small a projection. The robust large-subset conclusion gives the contradiction. $\square$

3.5 Selecting angular scales and differentiating

The sum bound (3.11) does not say that one fixed marginal is nonconcentrated at every radius. The following finite choice of comparable scales supplies exactly what the block test needs.

Lemma 3.12 (Finite angular alternatives). *Fix $0 < d < 1$ and $d_0 > 0$. There is a fixed finite list of positive rational ratios, with minimum $c_* > 0$, such that for every sufficiently small rational b_0 , almost every path has a choice b' from this list times b_0 and a marginal $j \in \{2, 3\}$ satisfying the angular hypothesis of Lemma 3.11, with spare slack, for block lengths in $[b'/4, b']$. Each alternative has its own fixed positive κ_0 , tolerance, and finite relative mesh. All choices are independent of b_0 , and remain valid under sufficiently small variations of the spatial increment parameters.*

Proof. First obtain a sufficiently reduced block-test tolerance ζ_2 for $\kappa_0 = \kappa/4$. Choose a positive rational $c_1 \ll \kappa\zeta_2$, and then a sufficiently reduced tolerance ζ_1 for $\kappa_0 = c_1/2$. Test $h_2(u) \geq c_1u$ on a sufficiently fine finite rational grid from $c\zeta_1b_0$ to b_0 , where $c > 0$ is fixed and small. If all tests pass, monotonicity fills the mesh gaps; below the first grid point nonnegativity gives the required inequality with the allowed additive error. Take $b' = b_0$ and $j = 2$.

Otherwise some grid point v satisfies $h_2(v) < c_1v$. For rational $u \leq v$, monotonicity and (3.11) yield

$$h_3(u) \geq \kappa u - h_2(u) > \kappa u - c_1v \geq \frac{\kappa}{2}u \quad \left(\frac{2c_1v}{\kappa} \leq u \leq v \right).$$

Take $b' = v$, $j = 3$. Because $c_1 \ll \kappa\zeta_2$, the missing smaller radii meet the weaker $\kappa/4$ bound with additive error ζ_2b for every $b \in [v/4, v]$. Small enough mesh and cutoff constants leave spare slack in both cases. The possible v/b_0 form a fixed finite list bounded away from zero. Reduce all constants once more to permit the stated small spatial variations. $\square$

Lemma 3.13 (Binary derivative). *For almost every path, $\dot{G}(s) \in \{0, 1\}$ for almost every $s \in [0, 1]$.*

Proof. Suppose otherwise. Choose $0 < d < 1$ and $0 < d_0 < d$ such that a sufficiently small neighborhood of d contains $\dot{G}(s)$ on a set of positive path-times-exponent measure. This is possible by choosing an interior point of the essential support. Apply Lemma 3.12 for these parameters. For dyadic $b_0 \downarrow 0$, and each of its finite alternatives, form a rational mesh of test blocks of comparable length, so that each interior exponent is contained in an appropriate block, and include each block's finite prefix grid. Use direction precision $b_* = b_0$ for all

alternatives. Within each size and prefix type, the candidate intervals have bounded overlap, with constants independent of b_0 . Explicitly, choose endpoints on one lattice of spacing qb_0 , where the positive rational q is fixed below all the relative mesh tolerances, and allow all candidate lengths between $c_*b_0/4$ and b_0 . A point belongs to at most $O(q^{-2})$ such candidates. Every tested prefix is contained in its candidate, and the number of prefix types is fixed. Color each of these interval families into finitely many disjoint lists.

Let $\mathcal{C}_I$ be the sum of all reference, replacement, and extra-bin costs used by candidate block I . By Lemma 3.6,

$$\sum_I \mathbb{E} \mathcal{C}_I \lesssim b_0. \quad (3.19)$$

All candidate lengths lie between fixed positive multiples of b_0 . For the nonnegative function

$$R_{b_0}(\omega, s) = \sum_{I \ni s} \mathcal{C}_I(\omega) / b_0$$

we have $\mathbb{E} \int_0^1 R_{b_0}(\omega, s) ds \lesssim b_0$: integration cancels the normalizing block length. Dyadic summability therefore gives $R_{b_0} \rightarrow 0$ almost everywhere. In particular, all tested costs in a block containing such an s are $o(b_0)$.

At a differentiability point s , the approximation $G(y) = G(s) + \dot{G}(s)(y - s) + o(b_0)$ is uniform for $|y - s| \leq Cb_0$. Hence all increments and finitely many prefix increments in these nearby blocks are approximated to $o(b_0)$. If $\dot{G}(s)$ lies in the chosen sufficiently small neighborhood of d , the spatial hypotheses of the block test hold eventually, with its reduced tolerance. An angular alternative supplies its angular hypotheses, and the preceding cost estimate supplies the remaining ones. Only countably many fixed rational blocks and finite tests are involved. Positive path-times-exponent measure therefore implies that one fixed test holds on a positive-probability event, contradicting Lemma 3.11. A countable cover of interior derivative values proves the claim. $\square$

3.6 Hills and finitely many jumps

Write $E = \{s : \dot{G}(s) = 1\}$ for a path satisfying Lemma 3.13. Equations (3.7) and (3.10) give

$$|E| = \frac{2}{3}, \quad 3|E \cap [0, s]| \geq 2s \quad (0 \leq s \leq 1). \quad (3.20)$$

We next exclude infinitely many changes from a full scale interval to an empty one. The block test applies to the balanced intervals created at such changes, even though their derivatives themselves are binary.

Lemma 3.14 (Balanced subintervals inside a hill). *Let G be Lipschitz with derivative $\mathbf{1}_E$, and let $a < b$ be respectively a density point of E and of its complement. There is a positive number b_{hill} such that, for every $0 < r < b_{\text{hill}}$, some $[x, y] \subset (a, b)$ has*

$$r \leq y - x \leq 2r, \quad G(y) - G(x) = \frac{1}{2}(y - x), \quad G(x + u) - G(x) \geq \frac{1}{2}u \quad (0 \leq u \leq y - x).$$

For any fixed finite collection of disjoint ordered density pairs, these intervals can be chosen disjointly at every sufficiently small common r .

Proof. Put $H(s) = G(s) - s/2$. The density conditions give positive right slope at a and a positive drop towards b from the left. Thus $\max_{[a, b]} H$ exceeds both $H(a)$ and $H(b)$: if the larger endpoint is a use its right increase, and if it is b use its left increase. A component of a suitable strict superlevel set therefore gives a closed hill $[a', b'] \subset (a, b)$ with equal endpoint heights and strictly larger values inside.

Fix $r < b' - a'$. Among levels having a closed subinterval of this hill of length at least r , with both endpoints at the level and values no smaller inside, take the highest. Such a level exists: the admissible triples of level and endpoints form a nonempty compact set. At the maximizing level h take an admissible interval $[c, d]$. No component on which $H > h$ inside it has length greater than r , since such a component contains an interval of length at least r closed off at a slightly higher level. The closed set $\{H = h\} \cap [c, d]$ thus contains both endpoints and has no gap longer than r . Starting at c , take its first point at or after $c + r$. Its distance from c is between r and $2r$. Between these two points $H \geq h$, which gives the asserted total and prefix identities for G . Apply this construction separately inside finitely many disjoint hills to obtain the final assertion. $\square$

Lemma 3.15 (Finite alternation and a full-to-empty jump). *Almost surely E agrees modulo null sets with a finite union of intervals and has an interior transition from an interval of full measure to an interval of zero measure. For every fixed $\eta > 0$ there is a rational block $[s, s + b]$, $0 < b < u_0$, on an event of positive probability, such that*

$$G(s + b) - G(s) = \frac{b}{2} + O(\eta b), \quad G(s + b/2) - G(s) = \frac{b}{2} + O(\eta b). \quad (3.21)$$

The absolute error constants can be fixed independently of η .

Proof. Suppose a path has arbitrarily many disjoint ordered density pairs of the type in Lemma 3.14. Use the angular alternatives for $d = 1/2$, $d_0 = 1/4$ at dyadic $b_0 \downarrow 0$. For its successful angular size $b' \asymp b_0$, choose the balanced intervals with lengths between $b'/3$ and $2b'/3$. Round their endpoints to a sufficiently fine fixed relative rational mesh. Lipschitz continuity preserves the endpoint and finite prefix requirements of the block test with spare slack. For any prescribed number of disjoint density pairs this produces that many distinct spatially and angularly successful candidates at every sufficiently small b_0 ; the rounding preserves distinctness inside their separate hills.

Consider all mesh candidates for the fixed finite list of angular alternatives, and all their prefix and direction costs, with $b_* = b_0$. Equation (3.19) still holds. A failure of a candidate's cost test requires one nonnegative cost at least a fixed positive multiple of b_0 . Hence the expected number N_{b_0} of such failures is bounded uniformly in b_0 . Every spatially and angularly successful candidate must fail a cost test almost surely, by Lemma 3.11 and the countability of the mesh collection. On the paths just considered, $N_{b_0} \rightarrow \infty$. Fatou's lemma contradicts their having positive probability, because $\liminf \mathbb{E} N_{b_0} < \infty$.

Thus the number of alternations among density points of E and of its complement is bounded on almost every path. To see the finite-interval conclusion directly, assign to each ordered density point the maximum length of an alternating sequence ending there. These integers are bounded, nondecreasing along the ordered density points, and strictly increase when the type changes. Each integer level therefore has one type and occupies an ordered interval. The Lebesgue density theorem gives the asserted finite union modulo null sets.

The prefix inequality in (3.20) rules out an initial interval of zero measure, while $|E| = 2/3 < 1$ forces a subsequent nonempty interval of zero measure. There is therefore an interior full-to-empty jump. Around that jump choose a short block whose midpoint differs from the jump by at most ηb , with its left and right halves inside the adjacent full and empty intervals except for that discrepancy. Rational endpoints can be chosen with $b < u_0$, giving (3.21). The countable family of rational blocks covers almost all paths; some fixed block has positive success probability. No direction cost is needed in this choice. $\square$

3.7 The partitioning contradiction

A full-to-empty information jump says that almost all spatial branching occurs in the first half of a block. At its endpoint, each midpoint cell contains very few eligible descendants.

We now combine this with the angular law by drawing two endpoints on the same tube.

Lemma 3.16 (No full-to-empty block). *For sufficiently small fixed $\eta > 0$, the positive-probability configuration in (3.21) is impossible.*

Proof. Fix such a rational block and put $K = L^b$. By (3.10) and the chain rule, its endpoint and midpoint information increments for X are both $3b/2 + O(\eta b)$, while the conditional endpoint increment given $T = T_1$ is $b/2 + O(\eta b)$. Pass to finite L with fixed slack, and fix a parent $C = X_s = c$ with conditional success probability bounded below by a constant $p_* > 0$ along the sequence. Write

$$p_c(z) = \mathbb{P}(X_{s+b} = z \mid C = c), \quad q_T(z) = \mathbb{P}(X_{s+b} = z \mid T, C = c).$$

Call an incidence (T, z) good if it has at least one realization in the success event. Let $\mathcal{Z}$ be all endpoint labels with a good incidence, represented by their centers in rescaled parent coordinates. Every such endpoint satisfies

$$K^{-3/2-O(\eta)} \leq p_c(z) \leq K^{-3/2+O(\eta)}, \quad q_T(z) \leq K^{-1/2+O(\eta)} \quad \text{for a good } (T, z). \quad (3.22)$$

Its midpoint cell has marginal mass at most $K^{-3/2+O(\eta)}$. These estimates depend only on the displayed labels, so they remain true after defining goodness by existence of a successful realization. In particular

$$|\mathcal{Z}| \leq K^{3/2+O(\eta)}, \quad \#(\mathcal{Z} \cap B(z, CK^{-1/2})) \leq K^{O(\eta)} \quad (3.23)$$

for every fixed C . For the second bound divide the midpoint mass upper bound by the endpoint mass lower bound, then cover a ball by boundedly many midpoint cells. The central line of a tube with a good incidence passes within $\delta = K^{-1}L^{o(1)}$ of the endpoint center. The parent has a fixed bounded diameter in these coordinates.

Choose a small fixed $\varepsilon > 0$, and set $P = K^{1/2-\varepsilon}$. Polynomial partitioning in $\mathbb{R}^3$ [15, Theorem 4.1] gives a nonzero polynomial of degree $D \lesssim P$ whose complementary open cells each contain at most

$$O(|\mathcal{Z}|/P^3) = K^{3\varepsilon+O(\eta)} \quad (3.24)$$

of the centers. Take a wall of thickness a sufficiently large constant times δ about its zero set. At midpoint resolution $r = K^{-1/2}$, the number of boxes meeting the wall is

$$O(Dr^{-2} + D^2r^{-1} + D^3) = O(PK). \quad (3.25)$$

Indeed $\delta = o(r)$. Intersect the zero set with a fixed ball strictly larger than the bounded parent; every wall point relevant to that parent is within the required distance of this restricted zero set. Wongkew's hypersurface neighborhood-volume bound [27, Main Theorem, p. 178], thickened to scale comparable to r in that fixed enlarged ball, is $O(Dr + D^2r^2 + D^3r^3)$. Divide by the disjoint box volume r^3 . The degree condition $D \lesssim K^{1/2-\varepsilon}$ makes all three terms in (3.25) at most $O(PK)$.

By (3.22)–(3.23), the marginal mass of good endpoints lost to the wall is at most

$$O(PK) K^{O(\eta)} K^{-3/2+O(\eta)} = K^{-\varepsilon+O(\eta)}. \quad (3.26)$$

Take η small compared with ε , so this tends to zero. The enlarged wall ensures that the nearby central-line point of each surviving incidence lies in the same open cell as its endpoint center. A line outside the polynomial zero set crosses it at most D times, and hence visits at most $D + 1 = O(P)$ open cells. A line contained in the zero set has no off-wall incidence. Thus every tube's good off-wall endpoints lie in at most $O(P)$ open cells.

Draw T with its original marginal given $C = c$, then draw z, z' independently with law q_T . Let $g(T)$ be the q_T mass of good off-wall endpoints. Its mean is at least $p_*/2$ for large K . Conditional on T , Cauchy–Schwarz over its at most $O(P)$ open cells gives probability at least $g(T)^2/O(P)$ that both draws are good off-wall and belong to the same open cell. Jensen’s inequality therefore gives the unconditional lower bound

$$\mathbb{P}\{z, z' \text{ good off-wall in the same cell} \mid C = c\} \gtrsim P^{-1} = K^{-1/2+\varepsilon}. \quad (3.27)$$

We bound this probability from above by separating near and far pairs.

For $|z - z'| \lesssim K^{-1/2}$, the second bound in (3.23) gives at most $K^{O(\eta)}$ possible good second endpoints. Each costs $K^{-1/2+O(\eta)}$ under q_T , so near pairs contribute at most

$$K^{-1/2+O(\eta)}. \quad (3.28)$$

For a farther pair, a tube meeting both endpoint centers to error δ must have direction within $O(\delta K^{1/2}) = K^{-1/2+o(1)}$ of their joining line, allowing two signs.

The conditioning at this point is essential. Under the double sampling,

$$\mathbb{P}(T, z, z' \mid C = c) = \mathbb{P}(T \mid C = c) q_T(z) q_T(z').$$

Thus its (T, z) marginal is the original joint distribution given $C = c$. Conditional on the first endpoint z , the tube law is

$$\mathbb{P}(T \in \mathcal{A} \mid z, C = c) = \sum_{p: X_{s+b}(p)=z} \mathbb{P}(X = p \mid z, C = c) \mathbb{P}(T \in \mathcal{A} \mid X = p). \quad (3.29)$$

The parent is determined by z , and X determines all these spatial labels. Equation (3.5) therefore applies to every inner law. Use the slightly coarser rational angular scale $L^{-b/3} = K^{-1/3}$, which covers the above cap for large L and has $b/3 < u_0$. For every fixed far pair the resulting angular probability is at most

$$K^{-\kappa/3+o(1)}.$$

For each first endpoint, (3.24) allows at most $K^{3\varepsilon+O(\eta)}$ same-open-cell second endpoints. Each good second incidence costs an additional $K^{-1/2+O(\eta)}$ under its original q_T law. For each first z this list is fixed before the tube is drawn. In the integral over the original law of T given z, C , apply the bound on $q_T(z')$ only where (T, z') is good, and apply the angular cap bound to the resulting indicator. Sum over the fixed list of z' and then over the first-endpoint law in (3.29). The total far-pair contribution is at most

$$K^{-1/2-\kappa/3+3\varepsilon+O(\eta)+o(1)}. \quad (3.30)$$

No angular regularity after conditioning on the success event has been used; success only restricts the pairs to which the original mass bounds apply.

Divide (3.28) and (3.30) by (3.27). The respective ratios are bounded by

$$K^{-\varepsilon+O(\eta)}, \quad K^{-\kappa/3+2\varepsilon+O(\eta)+o(1)}.$$

Choose $\varepsilon > 0$ sufficiently small relative to κ , then $\eta > 0$ sufficiently small relative to ε . Both ratios tend to zero, contradicting (3.27). $\square$

Remark 3.17 (A grid proof of the wall count). Retain the notation of the proof of Lemma 3.16. For comparison, the same wall count follows from a generic grid of spacing r . A component wholly contained inside a grid box costs at most $O(D^3)$ boxes by the real algebraic component

bound [27, Lemma (C), p. 181]. All other intersections are counted on grid faces. On each generic grid plane the curve meets $O(Dr^{-1} + D^2)$ squares: intersections with grid lines cost $O(Dr^{-1})$, and components wholly inside squares cost $O(D^2)$. There are $O(r^{-1})$ relevant planes, giving $O(Dr^{-2} + D^2r^{-1} + D^3)$. A bounded enlargement covers near-wall boxes. The volume argument above already includes singular zero sets and suffices for the estimate.

Proof of Proposition 3.1. A countersequence gives the weighted tube configuration by Lemma 3.3. Its limiting information profiles satisfy (3.10). Lemma 3.13 makes the common derivative binary, and Lemma 3.15 supplies a positive-probability full-to-empty block for any prescribed fixed small tolerance. Lemma 3.16 rules out such a block. Hence no countersequence exists, which proves the existence of the fixed power $c > 0$ and the large-scale threshold. $\square$

4 A sparse estimate for packet arrays

The transverse saving has a useful consequence for arrays whose individual packets see little measure. The broad–narrow decomposition follows the framework of Bourgain–Guth [6, Section 2] and its fractal restriction implementation in Du–Zhang [13, Section 3]; the parabolic rescaling below also preserves packet sparsity. Broad contributions gain a power by Proposition 3.1; when the iteration stops, the sparsity condition supplies the gain through the refined fractal estimate.

Proposition 4.1 (Sparse packet estimate). *There is an absolute constant $c_s > 0$ with the following property. Fix $\tau_0 > 0$, and let $L^{\tau_0} \leq D \leq L$. Let U be a finite packet array of the form (2.2), with coefficient vector c , and let μ be a nonnegative measure such that*

$$\mu(B(z, r)) \leq mr^2 \quad (r \geq 1), \quad \text{supp } \mu \subset \{|t| \leq CL^2\}.$$

Assume the packet sparsity condition (2.4), with $M = 100$, for every label with nonzero coefficient. Then

$$\sum_{q \text{ a unit lattice cube}} \mu(q)^{2/3} \sup_q |U|^2 \lesssim_{\tau_0} m^{2/3} L^{4/3} D^{-c_s} \|c\|_{\ell^2}^2. \quad (4.1)$$

Only finitely many uniform profile derivative bounds, depending on τ_0 , are required. Consequently the same assertion holds for any prescribed finite set of field derivatives, provided the original profiles have sufficiently many bounded derivatives. Fixed enlargements of the time range and restriction to a submeasure are allowed. The exponent c_s is independent of τ_0 ; the implicit constant and the required differentiability need not be.

The proof is divided into the exact-field reduction, a broad estimate, and the rescaling iteration. All bounds on packet classes and frequency boxes are fixed throughout. Zero measures and zero coefficient vectors are immediate, so normalization by their sizes is legitimate.

4.1 A local norm and an exact-field reduction

Choose a dyadic integer $K \asymp L^\beta$, with $\beta > 0$ to be specified at the end. For a measure ν define

$$\mathcal{T}_\nu(V) = \sum_{Q \text{ a } K^2\text{-cube}} \nu(Q)^{2/3} \|V\|_{L^6(Q)}^2. \quad (4.2)$$

The cubes form a fixed half-open grid; taking suprema on their closures does not change any estimate. The quantity $\mathcal{T}_\nu(V)^{1/2}$ is a seminorm, namely the ℓ^2 norm of $\nu(Q)^{1/3} \|V\|_{L^6(Q)}$.

This observation will allow us to use Minkowski's inequality for superpositions of exact extensions.

It suffices first to bound (4.2) by the right-hand side of (4.1), with μ restricted to a spatial box of side $O(L^2)$. Indeed, the three-dimensional unit-cube Sobolev inequality and Hölder's inequality give, for a fixed finite set of multiindices a ,

$$\begin{aligned} \sum_{q \subset Q} \mu(q)^{2/3} \sup_q |U|^2 &\lesssim \sum_a \sum_{q \subset Q} \mu(q)^{2/3} \left(\int_q |\partial^a U|^6 \right)^{1/3} \\ &\lesssim \sum_a \mu(Q)^{2/3} \left(\int_Q |\partial^a U|^6 \right)^{1/3}. \end{aligned} \quad (4.3)$$

The dyadic K^2 -grid is nested with the unit grid, and the Sobolev inequality on a cube controls the supremum on its closure, so no boundary enlargement is needed in this calculation. Differentiating an atom in (2.2) produces only bounded carrier-frequency factors and native profile derivatives multiplied by bounded powers of L^{-1} . It therefore gives another finite sum of atoms in the same class, with the same labels and comparable coefficient energy. Applying the $\mathcal{T}$ estimate to these differentiated arrays proves the desired unit-cube statement.

For this reduction, first localize the original compact packets to spatial boxes of side L^2 . On the stipulated time interval, one such packet meets only boundedly many enlarged boxes, since its velocity is bounded. Thus the energies of the retained arrays sum with bounded overlap. This justifies summing the spatially localized estimates and also the treatment of boundary cubes in (4.3).

The compact profiles are convenient for this first localization but are not themselves exact Schrödinger extensions. The following construction passes to exact fields once, before the iteration.

Lemma 4.2 (Exact Fourier layers). *Fix a bounded native time interval $|\sigma| \leq C_{\text{nat}}$, any finite number of profile seminorms on that interval, and any prescribed finite spatial decay orders there. After requiring sufficiently many derivatives of the compact profiles, an array of atoms in (2.2) is a sum-integral of exact extension arrays, indexed by $k \in \mathbb{Z}^2$ and $u \in \mathbb{R}$, times a common phase e^{iut/L^2} . There are nonnegative common weights $w(k, u)$, with*

$$\sum_k \int_{\mathbb{R}} w(k, u) du < \infty,$$

such that division by $w(k, u)$ gives all the stipulated uniform seminorm and decay bounds. For every subarray A the frequency input $v_{A,k,u}$ of its layer satisfies

$$\|v_{A,k,u}\|_2^2 \leq C w(k, u)^2 \sum_{\alpha \in A} |c_\alpha|^2. \quad (4.4)$$

Its exact frequencies lie within $O(L^{-1})$ of $\omega_\alpha + k/L$ for the corresponding labels.

Proof. Use the Fourier transform in all native variables $(y, \sigma) \in \mathbb{R}^3$, and choose a smooth partition of unity $\sum_{k \in \mathbb{Z}^2} \rho(\eta - k) = 1$, with ρ supported in $[-2, 2]^2$. With

$$a_{\alpha,k,u}(\eta) = \rho(\eta - k) \hat{H}_\alpha(\eta, u - |\eta|^2),$$

Fourier inversion followed by $u = \zeta + |\eta|^2$ gives the exact identity

$$H_\alpha(y, \sigma) = (2\pi)^{-3} \sum_k \int_{\mathbb{R}} e^{iu\sigma} \left[\int_{\mathbb{R}^2} e^{i(y \cdot \eta - \sigma |\eta|^2)} a_{\alpha,k,u}(\eta) d\eta \right] du. \quad (4.5)$$

The carrier phase combines with the inner phase according to

$$x \cdot \omega_\alpha - t|\omega_\alpha|^2 + \frac{x - b_\alpha - 2\omega_\alpha t}{L} \cdot \eta - \frac{t}{L^2} |\eta|^2 = x \cdot (\omega_\alpha + \eta/L) - t|\omega_\alpha + \eta/L|^2 - b_\alpha \cdot \eta/L. \quad (4.6)$$

Thus, with $Ev(x, t) = \int e^{i(x \cdot \xi - t|\xi|^2)} v(\xi) d\xi$, the exact datum for one packet in a fixed layer is

$$v_{\alpha, k, u}(\xi) = L a_{\alpha, k, u}(L(\xi - \omega_\alpha)) e^{-ib_\alpha \cdot (\xi - \omega_\alpha)}. \quad (4.7)$$

The factor L makes its L^2 norm equal to $\|a_{\alpha, k, u}\|_2$; the inverse Jacobian L^{-2} then produces the atom's prefactor L^{-1} in the extension field.

Compact support and sufficiently many derivatives of H_α give any prescribed finite collection of rapidly decaying bounds for its Fourier transform and its derivatives. Differentiating $\hat{H}_\alpha(\eta, u - |\eta|^2)$ introduces only polynomial factors in η . On $\eta \in k + [-2, 2]^2$ these, including the case $|u| \asymp |k|^2$, are absorbed by taking more original derivatives. Explicitly, on this support,

$$(1 + |k|)^P (1 + |u|)^P \leq C_P (1 + |\eta| + |u - |\eta|^2|)^{3P}.$$

After fixing the desired seminorms, the spatial decay orders, and C_{nat} , differentiation and integration by parts introduce only an additional factor $(1 + |k|)^B$ for a fixed finite B . Taking the original Fourier decay order larger than $3P + B$ therefore permits, for any subsequently specified large finite P , the common weight

$$w(k, u) = C_P (1 + |k|)^{-P} (1 + |u|)^{-P},$$

so that the desired derivatives of $a_{\alpha, k, u}/w(k, u)$ are uniformly bounded. On the fixed bounded native time interval, integration by parts in η also gives the required finite-order spatial decay for the bracket in (4.5), after division by the same weight. Derivatives of $e^{-i\sigma|\eta|^2}$ cost polynomial powers of k , already included in B . This also absorbs the displacement $y \simeq 2\sigma k$ of a layer's own propagated center: spatial decay is asserted after normalization by the weight chosen for the prescribed decay order. In particular these exact packet fields decay about the original label trajectories to any prescribed finite order, uniformly in the layer after normalization.

For completeness, the energy bound uses the full phase-space separation, not bounded multiplicity of frequencies alone. In the Gram matrix of (4.7), the entry indexed by $\alpha, \tilde{\alpha}$ vanishes unless $L|\omega_\alpha - \omega_{\tilde{\alpha}}| \leq C$, since the same k occurs in both frequency cutoffs. On an overlap, integration by parts in the rescaled frequency variable gives, for any fixed large N ,

$$|\langle v_{\alpha, k, u}, v_{\tilde{\alpha}, k, u} \rangle| \leq C_N w(k, u)^2 \mathbf{1}_{\{L|\omega_\alpha - \omega_{\tilde{\alpha}}| \leq C\}} \left(1 + \frac{|b_\alpha - b_{\tilde{\alpha}}|}{L}\right)^{-N}. \quad (4.8)$$

The bounded number of labels in every unit phase-space ball makes each row and column summable. Schur's test proves (4.4), for an arbitrary subarray as well as for the entire array. $\square$

We may therefore prove the $\mathcal{T}$ estimate for the normalized exact arrays of Lemma 4.2. The phase e^{iut/L^2} is common to an entire layer and has modulus one, so it does not change $\mathcal{T}$ at the root. Minkowski's inequality and the summability of w return the estimate for the original array. If derivatives are needed in (4.3), differentiate the compact atoms *before* this reduction; no differentiation of an exterior layer phase inside the iteration is needed.

Fix a very small $\gamma > 0$ later. Layers with $|k| > L^\gamma$ can be discarded with an arbitrarily small negative-power error, by increasing P and the number of original derivatives. To see

uniformity, normalize $m = \|c\|_{\ell^2} = 1$ and use the spatial restriction already made. The number of relevant cells, their total mass, and a trivial bound for the functional are polynomial in L ; (4.4) and the bounded area of each layer's frequency support give a uniform field supremum bound after normalization. The tail sum of w beats any of these fixed polynomial factors. Homogeneity restores general m, c . Henceforth fix one retained layer and suppress k, u, w from the notation.

4.2 The iteration and broad saving

We first specify the scales at which the next estimates will be used. A child retains one frequency square of side K^{-1} and applies the cap rescaling in (2.5). At depth j set $h = K^j$ and $s = L/h$, and divide out the accumulated field factor h^{-1} . The transformed labels occupy a fixed bounded square. The native variables, native time bounds, and uniform profile bounds are unchanged.

Stop at the last scale before the next division by K would go below $D^{1/2}$. Thus every scale satisfies $s \geq D^{1/2}$, and terminal scales satisfy

$$D^{1/2} \leq s < KD^{1/2}. \quad (4.9)$$

The depth J is bounded by a constant depending on β . The broad estimate below is stated independently of this stopping rule; the terminal estimate will use it together with packet sparsity.

At current packet size s , partition the bounded square of label frequencies into K^{-1} -squares θ , and write $V = \sum_{\theta} V_{\theta}$. Each V_{θ} is exact. Its frequencies are in its label square, enlarged by $O(s^{-1})$, plus the common shift k/s . We will always have $s \gg K^2$ and $|k|/s = o(1)$. For a fixed sufficiently large absolute A , call θ active on Q if

$$\|V_{\theta}\|_{L^6(Q)} \geq K^{-A} \|V\|_{L^6(Q)}. \quad (4.10)$$

There are $O(K^2)$ squares. Thus the inactive sum has norm at most $CK^{2-A} \|V\|_{L^6(Q)}$, which is absorbed by choosing A once and for all.

There is a fixed large C_0 such that either all active centers lie within C_0/K of a line, or three active squares have direction determinant at least cK^{-2} throughout their exact supports. Indeed, choose a farthest pair of active centers. If another center lies outside the indicated strip about their line, the resulting triangle has area at least a constant times K^{-2} . More precisely its height is at least C_0/K , and its base is at least as long as this height. Changing each vertex by $O(K^{-1})$ changes its doubled area by at most a fixed constant times the base length divided by K ; large C_0 preserves a fixed portion of the area. The identity between triangle area and the determinant of $(2\xi_i, 1)$, and the bounded frequency range, give the claim also for normalized directions. A common frequency translation does not change the unnormalized determinant. Call the two alternatives *narrow* and *broad*, respectively.

Lemma 4.3 (Broad saving). *There is an absolute $c_b > 0$ such that, whenever $K \leq s^{c_b}$, the broad part of the functional in a ball of radius $O(s^2)$ satisfies*

$$\sum_{Q \text{ broad}} \nu(Q)^{2/3} \|V\|_{L^6(Q)}^2 \lesssim s^{-c_b} m_0^{2/3} s^{4/3} \mathcal{E}. \quad (4.11)$$

Here $\nu(B(z, r)) \leq m_0 r^2$ for $r \geq 1$, and the exact frequency data for V and each V_{θ} have L^2 norms at most $C\mathcal{E}^{1/2}$. The implicit constant can depend on this fixed C but c_b does not depend on packet differentiability.

Proof. Suppose no positive c_b works. After normalizing $m_0 = \mathcal{E} = 1$ and dividing all fields by a fixed bound for their input norms, this gives a sequence $s \rightarrow \infty$, with $K = s^{o(1)}$, for which the broad sum is at least $s^{4/3-o(1)}$. Fixed constants in the radius, support box and norm bounds are harmless. There are polynomially many cubes, and the fields have bounded supremum by Cauchy–Schwarz in frequency. We may discard polynomially tiny mass and field levels, and then pigeonhole one active triple and dyadic values

$$\nu(Q) \asymp \chi, \quad \|V\|_{L^6(Q)} \asymp b$$

on N cubes. The number of triples and all factors lost here are $K^{O(1)}$ or logarithmic, hence $s^{o(1)}$. We obtain

$$\chi^{2/3} b^2 N \geq s^{4/3-o(1)}, \quad \chi \leq s^{o(1)}, \quad \chi N \lesssim s^4. \quad (4.12)$$

The last two statements follow respectively from the ball bound on a K^2 -cube and on the ambient ball.

We explain carefully the fourth inequality, furnished by multilinear restriction. Each chosen factor has $L^6(Q)$ norm at least $K^{-A}b$, so some unit cube in Q has supremum at least $K^{-A-1}b$: here $|Q|^{1/6} = K$. For each of the three factors, choose one such unit cube. Pigeonholing their three relative integer positions within Q leaves a common triple of offsets, with cost $K^{O(1)}$. Translate the three fields separately by these fixed offsets. They now all have supremum at least $b s^{-o(1)}$ on the same unit cube attached to each surviving Q . These attached cubes are disjoint. Translation only modulates the frequency inputs and preserves both their norms and their supports.

The separate-supremum version of Lemma 3.2 therefore yields

$$b^3 N \leq s^{o(1)}. \quad (4.13)$$

Equivalently, this use of the lemma follows by applying a rapidly decreasing reproducing kernel to each factor separately, integrating the product on each attached cube, and averaging the trilinear integral bound over the three independent shifts. For every fixed triple of shifts the integration domain is the same and the inputs are exact with unchanged supports and norms. The determinant lower bound is $K^{-2} = s^{-o(1)}$, precisely the regime of that lemma. Thus an unspecified dependence of a multilinear constant on transversality is not being used here.

The four inequalities force saturation of all three sizes. For clarity, write (x, y, n) for limiting base- s logarithms of (χ, b, N) , passing to a subsequence if needed. The retained levels and the first inequality give polynomial bounds, so such subsequences exist. Equations (4.12) and (4.13) imply

$$x \leq 0, \quad x + n \leq 4, \quad 3y + n \leq 0, \quad 2x + 6y + 3n \geq 4.$$

Consequently

$$4 \leq 2x + 6y + 3n \leq 2x + n \leq 4 + x \leq 4.$$

Equality holds throughout, giving $x = 0$, $n = 4$, and $y = -4/3$. In the original sequence this says

$$\chi = s^{o(1)}, \quad N = s^{4+o(1)}, \quad b = s^{-4/3+o(1)}. \quad (4.14)$$

In particular the needed lower bounds are $\chi \geq s^{-o(1)}$, $N \geq s^{4-o(1)}$ and $b \geq s^{-4/3-o(1)}$.

The attached unit cubes also have the required ball counting bound. If their locations lie in a radius- r ball, their associated full K^2 -cubes lie in a ball of radius $r + CK^2$. Each associated cube has mass at least $s^{-o(1)}$, and they are disjoint. For $r \geq 1$ their number is therefore at most

$$s^{o(1)}(r + CK^2)^2 \leq s^{o(1)}r^2.$$

The three translated exact fields, the cubes in (4.14), and the determinant bound thus contradict Proposition 3.1. A fixed enlargement of the ambient ball can be removed by replacing s by a fixed multiple and absorbing the constants into the subpower factors.

This contradiction supplies an absolute positive c_b , reducing it if necessary so that the same exponent appears in both its hypothesis and its conclusion. The argument used only the maximum of the total input norm and the individual subfield input norms. Normalizing by that maximum preserves activity ratios. Thus it is uniform for exact fields and its exponent is independent of the high decay orders used to localize packet arrays later. $\square$

4.3 Narrow decoupling and the child measure

We record the narrow estimate with its measure transformation, since the latter is what allows sparsity to survive the iteration. For any fixed sufficiently large decay order N_0 , write

$$w_Q(z) = \left(1 + \frac{|z - z_Q|}{K^2}\right)^{-N_0}.$$

On a narrow cube, parabola decoupling gives

$$\|V\|_{L^6(Q)}^2 \lesssim_\epsilon K^\epsilon \sum_\theta \|V_\theta\|_{L^6(w_Q)}^2. \quad (4.15)$$

Here the sum can include all children; only active children are needed in deriving the inequality.

To check the exact decoupling geometry, rotate frequency coordinates so that the active strip is $|\xi_2 - v_2| \leq C/K$, and put $\eta_2 = \xi_2 - v_2$. A bounded physical shear $y_2 = x_2 - 2v_2t$ and a modulation remove the constant and linear normal terms. At each fixed y_2 , the resulting function of (x_1, t) has Fourier support in

$$\{(\xi_1, \tau) : |\tau + \xi_1^2| \leq CK^{-2}\},$$

because the remaining normal contribution is $-\eta_2^2$. Its tangential cap scale is K^{-1} . The K^{-1} grid of original patch centers has boundedly many centers in each such tangential bin inside a strip of width C/K : the corresponding planar region has area $O(K^{-2})$. The fixed enlargement $O(s^{-1})$ preserves this bounded overlap. Also, each original patch meets only boundedly many tangential bins.

For localization, multiply by a Schwartz cutoff ψ_Q bounded below on Q whose Fourier transform is supported in a ball of radius $O(K^{-2})$. After the bounded shear and rotation, its restriction to each y_2 slice has the same order of tangential/time Fourier thickness. The thick-support hypotheses of the parabola theorem therefore remain valid. Write $F_\theta = \psi_Q V_\theta$ in these coordinates, and $F = \sum_{\theta \text{ active}} F_\theta$. On each fixed y_2 slice, use the smooth tangential cap projections P_I at interval length K^{-1} in the usual smooth-cap form of Bourgain–Demeter decoupling [5, Theorem 1.1]. Their multipliers depend only on ξ_1 ; their one-dimensional convolution kernels have uniformly bounded L^1 norms. Thus, with $\delta \asymp K^{-2}$,

$$\|F(\cdot, y_2, \cdot)\|_{L_{x_1, t}^6}^2 \lesssim_\epsilon K^\epsilon \sum_I \|P_I F(\cdot, y_2, \cdot)\|_{L_{x_1, t}^6}^2.$$

This decouples frequency projections of the localized sum. To recover the original, possibly overlapping subfields, let $\mathcal{N}(I)$ consist of the patches whose enlarged tangential supports meet I . The preceding geometry gives $|\mathcal{N}(I)| \leq C$, and each patch belongs to at most C such sets; the cutoff's extra $O(K^{-2})$ thickness does not change these bounds. Since $P_I F = \sum_{\theta \in \mathcal{N}(I)} P_I F_\theta$, the triangle inequality and the uniform convolution bound give

$$\sum_I \|P_I F(\cdot, y_2, \cdot)\|_6^2 \leq C \sum_I \sum_{\theta \in \mathcal{N}(I)} \|F_\theta(\cdot, y_2, \cdot)\|_6^2 \leq C' \sum_\theta \|F_\theta(\cdot, y_2, \cdot)\|_6^2.$$

In particular no lower bound on any component of a possibly cancelling sum is required. Taking the L^6 norm in y_2 and using Minkowski now gives

$$\left\| \left(\sum_{\theta} \|F_{\theta}(\cdot, y_2, \cdot)\|_{L^6_{x_1, t}}^2 \right)^{1/2} \right\|_{L^6_{y_2}}^2 \leq \sum_{\theta} \|F_{\theta}\|_{L^6}^2 \lesssim \sum_{\theta} \|V_{\theta}\|_{L^6(w_Q)}^2.$$

Here $|\psi_Q|^6 \lesssim w_Q$; all projections above were estimated in global slice norms after localization, so no inverse of the physical cutoff is needed. Choosing the cutoff with arbitrarily high fixed decay gives any required N_0 . Finally the inactive sum is absorbed using (4.10). This proves (4.15) for the stated subfields.

Fix a small padding exponent $\eta > 0$. The weighted norms can be truncated to $L^{\eta}Q$, with an arbitrarily small negative-power error, by taking N_0 sufficiently large. Exact fields here have bounded frequencies and controlled input norms, hence polynomial global supremum bounds. Their measures and domains have polynomial size in L , so this truncation is uniform after energy and mass normalization.

Consider a child square centered at v and its parabolic map

$$\Phi_v(x, t) = \left(\frac{x - 2vt}{K}, \frac{t}{K^2} \right), \quad s' = s/K, \quad b'_{\alpha} = b_{\alpha}/K, \quad \omega'_{\alpha} = K(\omega_{\alpha} - v). \quad (4.16)$$

After removing a modulation, $V_{\theta} = K^{-1}V'_{\theta} \circ \Phi_v$. This transformation leaves the native packet variables and the phase-space counting condition unchanged. If $\nu' = (\Phi_v)_{\#}\nu$, then

$$\nu'(B(z, r)) \leq CK^3 m_0 r^2 \quad (r \geq 1), \quad \int \left(1 + \frac{|x' - b'_{\alpha} - 2\omega'_{\alpha} t'|}{s'} \right)^{-M} d\nu' \leq \frac{m_0 s^3}{D}. \quad (4.17)$$

Indeed, the inverse image of a radius- r ball has dimensions $O(Kr), O(Kr), O(K^2r)$ and is covered by $O(K)$ balls of radius $O(Kr)$. This proves the first bound. The normalized transverse distance in the second integral equals its parent value pointwise, proving the second bound. Its right-hand side is exactly $(K^3 m_0)(s')^3/D$.

The measure directly pushed forward is slightly modified to account for the weighted norms. Group parent cubes Q by the child K^2 -cube P containing $\Phi_v(z_Q)$. Their total mass is at most $\nu'(CP)$, since their images lie in CP . The sets $\Phi_v(L^{\eta}Q)$ lie in $CL^{\eta}P$ and have overlap at most $CL^{3\eta}$. The Jacobian and field normalization give

$$\left(\int_{L^{\eta}Q} |V_{\theta}|^6 \right)^{1/3} = K^{-2/3} \left(\int_{\Phi_v(L^{\eta}Q)} |V'_{\theta}|^6 \right)^{1/3}, \quad (4.18)$$

because $dx dt = K^4 dx' dt'$ and $|V_{\theta}|^6 = K^{-6}|V'_{\theta}|^6$. Hölder with exponents $3/2$ and 3 , first within each group, yields

$$\begin{aligned} \sum_Q \nu(Q)^{2/3} \left(\int_{L^{\eta}Q} |V_{\theta}|^6 \right)^{1/3} \\ \leq CL^{O(\eta)} K^{-2/3} \sum_P \nu'(CP)^{2/3} \left(\int_{CL^{\eta}P} |V'_{\theta}|^6 \right)^{1/3}. \end{aligned} \quad (4.19)$$

There is no factor for the number of parent cubes in a group.

To put the last sum on the child grid, cover CP by child cubes $P + a$ with a in a fixed finite set of lattice shifts, and cover $CL^{\eta}P$ by $P + b$ with b in a set of $O(L^{3\eta})$ shifts. Both shift lengths are at most $CL^{\eta}K^2$. Subadditivity of the powers $2/3$ and $1/3$ bounds a summand by the sum of

$$\nu'(P + a)^{2/3} \left(\int_{P+b} |V'_{\theta}|^6 \right)^{1/3}.$$

For a vector d , let $(T_d)_\# \nu'$ denote translation of the measure by d , and define

$$\nu_\theta = \sum_{d \in \mathcal{D}} (T_d)_\# \nu', \quad \mathcal{D} = \{b - a \text{ for the shifts above}\}. \quad (4.20)$$

For $d = b - a$, the mass of $P + b$ under $(T_d)_\# \nu'$ equals $\nu'(P + a)$. Summing with the possible multiplicities, which are $L^{O(\eta)}$, transforms (4.19) into

$$\sum_Q \nu(Q)^{2/3} \left(\int_{L^\eta Q} |V_\theta|^6 \right)^{1/3} \leq C L^{O(\eta)} K^{-2/3} \mathcal{T}_{\nu_\theta}(V'_\theta). \quad (4.21)$$

The number of translates is $L^{O(\eta)}$, so their ball bounds hold with parameter

$$m_1 = C K^3 L^{O(\eta)} m_0. \quad (4.22)$$

They also preserve the sparsity bound with this same enlarged parameter. In fact, for a shift $d = (d_x, d_t)$, the child's transverse displacement changes by $d_x - 2\omega'_\alpha d_t$. Child velocities are bounded, and we impose

$$L^\eta K^2 \ll s' \quad (4.23)$$

at every existing child scale. The weight $(1 + |x' - b'_\alpha - 2\omega'_\alpha t'|/s')^{-M}$ then changes by at most a fixed factor under any translation in (4.20). Summing (4.17) over these shifts proves

$$\int \left(1 + \frac{|x' - b'_\alpha - 2\omega'_\alpha t'|}{s'} \right)^{-M} d\nu_\theta \leq m_1 (s')^3 / D.$$

Their time shifts are also $o(s')$, so a time range $O((s')^2)$ remains of that order. Combining (4.15) and (4.21) proves the narrow recursion, with factor $C K^{-2/3+\epsilon} L^{O(\eta)}$ per generation. Its principal scale factors satisfy the exact cancellation

$$K^{-2/3} (K^3 m_0)^{2/3} (s/K)^{4/3} = m_0^{2/3} s^{4/3}. \quad (4.24)$$

4.4 Terminal estimate and summation

Proof of Proposition 4.1. We apply the preceding estimates on the scales just specified. At any node the preceding transport calculation gives a measure ν with time range $O(s^2)$, ball parameter m_j , and packet moment bound $m_j s^3 / D$, where

$$m_j \leq C^j h^3 m L^{Cj\eta}. \quad (4.25)$$

All support boxes and cell counts are polynomial in L . We may apply the estimates at radius $O(s^2)$ spatial box by spatial box. Here exact packets replace the compact packets used at the root: retain only packets whose trajectories meet a fixed enlargement of the box, and discard the others using their prescribed rapid decay. On the time range $O(s^2)$, each retained packet is used in boundedly many boxes. Its coefficient energy therefore has bounded overlap, also for every child subarray. A discarded packet has native transverse distance $\gtrsim s$ on the box. Cauchy–Schwarz, the phase-space counting condition and a sufficiently high decay order make the total error an arbitrarily small negative power of L , since $s \geq L^{\tau_0/2}$. This is the localization mechanism of Lemma 2.2, applied to the exact arrays constructed above.

Whenever this localization is performed, use the same retained labels in the total field and in all its subfields. Negligible field-level cubes are absorbed in the additive error; on the remaining cubes the activity tests tolerate that error, or can be made afresh for the localized exact array. Lemma 4.3 then uses the maximum of its total and subfield input

norms, bounded by a fixed multiple of the retained coefficient norm by (4.4). In particular increasing the later decay orders cannot reduce the absolute broad saving.

At every nonterminal node the broad contribution is bounded by (4.11), and the narrow contribution is passed to children by (4.21). Child arrays partition the parent labels. Hence their squared coefficient norms sum to the parent energy; there is no loss from the number of children.

It remains to estimate a terminal node. By rapid decay keep only K^2 -cubes lying within transverse distance $L^\eta s$ of at least one label trajectory. Taking the decay order sufficiently high again makes the omitted contribution negligible. Restrict ν to these retained cubes. If a ball B of radius s meets the retained support, choose one retained cube it meets and a packet near that cube. Since $K^2 \ll s$ and velocities are bounded, every point of B is at normalized transverse distance $O(L^\eta)$ from that one packet. Its moment bound therefore gives

$$\nu(B) \leq CL^{CM\eta} \frac{m_j s^3}{D}. \quad (4.26)$$

The same estimate holds for fixed enlargements of B . No sum over packets occurs: one packet suffices for each ball.

Normalize the node coefficient energy to one and split retained cubes into dyadic classes $\nu(Q) \asymp m_j \chi$. The upper range is $\chi \lesssim K^4$ by the ball bound. Polynomially tiny levels are negligible: there are polynomially many tested cells, and a uniform polynomial supremum bound follows from the exact input energy. Thus only $O(\log L)$ levels need be kept. In each Q choose a unit cube q_Q with the largest field supremum. Then

$$\sum_{Q \text{ in the class}} \nu(Q)^{2/3} \|V\|_{L^6(Q)}^2 \lesssim K^2 (m_j \chi)^{2/3} \sum_Q \sup_{q_Q} |V|^2. \quad (4.27)$$

The selected unit cubes need not themselves carry measure. Their counts are instead controlled by their disjoint full Q -cubes, each of mass comparable to $m_j \chi$. For $r \geq 1$ this gives

$$\#\{q_Q : q_Q \cap B(z, r) \neq \emptyset\} \lesssim \chi^{-1} (r + CK^2)^2 \lesssim K^4 \chi^{-1} r^2.$$

At radius s , using (4.26) and $K^2 \ll s$ instead gives occupancy at most $CL^{CM\eta} \chi^{-1} s^3/D$. Thus the parameters in Lemma 2.3, with ambient radius $S \asymp s^2$, can be taken as

$$\gamma_0 \lesssim K^4 \chi^{-1}, \quad \lambda_0 \lesssim L^{CM\eta} \chi^{-1} s^3/D.$$

Its unit-cube supremum version and (4.27) yield

$$\begin{aligned} \mathcal{T}_\nu(V) &\lesssim_\epsilon L^{CM\eta} K^{10/3} L^\epsilon m_j^{2/3} s^{5/3} D^{-1/3} \|c_{\text{node}}\|_{\ell^2}^2 \\ &= L^{CM\eta} K^{10/3} O_\epsilon(L^\epsilon) m_j^{2/3} s^{4/3} (s/D)^{1/3} \|c_{\text{node}}\|_{\ell^2}^2. \end{aligned} \quad (4.28)$$

Here the logarithmic number of weight classes is absorbed in L^ϵ . The cancellation of the weight parameter is complete:

$$\chi^{2/3} (\chi^{-1} \chi^{-1})^{1/3} = 1.$$

The displayed $K^{10/3}$ consists of K^2 in (4.27) and $K^{4/3}$ from the counting parameter. Allowing additional fixed cube or grid enlargements merely replaces it by K^{C_1} for an absolute C_1 . Spatial localization is as above, so this argument applies even when the node's spatial support extends beyond a single s^2 -box.

We finish with an explicit order of choices. Bounded L causes no difficulty, and $\tau_0 > 1$ permits no unbounded admissible scales, so assume $0 < \tau_0 \leq 1$. Fix first the absolute numbers

$$a_* = \min\{c_b/2, 1/6\}, \quad c_s = a_*/8. \quad (4.29)$$

At a broad leaf $s \geq D^{1/2}$ gives $s^{-c_b} \leq D^{-c_b/2}$, while (4.9) gives the terminal gain

$$(s/D)^{1/3} \leq K^{1/3} D^{-1/6}.$$

Thus both gains are at least D^{-a_*} before the losses just recorded.

Now choose $\beta > 0$ sufficiently small in terms of τ_0, c_b, a_* . It can satisfy, with fixed strict margins,

$$\beta < \frac{\tau_0 c_b}{4}, \quad 2\beta < \frac{\tau_0}{4}, \quad (C_1 + 1/3)\beta < \frac{\tau_0 a_*}{16}.$$

For all large L , these imply $K \leq s^{c_b}$ at every broad node and allow the terminal K factors to consume at most $D^{a_*/8}$, after fixed constants are absorbed. The depth bound $J = O(\beta^{-1})$ is now fixed. Choose the small losses in decoupling and the refined fractal bound so that their total product throughout a path is at most $D^{a_*/8}$. Next choose $\eta > 0$, depending also on J and M , so that all padding losses together cost at most $D^{a_*/8}$ and $2\beta + \eta < \tau_0/2$. This last inequality implies (4.23) at every existing child scale, since each such scale is at least $D^{1/2} \geq L^{\tau_0/2}$. Finally choose $\gamma < \tau_0/4$, so that $|k|/s = o(1)$ for every retained layer, and then choose the frequency-tail exponent, the physical decay orders, and the original profile differentiability. The latter choices make every discarded tail as small a negative power as required; they do not change c_b or c_s .

To see that these choices close the entire tree, a depth- j path contributes, before its broad or terminal gain, at most

$$C^j L^{Cj\eta + O(\epsilon)} h^{-2/3}, \quad h = K^j,$$

times its node estimate. On the other hand (4.25) gives

$$m_j^{2/3} s^{4/3} \leq C^j L^{Cj\eta} h^{2/3} m^{2/3} L^{4/3}.$$

The h powers cancel, as anticipated in (4.24). At a fixed depth the total branch energy is bounded by a fixed constant to that depth times the original energy, including the spatial localizations. Summing over the bounded depth therefore costs only a constant depending on τ_0 . The allotted losses leave more than the saving D^{-c_s} in (4.29). Finally, all tail errors can be made negligible relative to $m^{2/3} L^{4/3} D^{-c_s} \|c\|_{\ell^2}^2$, even after summing the tree: after normalization there are only polynomially many supported cells, boxes and branches, while the tail exponent is chosen last.

This proves the desired bound for $\mathcal{T}$. Minkowski in Lemma 4.2, the Sobolev reduction (4.3), and the initial spatial energy summation give (4.1), with the stated derivative and uniformity properties. $\square$

5 A bilinear frame estimate without a scale loss

This section supplies the estimate for long, nearly flat frequency blocks. Its inputs need not solve the Schrödinger equation: their profiles may vary smoothly on the packet time scale. The proof successively merges frequency caps and removes large pieces. A positive energy recurrence records the removals. Low-frequency flux bounds and a quartic potential control its errors uniformly in the number of merges.

The cap-merging and amplitude-pruning scheme is related to the high-low method of Guth–Maldague–Wang [16, Sections 2.2–2.4 and 3.3] and its p -adic implementation by Guo–Li–Yung [14, Section 7]. Retaining discarded quadratic energy and controlling the resulting square function by quartic telescoping are central in Deng–Fan–Guo–Guo–Luo [10, Sections 4.2 and 6.2] and the related contemporary work of Li–Yung [19, Sections 5.2 and 5.5]. Those latter arguments exploit exact truncation and Fourier-support identities. Here smooth real-variable masks introduce frequency leakage; the low-frequency flux estimate and Gaussian quartic

potential below control the resulting errors uniformly in the number of scales. The smooth-mask, energy-flux and quartic-potential construction and the scale-uniform estimate are proved locally; they are not conclusions imported from the Guth–Maldague–Wang decoupling theorem.

Theorem 5.1 (Bilinear frame estimate). *Fix bounded frequency and profile boxes, a bound on phase-space multiplicity, and a sufficiently large envelope exponent P . There is a fixed finite number of profile derivatives for which the following holds uniformly. Let $h \geq 1$ and $h^{-1/2} \leq \delta \lesssim 1$. For $i = 1, 2$, let*

$$V_i(x, t) = \sum_{\ell \in \mathcal{I}_i} s_\ell e^{i(x\xi_\ell - t\xi_\ell^2)} H_\ell \left(\frac{x - b_\ell - 2\xi_\ell t}{h}, \frac{t}{h^2} \right)$$

be a finite packet sum. The profiles are supported in the fixed box and have the stipulated uniform derivative bounds. The centers b_ℓ may be arbitrary real numbers, and the pairs $(b_\ell/h, h\xi_\ell)$ have bounded multiplicity in every unit phase-space ball. The frequencies of the two fields lie in intervals of length δ with distance at least a fixed positive multiple of δ . Put

$$S_i(x, t)^2 = \sum_{\ell \in \mathcal{I}_i} |s_\ell|^2 \left(1 + \frac{|x - b_\ell - 2\xi_\ell t|}{h} + \frac{|t|}{h^2} \right)^{-P}, \quad H_i^* = h^3 \sum_{\ell \in \mathcal{I}_i} |s_\ell|^2.$$

For $A_i, B_i > 0$, write $\alpha_i = A_i/B_i$. If $\alpha_i \geq C$, for a sufficiently large fixed C , the number of unit lattice squares containing a point where simultaneously $|V_i| \geq A_i$ and $S_i \leq B_i$ for $i = 1, 2$ satisfies

$$\# \leq C' \delta^{-C'} \alpha_1^{-2} \alpha_2^{-2} \left(\frac{H_1^*}{A_1^2} + \frac{H_2^*}{A_2^2} \right). \quad (5.1)$$

The constants depend only on the fixed bounds just specified. In particular, they are independent of h , the number of packets, and the number of scales in the proof. The loss in the separation parameter has at most a fixed polynomial power.

We always take one witness per counted unit square; the witnesses for all four inequalities in a square must be the same point. All norms below, unless a variable is specified, are on $(x, t) \in \mathbb{R}^2$. Spatial Fourier multipliers act only in x . Fourier normalization constants in positive potentials are absorbed into their definitions. Set

$$\mathcal{L} = i\partial_t + \partial_x^2, \quad D_a = \partial_t + 2a\partial_x - ia^2.$$

The latter derivative annihilates the carrier at frequency a ; useful identities are

$$\mathcal{L} = iD_a + (\partial_x - ia)^2, \quad D_a - D_b = 2(a - b)(\partial_x - ib) - i(a - b)^2. \quad (5.2)$$

5.1 Reduction to small, exactly localized frequency pieces

Lemma 5.2 (Frequency layers and initial normalization). *It suffices to prove (5.1) for packets whose spatial frequencies are supported in caps of width $O(h^{-1})$ about their assigned centers, whose spatial profiles have sufficiently rapid polynomial decay, and whose native time supports remain bounded. After dividing a field by its envelope bound B , its witness level is $\alpha = A/B$ and its envelope is at most one. Given*

$$\lambda = C_*/\alpha \leq 1,$$

deleting all packets of normalized amplitude greater than λ changes the field at every witness by at most $C\lambda^{-1}$. The remaining field g_0 has an energy parameter $D \lesssim H^/B^2$ such that*

$\|g_0\|_2^2 \leq D$. At a dyadic scale $\Delta_0 \asymp h^{-1}$, its fine cap pieces $g_{0,b}$ satisfy, for all fixed orders needed below,

$$\|(\partial_x - ib)^s D_b^r g_{0,b}\|_\infty \leq C_{r,s} \lambda \Delta_0^{s+2r}, \quad (5.3)$$

$$\sum_b \|(\partial_x - ib)^s D_b^r g_{0,b}\|_2^2 \leq C_{r,s} \Delta_0^{2s+4r} D. \quad (5.4)$$

In particular $\|\mathcal{L}^r g_0\|_2 \lesssim_r \Delta_0^{2r} \sqrt{D}$.

Proof. Smoothly partition the spatial Fourier transform of each native profile into layers $k + [-2, 2]$, $k \in \mathbb{Z}$, writing $\langle k \rangle = 1 + |k|$. A fixed, sufficiently large number of derivatives gives any prescribed fixed polynomial layer decay together with the finitely many spatial decay and derivative estimates that we will use. Assign the new carrier center $\xi_\ell + k/h$. In its native coordinates the profile undergoes a translation by $2k$ times native time and acquires a temporal oscillation of frequency k^2 . Every required derivative and envelope comparison therefore costs only a fixed polynomial in $\langle k \rangle$. Choose the original decay orders larger than these costs and put the remaining decay into the layer amplitudes. A common frequency shift preserves the phase-space counts.

At a witness the sum of all layers with $|k| \geq c\delta h$ is bounded by

$$C_N(\delta h)^{-N} \sqrt{h} B.$$

Indeed, in each original frequency bin of width h^{-1} , the spatial tails sum uniformly by bounded offset counts; weighted Cauchy–Schwarz over the $O(h)$ frequency bins gives $\sqrt{h}$. Since $\delta h \geq \sqrt{h}$, the displayed error is negligible for large h . Bounded scales are vacuous once the required ratio constant is large, by the same envelope Cauchy–Schwarz estimate. Choose the fixed $c > 0$ small enough that the retained frequency layers still lie in separated intervals of length $O(\delta)$.

If the retained sum is large, some layer has modulus at least $c'\langle k \rangle^{-2}A$. For any prescribed fixed large N its envelope bound and energy can be taken to be $C_N \langle k \rangle^{-N}B$ and $C_N \langle k \rangle^{-2N}H^*$, respectively. Thus the ratio condition persists, and applying the desired estimate at these thresholds and summing the two layer indices converges. For example, the ratio factors gain $\langle k \rangle^{-2(N-2)}$ and the corresponding energy-to-level ratio gains the same power. Constants and an initial finite set of layers are absorbed into the fixed ratio threshold.

We may consequently work with one exactly cap-supported layer, divided by B . If $|s_\ell| > \lambda$, then $|s_\ell| \leq \lambda^{-1}|s_\ell|^2$. The pointwise profile decay, chosen stronger than the envelope decay, bounds the deleted sum by $C\lambda^{-1}S^2 \leq C\lambda^{-1}$ at witnesses.

For a fixed fine frequency bin, transported spatial intercepts still have bounded counts per interval of length h on $|t| \lesssim h^2$: velocity variation inside that bin changes positions by only $O(h)$. Summing profile tails over these offsets gives (5.3), since all retained amplitudes are at most λ . Each derivative D_b costs Δ_0^2 , and each $\partial_x - ib$ costs Δ_0 , including the harmless change from an individual center to the bin center. Weighted Cauchy–Schwarz within a bin gives (5.4); a single packet has squared spacetime measure scale h^3 . Spatial frequencies of distinct bins have bounded overlap. Enlarging D by a fixed factor proves all the stated assertions. $\square$

5.2 A positive frame iteration and smooth deletion masks

Choose a large fixed dyadic integer K , whose size will be fixed in Lemma 5.3. Set $\Delta_j = K^j \Delta_0$, ending at $\Delta_n \asymp_K c_0 \delta$, where $c_0 > 0$ is a sufficiently small fixed constant. A shorter range with only boundedly many fine caps per frequency interval again makes large ratios impossible.

The terminal cap size is chosen small enough to preserve the frequency gap under all the enlargements below.

Fix a real $p \in C_c^\infty(\mathbb{R})$ with $\int p^2 = 1$. At stage j , write $\Delta = \Delta_j$ and

$$P_a = p((\xi - a)/\Delta), \quad \int_a = \int_{\mathbb{R}} \frac{da}{\Delta}, \quad \int_a P_a^2 = I.$$

With $g = g_{j-1}$ define

$$\begin{aligned} c_a &= P_a g, & b_a &= \rho_a c_a, & d_a &= c_a - b_a, \\ g_j &= \int_a P_a b_a, & R_j &= g_{j-1} - g_j = \int_a P_a d_a, \\ L_a &= |c_a|^2 - |b_a|^2, & L_j &= \int_a L_a, & l_j &= \int_{\mathbb{R}^2} L_j. \end{aligned} \tag{5.5}$$

The masks will obey $0 \leq \rho_a \leq 1$. Analysis by the P_a is an isometry and synthesis is its adjoint, hence a contraction. Therefore

$$\|g_j\|_2^2 \leq \|g_{j-1}\|_2^2 - l_j, \quad \sum_j l_j \leq D, \quad \int_a \|d_a\|_2^2 \leq l_j. \tag{5.6}$$

Here $|d_a|^2 \leq L_a$ follows from $(1 - \rho_a)^2 \leq 1 - \rho_a^2$. The outer projection in synthesis ensures that each step inflates spatial frequency support by only $O(\Delta_j)$, even though masking itself creates frequencies. The total inflation is $O(\sum_j \Delta_j) = O(\Delta_n)$, preserving the gap. All states retain the original bounded time support.

We also fix positive spatial convolutions W_j with kernels comparable to $\Delta_j(1 + \Delta_j|x|)^{-P_0}$, where $P_0 > P$ is a sufficiently large fixed even integer. Their common rescaled multiplier w is even, compactly supported, sufficiently smooth, and satisfies $w(0) = 1$. One construction takes the normalized autocorrelation of a high convolution power of $e^{-x}\mathbf{1}_{[0,1]}(x)$ as multiplier. The Fourier modulus of the truncated exponential is comparable to $(1 + |\xi|)^{-1}$, so the resulting positive kernel has the asserted two-sided polynomial bounds. Increasing that fixed convolution power supplies the required multiplier smoothness.

At every normalized witness z_0 ,

$$W_0|g_0|^2(z) \lesssim (1 + |z - z_0|/h)^P, \tag{5.7}$$

with a harmless enlargement of the fixed envelope powers if needed. To verify this, compact support of the multiplier of W_0 eliminates products of nonneighboring fine caps. Bound each remaining product by the sum of the squared cap moduli, and then use weighted Cauchy-Schwarz and the transported offset counts within each cap. Positive convolution preserves the resulting sum of squared packet amplitudes with decaying spatial tails. Bounded velocities give its polynomial comparison with the envelope at z_0 ; the original native time support is bounded. Choosing all decay powers in advance larger than P proves (5.7).

The global inequality $\sum_j l_j \leq D$ does not by itself control how much field is removed at a particular witness. The masks below will satisfy the pointwise bound $|R_j| \lesssim_K \lambda^{-1} W_j L_j$. We therefore need a positive recurrence for the smoothed losses $W_j L_j$. The following identity holds for any masks $0 \leq \rho_a \leq 1$ in (5.5); it does not use their construction.

Define

$$S_j^0 = W_j |g_j|^2, \quad B_j^0 = \int_a |d_a - P_a R_j|^2 \geq 0.$$

The term B_j^0 compares the removed cap family d_a with the analysis $P_a R_j$ of its synthesized field. Define the two error fields and their accumulated sum by

$$H_j = (W_j - W_{j-1})|g_{j-1}|^2, \quad \Phi_j = |g_j|^2 - |g_{j-1}|^2 + L_j + B_j^0, \quad Y_k = \sum_{j \leq k} (H_j + W_j \Phi_j).$$

The exact telescoping identity is

$$S_k^0 + \sum_{j \leq k} W_j(L_j + B_j^0) = S_0^0 + Y_k. \quad (5.8)$$

The term H_j records the change of smoothing scale; Φ_j records the local error in passing between cap analysis and synthesis. Every term on the left is nonnegative. In particular, the witness baseline (5.7) gives $\sum_{j \leq k} W_j L_j \lesssim 1 + |Y_k|$ at a witness. Thus the accumulated deletions remain controlled until Y_k becomes large. We will prove the uniform maximal estimate

$$\left\| \sup_{0 \leq k \leq n} |Y_k| \right\|_2 \leq C_K \lambda \sqrt{D}, \quad Y_0 = 0. \quad (5.9)$$

Lemma 5.3 (Masks, persistence, and loss bounds). *The masks can be chosen so that*

$$|b_a| \leq \lambda, \quad \|\mathcal{L}P_a b_a\|_\infty \leq C_1 \lambda \Delta^2, \quad (5.10)$$

where C_1 is independent of the large fixed K . For every required fixed derivative order,

$$\|D_a^r P_a b_a\|_\infty \leq C_{r,K} \lambda \Delta^{2r}, \quad \int_a \|D_a^r P_a b_a\|_2^2 \leq C_{r,K} \Delta^{4r} D. \quad (5.11)$$

The removed pieces obey

$$\begin{aligned} \|(\partial_x - ia)^s D_a^r d_a\|_1 &\leq C_{K,r,s} \Delta^{s+2r} \lambda^{-1} \int L_a, \\ \|(\partial_x - ia)^s D_a^r d_a\|_2^2 &\leq C_{K,r,s} \Delta^{2s+4r} \int L_a, \end{aligned} \quad (5.12)$$

and, pointwise,

$$|R_j| \leq C_K \lambda^{-1} W_j L_j. \quad (5.13)$$

All derivative constants are uniform in the stage number.

Proof. Construction of the masks. Summing the preceding caps that meet P_a gives

$$\|c_a\|_\infty \leq CK\lambda, \quad \|\mathcal{L}c_a\|_\infty \leq CC_1 \lambda \Delta^2 / K. \quad (5.14)$$

There are $O(K)$ preceding normalized centers, and their curvature scale is $(\Delta/K)^2$. The initial bounds give the same estimates at the first merge.

Take a nonnegative smooth partition of unity $\sum_e \chi_e(\Delta^2 t) = 1$ by translates with bounded support and overlap, each bounded below on a fixed inner interval. In the co-moving coordinate $r_x = \Delta(x - 2at)$, mark every unit cell on which $|c_a| > \lambda$ somewhere during the open positive support of the e th time cutoff, and let A_e be their union. With a fixed nonnegative $\psi \in C_c^\infty((-1, 1))$ of integral one, take

$$m_e = \psi * \mathbf{1}_{\{r: \text{dist}(r, A_e) < K+2\}}.$$

For $K \geq 4$, this function is one within distance K of A_e , zero outside its $2K$ enlargement, and lies in $[0, 1]$; its fixed-order derivative bounds are independent of K . Define

$$1 - \rho_a = \sum_e \chi_e(\Delta^2 t) m_e(r_x).$$

Choose the time cutoffs flat to all required orders at the endpoints of their positive supports. Marking can be tested on dense countable sets in the open slabs, using continuity, so the construction is measurable also in a . Half-open cell conventions, or fixed slight enlargements,

leave all bounds unchanged. At a point with $|c_a| > \lambda$ every active index is marked, giving $\rho_a = 0$ and therefore $|b_a| \leq \lambda$.

The curvature bound. Where a derivative of ρ_a is nonzero, there is no same-time peak within co-moving distance $K/2$. Otherwise every active index would be identically fully marked locally, and all derivatives of the mask would vanish. Local Bernstein, using $|c_a| \leq \lambda$ in this neighborhood and the global bound $CK\lambda$ against rapidly decreasing reproducing-kernel tails outside it, gives $|(\partial_x - ia)c_a| \lesssim \Delta\lambda$ there. Now

$$\mathcal{L}(\rho_a c_a) = \rho_a \mathcal{L}c_a + (i(\partial_t + 2a\partial_x)\rho_a + \partial_x^2 \rho_a)c_a + 2(\partial_x \rho_a)(\partial_x - ia)c_a.$$

Scaled derivatives of the masks cost Δ spatially and Δ^2 in co-moving time. Thus $\|\mathcal{L}b_a\|_\infty \lesssim \lambda\Delta^2(1 + C_1/K)$. The outer projection has bounded L^1 kernel. Choose C_1 first sufficiently large independently of K , and then K sufficiently large, to close (5.10).

Uniform higher derivatives. Order zero follows from bounded masks and frame contraction; spatial derivatives of projected pieces follow from Bernstein. For higher D_a orders use (5.2) to express the incoming piece in preceding centers $\tilde{a}$. After normalization by Δ^{2r} , the coefficient of the preceding top-order supremum bound is at most CK^{1-2r} . In integrated center L^2 norm it is at most $CK^{1/2-2r}$: Cauchy–Schwarz in the preceding-center window costs $O(\sqrt{K})$, while the current-center window for each fixed preceding center has da/Δ measure $O(1)$. Every other term in the center change uses a lower affine time order; its spatial derivatives cost the preceding Bernstein scale. In $D_a^r(\rho_a c_a)$ the top term is $\rho_a D_a^r c_a$, with $|\rho_a| \leq 1$, and all remaining terms use lower orders and controlled co-moving derivatives of ρ_a . Fix the maximum required derivative order in advance. The top coefficients are contractive for every $r \geq 1$ in this finite list after increasing K once. This enlargement leaves C_1 unchanged and only improves both the curvature closure and the Duhamel source smallness. Induction on r , and then on stages with that contraction, proves (5.11). The first step uses the fine lattice-bin estimates. The same reasoning gives mixed spatial and affine supremum bounds for c_a, d_a , with constants depending on K .

Persistence of a marked peak. A peak in a marked cell forces spatial squared mass at least $c\lambda^2/\Delta$ within co-moving distance $K/2$ of that cell at *every* comparison time in its time slab. To see this, propagate from the comparison time t to the peak time t_0 by Duhamel, inserting a smooth cap cutoff equal to one on the spatial Fourier supports of both c_a and $\mathcal{L}c_a$. Since $|t - t_0| = O(\Delta^{-2})$, the co-moving propagator kernels are uniformly rapidly decreasing at scale Δ^{-1} , with L^1 norm $O(1)$ and L^2 norm $O(\sqrt{\Delta})$. The source contribution is at most $CC_1\lambda/K$ by (5.14). The main term outside distance $K/2$ is negligible by its kernel tails and the global $CK\lambda$ bound. With K fixed sufficiently large, the remaining main term has modulus at least $c\lambda$. Cauchy–Schwarz with its $O(\sqrt{\Delta})$ kernel norm proves the asserted mass lower bound. This argument explicitly permits nonzero $\mathcal{L}c_a$.

Charging the activated sets. We have $L_a \geq \sum_e \chi_e m_e |c_a|^2$. For each marked cell and slab, its enlarged activated region has spacetime volume $O_K(\Delta^{-3})$. Integrating the persistence bound over the inner times where χ_e is bounded below gives a charge $c\lambda^2\Delta^{-3}$ in that cell’s fully marked neighborhood. These neighborhoods have overlap $O_K(1)$, including across slabs. Consequently the total activated volume, including the supports of mask derivatives counted slab by slab, is at most $C_K\lambda^{-2} \int L_a$. The mixed derivative supremum bounds for d_a are $C_{K,r,s}\lambda\Delta^{s+2r}$ on this activated set and vanish off it. Multiplying by its volume proves both inequalities in (5.12).

Finally fix a time with $\chi_e > 0$. The spatial projection of $\chi_e m_e c_a$ is bounded by summing its rapidly decreasing kernel over activated cells, using $|c_a| \leq CK\lambda$. Persistence gives at that same time a lower squared-mass bound in each corresponding fully marked neighborhood. Since the positive kernel of W_j decays more slowly than the projection kernel, and the

enlarged neighborhoods have bounded overlap, this gives

$$|P_a(\chi_e m_e c_a)| \leq C_K \lambda^{-1} W_j(\chi_e m_e |c_a|^2).$$

The common factor $\chi_e(t)$ causes no difficulty even near slab endpoints. More explicitly, with $u = \Delta(x - 2at)$ and marked cell indices q , the kernel upper bound is $C_K \lambda \sum_q (1 + |u - q|)^{-P_0}$. Persistence, kernel comparability on the fully marked neighborhoods, and their $O_K(1)$ overlap give

$$\lambda^2 \sum_q (1 + |u - q|)^{-P_0} \lesssim_K W_j(m_e |c_a|^2)(x, t).$$

Multiplying by $\chi_e(t)$, summing e , and integrating a proves (5.13). $\square$

5.3 Energy flux and the quartic potential

We now bound the two errors in (5.8). The deletion flux $W_j \Phi_j$ will be charged to l_j ; the changes H_j of smoothing scale require a quartic potential.

Lemma 5.4 (Low-frequency flux). *For a smooth spatial projection Proj_r to $|\xi| \asymp r$, where $r \lesssim \Delta_j$,*

$$\|\text{Proj}_r W_j \Phi_j\|_2^2 \leq C_K \lambda^2 (r/\Delta_j) l_j. \quad (5.15)$$

Proof. Write $g = g_{j-1}$, $R = R_j$, and $\Delta = \Delta_j$. Expanding B_j^0 and using $g_j = g - R$ gives

$$\begin{aligned} \Phi_j = & -2 \operatorname{Re} \left(g \bar{R} - \int_a (P_a g) \bar{d}_a \right) \\ & + 2 \operatorname{Re} \left(|R|^2 - \int_a d_a \overline{P_a R} \right) + \int_a |P_a R|^2 - |R|^2. \end{aligned}$$

For either of the first two differences the product multiplier contains, up to sign, $p((\xi_1 - a)/\Delta) - p((\xi_2 - a)/\Delta)$. On the output band $|\xi_1 - \xi_2| \asymp r$ it costs $O(r/\Delta)$. Both inputs may be restricted to $O(\Delta)$ about a : the cutoff on one factor and the output support force the same localization on the other, including the otherwise unprojected d_a . The last difference has symbol

$$\int_{\mathbb{R}} p((\xi_1 - a)/\Delta) p((\xi_2 - a)/\Delta) \frac{da}{\Delta} - 1,$$

which vanishes at $\xi_1 = \xi_2$ and has the same bound. After bin and band localization, all fixed rescaled derivatives in $(\xi_1 - a)/\Delta$ and $(\xi_1 - \xi_2)/r$ satisfy that bound. Fourier expansion therefore represents each multiplier by spatially translated products with total coefficient or kernel mass $O_K(r/\Delta)$.

Here is the loss-sensitive norm bound for those products. Let Q_b be smooth lattice projections of width Δ , allowing fixed enlargements. From (5.11) and the center-change identity,

$$\|D_b^s Q_b g\|_\infty + \|D_b^s Q_b R\|_\infty \leq C_{s,K} \lambda \Delta^{2s}.$$

Moreover,

$$\sum_b \|D_b^s Q_b R\|_2^2 \leq C_{s,K} \Delta^{4s} l_j. \quad (5.16)$$

Indeed $Q_b R = \int_a Q_b P_a d_a$ uses only $|a - b| \lesssim \Delta$; apply Cauchy–Schwarz in this normalized window and (5.12), observing boundedly many b per a . The terms with an explicit d_a are handled in the same way after grouping the continuous centers into lattice bins and inserting the permitted spatial projections. All these estimates persist under the spatial translations in the multiplier expansion. Indeed, for $T_y f(x, t) = f(x - y, t)$, one has $D_b T_y = T_y D_b$;

translations preserve both spatial cap support and the global L^∞, L^2 norms used here. Thus even translations of size r^{-1} in the output-band variable have no cost depending on Δ/r .

Thus the output from a bin centered at b has L^2 norm at most $C_K \lambda(r/\Delta) l_b^{1/2}$, with $\sum_b l_b \lesssim_K l_j$. It has the same bound after any of the fixed normalized product derivatives $((\partial_t + 2b\partial_x)/\Delta^2)^s$ that we need. On a product with a conjugated factor this derivative splits as D_b on the factors, with the carrier constants canceling. On spacetime Fourier variables (ξ, τ) , the weighted overlap of the bin strips obeys

$$\sum_b \left(1 + \frac{|\tau + 2b\xi|}{\Delta^2}\right)^{-4} \lesssim 1 + \frac{\Delta}{r}.$$

The centers b have spacing Δ , so their strip centers in τ have spacing comparable to Δr and width Δ^2 . Weighted Cauchy–Schwarz, followed by Plancherel and the differentiated product bounds, gives

$$\|\text{Proj}_r W_j \Phi_j\|_2^2 \lesssim_K \lambda^2(r/\Delta)^2 (1 + \Delta/r) \sum_b l_b,$$

which is (5.15). $\square$

Flux alone does not control the change in the spatial smoothing scale. We obtain that control from a positive quartic expression, whose Gaussian multiplier permits an explicit calculation.

Lemma 5.5 (Quartic control of the growth bands). *Let $G_j(u) = \exp(-cu^2/\Delta_j^2)$, for a fixed $c > 0$. Then*

$$\sum_{j=1}^n \int_{\mathbb{R}^2} (G_j - G_{j-1})(u) |\widehat{|g_{j-1}|^2}(u, \tau)|^2 du d\tau \lesssim_K \lambda^2 D. \quad (5.17)$$

Proof. For a spatial Gaussian multiplier $q_{j,a}$ of width Δ_j centered at a , define the normalized potential

$$\mathcal{P}_j(f) = \int_{\mathbb{R}} \int_{\mathbb{R}^2} |q_{j,a} f(z)|^4 dz \frac{da}{\Delta_j}.$$

On quartets with alternating sums of both spatial and temporal frequencies equal to zero, put $u = \xi_1 - \xi_2$ and $v = \xi_1 - \xi_4$. The common midpoint is $(\xi_1 + \xi_3)/2 = (\xi_2 + \xi_4)/2$; the sum of the four squared deviations from it is $u^2 + v^2$. Integrating the Gaussian center therefore gives exactly the multiplier $G_j(u)G_j(v)$, after fixed normalization. Expanding this product yields

$$\mathcal{P}_j(g) - \mathcal{P}_{j-1}(g) = 2 \int (G_j - G_{j-1})(u) |\widehat{|g|^2}(u, \tau)|^2 du d\tau + \text{Err}_j, \quad (5.18)$$

where $g = g_{j-1}$ and the error multiplier is

$$E_j(u, v) = (1 - G_j(u))(1 - G_j(v)) - (1 - G_{j-1}(u))(1 - G_{j-1}(v)).$$

The two quadratic terms in the expansion are equal by interchanging u and v .

We establish the quantitative error estimate

$$|\text{Err}_j| \lesssim_K \lambda^2 \left[\left(\frac{\Delta_0}{\Delta_{j-1}} \right)^2 D + \sum_{m < j} \left(\frac{\Delta_m}{\Delta_{j-1}} \right)^2 l_m \right]. \quad (5.19)$$

Write $\Delta' = \Delta_{j-1}$ and partition every input into smooth spatial caps Q_b of width Δ' . The quartet identity

$$2uv = \xi_1^2 - \xi_2^2 + \xi_3^2 - \xi_4^2 = \sum_{\nu=1}^4 (-1)^{\nu+1} (\tau_\nu + \xi_\nu^2) \quad (5.20)$$

allows division of E_j by $2uv/(\Delta')^2$, placing a normalized curvature derivative $\mathcal{L}/(\Delta')^2$ on one input. The quotient is smooth at $u = 0$ or $v = 0$, since E_j vanishes quadratically in each variable. The sign of the multiplier $\mathcal{L}$, namely $-(\tau + \xi^2)$, is irrelevant to the bounds.

Norm bookkeeping for a distinguished derivative. On the input that receives this first derivative write $g = g_0 - \sum_{m < j} R_m$. For every required fixed $q \geq 1$,

$$\left(\sum_b \|Q_b \mathcal{L}^q g_0 / (\Delta')^{2q}\|_2^2 \right)^{1/2} \lesssim_K (\Delta_0 / \Delta')^{2q} \sqrt{D}, \quad (5.21)$$

$$\sum_b \|Q_b \mathcal{L}^q R_m / (\Delta')^{2q}\|_1 \lesssim_K (\Delta_m / \Delta')^{2q} \lambda^{-1} l_m. \quad (5.22)$$

The first assertion is the initial mixed-derivative estimate and bounded spatial Fourier overlap. For the second use $R_m = \int_a P_a d_a$ at scale Δ_m and $\mathcal{L} = iD_a + (\partial_x - ia)^2$. Each derivative costs Δ_m^2 in the L^1 bound (5.12). The projection kernels have bounded L^1 norm, and each earlier synthesis center meets only boundedly many current bins, proving (5.22). On every other input keep $g = g_{j-1}$ itself. Its normalized curvature derivatives in Δ' caps have supremum $C_K \lambda$ and square-summed L^2 norm $C_K \sqrt{D}$, by (5.11), spatial Bernstein, and the bounded preceding-center window. These bounds hold for the fixed extra derivative orders introduced below.

For one fixed cap-offset pattern, Fourier expansion of a smooth rescaled multiplier produces integrals of spatially shifted products. Hölder with a distinguished L^2 input, another L^2 input and two L^∞ inputs gives the base term in (5.19). For an earlier loss, use its distinguished L^1 input and three L^∞ inputs. In summing the common cap vertex, Cauchy–Schwarz handles the two L^2 sequences, or direct summation handles the L^1 sequence. There is no additional factor counting free caps. Any further derivative falling on the distinguished input preserves its gain, since $\Delta_m \leq \Delta'$.

The off-diagonal multiplier summation. On $|u|, |v| \lesssim_K \Delta'$ the divided symbol, with smooth cutoffs, has bounded Fourier-kernel L^1 norm in its two rescaled variables, and only boundedly many cap offsets occur. For explicit rescaled bounds, write $s = u/\Delta'$, $t = v/\Delta'$, $F_A(s) = 1 - e^{-cs^2/A^2}$, and $E(s, t) = F_K(s)F_K(t) - F_1(s)F_1(t)$. For the remaining regions, by symmetry let $|v| \asymp Z\Delta'$ be the larger variable, with Z dyadic and large. If $|u| \lesssim \Delta'/Z$, the once-divided symbol has kernel cost $O_K(Z^{-2})$ at these derivative scales. There are $O(Z)$ possible offsets for each cap vertex, giving total cost $O_K(Z^{-1})$. In fact $E(s, t) = s^2 e(s, t)$, with every required derivative of e in coordinates $(Zs, t/Z)$ bounded uniformly in Z . Consequently $E(s, t)/(2st) = (s/(2t))e(s, t)$ has fixed rescaled derivative bounds $O_K(Z^{-2})$ after the cutoffs. Anisotropic rescaling preserves the L^1 norm of its Fourier kernel.

In the remaining dyadic regions $|u| \asymp U\Delta'$, $1/Z \lesssim U \lesssim Z$, use (5.20) three additional times. After four divisions the symbol has kernel cost

$$O_K((U/Z)(UZ)^{-3}) = O_K(U^{-2}Z^{-4}).$$

Indeed the undivided error and its fixed derivatives in coordinates $(s/U, t/Z)$ cost $O_K(\min(U^2, 1)) \leq O_K(U^2)$, while each division costs $(UZ)^{-1}$. Smooth Gaussian derivatives give these estimates uniformly on overlapping rectangles; the very small u region uses a single cutoff at scale Δ'/Z . The spatial sum constraint on the quartet leaves at most $O(Z \max(1, U))$ offset patterns per fixed vertex. Fixing the first cap leaves $O(\max(1, U))$ choices for the second and $O(Z)$ for the fourth, and then $O(1)$ for the third by the sum constraint. The multiplier-induced spatial translations only modulate Fourier factors and do not enlarge their supports, so they do not change these counts. Their combined cost is therefore $O_K(Z^{-3}U^{-2} \max(1, U))$. For fixed Z the dyadic sum over $1/Z \lesssim U \leq 1$ is $O_K(Z^{-1})$, and that over $1 \leq U \lesssim Z$ is $O_K(Z^{-3})$. Including the very small u part and then

summing dyadic Z converges. At least the first curvature derivative remains distinguished in every term of the expanded quartet identities. The norm bounds above thus prove (5.19), with no temporal cap-support assumption and no dependence on the number of scales.

Telescoping the potential. At fixed scale j ,

$$|\mathcal{P}_j(g_j) - \mathcal{P}_j(g_{j-1})| \lesssim_K \lambda^2 l_j. \quad (5.23)$$

Both Gaussian-filtered states have supremum $C_K \lambda$ by their nearby compact cap pieces and rapid Gaussian overlap. Also

$$\int_{\mathbb{R}} \|q_{j,a} R_j\|_1 \frac{da}{\Delta_j} \lesssim \int_a \|d_a\|_1 \lesssim_K \lambda^{-1} l_j.$$

The first inequality follows by composing the Gaussian with the compact synthesis projections: their L^1 kernels have rapidly decaying overlap in the normalized difference of centers. The inequality $||u|^4 - |v|^4| \lesssim (|u| + |v|)^3 |u - v|$ then proves (5.23). Similarly $\mathcal{P}_n(g_n) \lesssim_K \lambda^2 D$. In summing (5.18), insert and subtract $\mathcal{P}_j(g_j)$ at each stage. The remaining boundary terms are $\mathcal{P}_n(g_n) - \mathcal{P}_0(g_0)$, with the initial term favorable. The errors in (5.19) sum because

$$\sum_{j>m} (\Delta_m/\Delta_{j-1})^2 \lesssim_K 1, \quad \sum_j (\Delta_0/\Delta_{j-1})^2 \lesssim_K 1, \quad \sum_m l_m \leq D.$$

This proves (5.17).

All identities are global spacetime integrals. There are finitely many scales, and cap supremum bounds together with L^2 bounds give finite quartic integrals. Smooth localized approximations, if needed, justify Fourier expansions; the established fixed derivative and decay bounds pass to the limit. No derivative in the center variable a is used. $\square$

Lemma 5.6 (Maximal recovery of the error fields). *The partial errors obey (5.9). If $\mathcal{M}$ is the spacetime Hardy–Littlewood maximal operator, then also*

$$|Y_k(z')| \lesssim (1 + \Delta_k |z - z'|)^C \mathcal{M}(\sup_j |Y_j|)(z) + C_K \lambda^2. \quad (5.24)$$

Proof. Recovering partial sums without a logarithm. Decompose the spatial outputs of H_j and $W_j \Phi_j$ into smooth Littlewood–Paley bands $r \asymp 2^{-\ell} \Delta_j$, with ℓ bounded below by a fixed integer. Lemma 5.4 and $\sum_j l_j \leq D$ bound the square sum of flux band norms by $C_K \lambda^2 D 2^{-\ell}$. For the growth term, evenness of w at zero and its compact support give, on the corresponding band,

$$|w(u/\Delta_j) - w(u/\Delta_{j-1})|^2 \leq C_K 2^{-2\ell} (G_j - G_{j-1})(u).$$

Near zero the squared difference vanishes to fourth order, whereas the Gaussian gap vanishes to second order; on the rest of the fixed compact support their ratio is bounded. Hence (5.17) bounds the square sum of growth band norms by $C_K \lambda^2 D 2^{-2\ell}$.

For a fixed ℓ , successive stage bands have scale ratio K . Choose K larger than their fixed support ratio, so the spectra are disjoint and ordered. If F_ℓ is the sum over all stages in that relative band, a smooth spatial low pass between successive bands recovers each initial segment exactly and kills the later terms. Its pointwise maximal value is bounded by a constant times the spatial Hardy–Littlewood maximal function of F_ℓ . Plancherel for the disjoint spectra and the maximal L^2 bound, integrated also in t , therefore bound maximal initial segments by the square sum of their norms. Summing ℓ by the triangle inequality converges, with leakages $2^{-\ell/2}$ and $2^{-\ell}$. Zero spatial frequency contributes no separate L^2 piece. This proves (5.9); no maximal inequality for arbitrary orthogonal series is being used.

Local constancy. In each H_j and $W_j \Phi_j$, spatial output support and the flux expansion permit grouping into $O(1/\Delta_j)$ matched product bins at bounded centers b . The unprojected

d_a factors can again be restricted to matched bins by the other input and output supports. Every such product has

$$\|(\partial_t + 2b\partial_x)^s(\text{product})\|_\infty \leq C_{s,K}\lambda^2\Delta_j^{2s}.$$

This follows from (5.11), the mask derivative bounds, and bounded spatial L^1 kernels; the preceding caps per current bin cost only a fixed factor depending on K . Cut a product to affine temporal frequency $|\tau + 2b\xi| \lesssim \Delta_j$. The smooth high pass divided by the s th power of the rescaled affine frequency has integrable kernel, for a fixed sufficiently large s . The resulting error is $C_{s,K}\lambda^2\Delta_j^s$ per bin, hence at most $C_{s,K}\lambda^2\Delta_j^{s-1}$ for the stage. These errors sum geometrically for $s > 1$.

The sum $\tilde{Y}_k$ of the truncated products differs from Y_k by $O_K(\lambda^2)$ uniformly and has full spacetime Fourier support in a ball of radius $O(\Delta_k)$: its spatial frequencies are $O(\Delta_j)$ and its velocities are bounded. A smooth spacetime low pass at scale Δ_k recovers $\tilde{Y}_k$, so it also recovers Y_k up to $O_K(\lambda^2)$. Translating its Schwartz kernel from z' relative to a reference point z , and comparing dyadic kernel annuli with the maximal function, proves (5.24). $\square$

5.4 Stopping and sampling the two separated groups

Lemma 5.7 (A local bilinear sampling bound). *Consider the two states from the preceding construction with $0 < \lambda_i \leq 1$. Let Q be a square of side Δ_k^{-1} and let $\mathcal{Z}_Q \subset Q$ contain at most one designated point from each unit lattice square. Suppose both states $g_{k-1}^{(i)}$ have modulus at least $c\alpha_i$ at these points. There is a Schwartz weight η_Q , bounded below on Q and with Fourier support of radius $O(\Delta_k)$, for which*

$$\alpha_1^2\alpha_2^2\#\mathcal{Z}_Q \lesssim_K \delta^{-C} \int_{\mathbb{R}^2} |\eta_Q(z')|^2 \prod_{i=1}^2 \left(W_{k-1}|g_{k-1}^{(i)}|^2(z') + 1 \right) dz', \quad (5.25)$$

provided the ratio constants are sufficiently large. The two frequency intervals must retain a gap comparable to δ .

Proof. Put $\Delta = \Delta_k$. Partition each state by smooth spatial lattice caps Q_b of width Δ . Cut each piece to affine modulation

$$|\tau + 2b\xi - b^2| \lesssim \Delta.$$

Its D_b^r estimates show that this cutoff makes a uniform error $C_{r,K}\lambda_i\Delta^r$ per cap, by the same inverse-derivative high-pass argument as in Lemma 5.6. There are $O(1/\Delta)$ caps. For a fixed sufficiently large r , the total supremum error is $O_K(\lambda_i)$ and the sum of squared errors is $O_K(\lambda_i^2)$. Thus the truncated fields, denoted $\tilde{g}^{(i)} = \sum_b f_{i,b}$, still have modulus comparable to α_i at the designated points.

Choose η_Q as stated, with arbitrarily high fixed Schwartz decay at the scale of Q . Such a weight can be made by scaling a function with compact smooth Fourier support and positive lower bound on the unit square. The full spacetime Fourier support of $\eta_Q\tilde{g}^{(1)}\tilde{g}^{(2)}$ is bounded independently of scales, since the frequency centers and velocities are bounded. A Schwartz reproducing kernel at unit scale therefore bounds its squared value at any point of a unit square by a rapidly weighted local squared integral. Summing over the distinct unit squares gives

$$\alpha_1^2\alpha_2^2\#\mathcal{Z}_Q \lesssim \|\eta_Q\tilde{g}^{(1)}\tilde{g}^{(2)}\|_2^2.$$

This permits arbitrary designated points, including points on cell closures, with only a fixed multiplicity adjustment.

The support of $f_{i,b}$ lies within $O(\Delta)$ of $(b, -b^2)$ in spacetime frequency. Hence the pair products after multiplication by η_Q have Fourier overlap bounded by a fixed power of δ^{-1} . In detail, the map

$$(b, b') \mapsto (b + b', b^2 + (b')^2)$$

has determinant $2(b' - b)$ on the two ordered separated intervals. If two such mapped pairs are within $O(\Delta)$, their sums and squared differences imply that their original ordered centers are within $O(\delta^{-1}\Delta)$. Since the cap lattice spacing is Δ , the overlap is at most $C\delta^{-C}$, uniformly in the number of caps. Plancherel then gives

$$\|\eta_Q \tilde{g}^{(1)} \tilde{g}^{(2)}\|_2^2 \lesssim \delta^{-C} \int |\eta_Q|^2 \left(\sum_b |f_{1,b}|^2 \right) \left(\sum_{b'} |f_{2,b'}|^2 \right).$$

For completeness, the local spatial Bessel bound is

$$\sum_b |Q_b g(x, t)|^2 \lesssim \int_{\mathbb{R}} (1 + |y|)^{-P_0} |g(x - y/\Delta, t)|^2 dy \lesssim_K W_{k-1} |g|^2(x, t).$$

To obtain the first inequality, write the projection kernels as modulations of a common Schwartz kernel. The values $Q_b g(x, t)$ are Fourier coefficients on a fixed period of the periodization of $y \mapsto \check{\phi}(y) g(x - y/\Delta, t)$. Parseval and weighted Cauchy–Schwarz for the periodization tails give the displayed positive convolution. The second inequality compares its scale Δ^{-1} with $\Delta_{k-1}^{-1} = K\Delta^{-1}$. The bounded squared sum of the temporal-cutoff errors consequently gives $\sum_b |f_{i,b}|^2 \lesssim_K W_{k-1} |g_{k-1}^{(i)}|^2 + 1$, proving (5.25). $\square$

Proof of Theorem 5.1. Apply Lemma 5.2 and first work with one retained layer of each field, normalized by its fine envelope bound. Use the same scales and masks above, with $\lambda_i = C_*/\alpha_i$ and $D_i \lesssim H_i^*/B_i^2$. The constants are chosen in this order: the fixed profile and decay bounds, then C_1 and K as in Lemma 5.3, then a trigger level C_0 , then C_* , and finally the required lower bound for α_i . Choose C_0 sufficiently large relative to the uniform $O_K(\lambda_i^2)$ errors in (5.24), using $\lambda_i \leq 1$.

At a witness, until the first index k at which either $|Y_k^{(i)}| > C_0$, positivity in (5.8) and the witness baseline (5.7) give

$$\sum_{j < k} W_j L_j^{(i)} \lesssim 1 + C_0.$$

By (5.13), the total change in each field through those stages, including the initial deletion, is at most $C_K(1 + C_0)\lambda_i^{-1}$. Choose C_* large enough that this is a small fixed fraction of α_i , since $\lambda_i^{-1} = \alpha_i/C_*$. Both preceding states therefore retain modulus at least $c\alpha_i$ at the witness. A trigger must occur by stage n : otherwise both final states retain those values, whereas the terminal synthesis has only $O_K(1)$ normalized centers and hence supremum $O_K(\lambda_i)$. The last assertion uses the interval size $O(\delta)$ and $\Delta_n \asymp_K c_0\delta$.

Use nested dyadic square grids of side Δ_k^{-1} ; this is possible because K and Δ_0 are dyadic. Each witness selects the square of its first-trigger scale. Take the maximal squares Q among these selected squares. They are disjoint and cover all witnesses; assign a witness to the unique maximal square containing it. If such a square has scale k , an assigned witness cannot have triggered at a smaller index $j < k$: its larger dyadic square would strictly contain Q and contradict maximality. Thus both $g_{k-1}^{(i)}$ are large at every witness assigned to Q . At least one witness in Q triggers at index k . Applying (5.24) to that point yields, almost everywhere on Q ,

$$\mathcal{M}_1(z) + \mathcal{M}_2(z) \gtrsim_K 1, \quad \mathcal{M}_i = \mathcal{M}(\sup_j |Y_j^{(i)}|). \quad (5.26)$$

Apply Lemma 5.7 to the assigned witnesses. We explain how to bound its weighted integral using an arbitrary reference point $z \in Q$. Fix a witness $z_0 \in Q$, common to both fields. The recurrence at stage $k - 1$ gives

$$W_{k-1}|g_{k-1}^{(i)}|^2(z') \leq S_0^{0,(i)}(z') + |Y_{k-1}^{(i)}(z')|.$$

The baseline (5.7), and the fact that $h \gtrsim \Delta_k^{-1}$ up to a fixed constant, bound its first term by a fixed polynomial in $1 + \Delta_k \text{dist}(z', Q)$. For the second term use (5.24) at the reference point z , noting $\Delta_{k-1} \leq \Delta_k$; if $k = 1$, it is simply zero. We obtain

$$W_{k-1}|g_{k-1}^{(i)}|^2(z') + 1 \lesssim_K (1 + \Delta_k \text{dist}(z', Q))^C (1 + \mathcal{M}_i(z)).$$

Choose the fixed Schwartz decay of η_Q large enough to absorb this polynomial. Its integral scale is $|Q|$, and $(1 + \mathcal{M}_1)(1 + \mathcal{M}_2) \lesssim 1 + \mathcal{M}_1^2 + \mathcal{M}_2^2$. Thus the local sampling estimate is at most

$$C_K \delta^{-C} |Q| (1 + \mathcal{M}_1(z)^2 + \mathcal{M}_2(z)^2)$$

for almost every reference point $z \in Q$. Average over Q and absorb the constant using (5.26). This proves

$$\alpha_1^2 \alpha_2^2 \# \{\text{witnesses assigned to } Q\} \lesssim_K \delta^{-C} \int_Q (\mathcal{M}_1^2 + \mathcal{M}_2^2).$$

Summing the disjoint squares, then applying the maximal L^2 inequality and (5.9), gives

$$\# \lesssim_K \delta^{-C} \alpha_1^{-2} \alpha_2^{-2} (\lambda_1^2 D_1 + \lambda_2^2 D_2).$$

Finally $\lambda_i^2 D_i \lesssim H_i^* / A_i^2$ in the original units. The summable threshold splitting and excess amplitude decay in Lemma 5.2 allow summation over all initial layer pairs. This proves (5.1).

The construction used only a fixed finite list of profile derivatives and decay orders. The derivative inductions are contractive, the potential errors and relative bands sum geometrically, and all coverings have fixed overlap. Consequently no constant grows with the number of stages or the array cardinality. The only variable separation loss is the fixed polynomial overlap bound in Lemma 5.7. $\square$

Remark 5.8 (Use for composite tangential packets). The theorem uses only the stated derivative and phase-space bounds. It therefore applies to a smoothly localized sum of smaller packets treated as a single profile with amplitude equal to a bound for its required derivatives. Translating a time bin by t_0 replaces a tangential intercept by $b + 2\xi t_0$. For a generic family its phase-space counts remain bounded under this change when $|t_0| \lesssim h^2$. In the later composite-packet application, larger shifts are also permitted because the carrier centers have bounded count per h^{-1} frequency bin and the intercept lattice is translated separately at each fixed carrier. It is this extra spacing property that preserves the counts for those shifts. On a normal slab of width h , a Fourier expansion in the normalized normal variable produces mode amplitudes $O_N(\langle n \rangle^{-N})$ times that derivative bound; thresholds split with $\langle n \rangle^{-2}$ and sum as in the initial layer reduction. When omitted normal and time-cell coordinates are at bounded relative distance, the tangential envelope is bounded by a fixed multiple of the full cell envelope. Thus this use introduces neither a dependence on the number of component packets nor an additional count of slabs when their coefficient energies have bounded overlap.

6 Flat frequency arrays

Theorem 5.1 yields a linear distribution estimate in one space dimension. The useful output measure has a stronger bound along moving tubes than a general two-dimensional measure. Projecting a three-dimensional measure from a slab of width L produces exactly this bound. These two reductions account for the exponents $L^{-5/6}$ and $L^{-4/3}$ below.

6.1 A one-dimensional distribution estimate

Fix a bounded closed interval $I \subset \mathbb{R}$ of positive length. A one-dimensional packet array at scale $L \geq 1$ has the form

$$V(x, t) = L^{-1/2} \sum_{\ell} c_{\ell} e^{i(x\xi_{\ell} - t\xi_{\ell}^2)} H_{\ell} \left(\frac{x - b_{\ell} - 2\xi_{\ell}t}{L}, \frac{t}{L^2} \right), \quad \xi_{\ell} \in I. \quad (6.1)$$

The array is finite, the pairs $(b_{\ell}/L, L\xi_{\ell})$ have a fixed bounded number in every unit ball of $\mathbb{R}^2$, and the profiles have fixed compact support and uniformly bounded derivatives through a sufficiently large fixed order. The centers b_{ℓ} are arbitrary real numbers; there is no integrality assumption. Constants below may depend on these fixed bounds, on I , and on the fixed size of the output box.

Proposition 6.1 (One-dimensional flat estimate). *Let ν be a nonnegative measure supported in a spatial interval of length $O(L^2)$ and in $|t| \leq L^2$. Suppose that, for every $t_0, b \in \mathbb{R}$, every $v \in I$, and every $1 \leq \rho \leq r \leq L^2$,*

$$\nu\{(x, t) : |t - t_0| \leq r, |x - b - 2vt| \leq \rho\} \leq m_1 r \left(1 + \frac{\rho}{L}\right). \quad (6.2)$$

There is an exponent $p_0 > 2$, for which one may take $p_0 = 21/10$, such that every array (6.1) and every $g > 0$ satisfy

$$\nu\{|V| \geq gL^{-5/6}\} \leq Cm_1 L^3 g^{-p_0} \|c\|_2^{p_0}. \quad (6.3)$$

The constant is uniform in L , in the number of packets, and in their real centers. In particular, no factor growing with L is allowed.

Proof. The zero-coefficient case is immediate. Homogeneity reduces the proof to $\sum_{\ell} |c_{\ell}|^2 = 1$, but we retain the homogeneous form (6.3) in recursive applications to subarrays. We will obtain a broad bound with constant depending on K and a narrow contraction with a constant independent of K . We then choose K , the base scale, and the induction constant in that order.

First, the hypotheses give two elementary bounds:

$$\nu(\mathbb{R}^2) \leq Cm_1 L^3, \quad \|V\|_{\infty} \leq C \quad \text{on } |t| \leq L^2. \quad (6.4)$$

For the mass bound, cover the output box by a fixed number of tests (6.2) with $r = \rho = L^2$ and any fixed $v \in I$. For the field bound, a packet contributing at (x, t) must satisfy $|b_{\ell}/L + 2(t/L^2)L\xi_{\ell} - x/L| \leq C$. Because $\xi_{\ell} \in I$ and $|t|/L^2 \leq 1$, this part of phase space is covered by $O(L)$ unit balls. At most $O(L)$ labels contribute, and Cauchy–Schwarz cancels the normalization $L^{-1/2}$. It follows that a nonempty level set has

$$g \leq CL^{5/6}. \quad (6.5)$$

The mass bound settles $g < 1$, and also settles all bounded scales once the induction constant is chosen sufficiently large. The same tube test, with $r = \rho = 1$ and a fixed bounded covering when $v \neq 0$, shows that every unit lattice square has ν -mass at most Cm_1 .

The broad alternative. Choose an integer $K \geq 4$, eventually fixed sufficiently large, and partition I into K equal half-open intervals I_j . Make a consistent endpoint assignment at the two ends of I . Let V_j be the field of the labels in I_j , and write $A = gL^{-5/6}$. Call a point of $\{|V| \geq A\}$ broad if there are indices j_1, j_2 with $|j_1 - j_2| \geq 3$ and

$$|V_{j_i}(x, t)| \geq \frac{A}{10K}, \quad i = 1, 2.$$

The two frequency intervals have length $\delta = |I|/K$ and separation at least 2δ . We estimate each pair and then sum over at most K^2 pairs; all constants in this paragraph may depend on K .

To use the normalization of Theorem 5.1, put $W_i = LV_{j_i}$. Thus the frame amplitudes are $s_\ell = L^{1/2}c_\ell$, the frame scale is $h = L$, and

$$S_i(x, t)^2 = L \sum_{\xi_\ell \in I_{j_i}} |c_\ell|^2 \left(1 + \frac{|x - b_\ell - 2\xi_\ell t|}{L} + \frac{|t|}{L^2} \right)^{-P}, \quad (6.6)$$

$$H_i^* = L^3 \sum_{\xi_\ell \in I_{j_i}} |s_\ell|^2 \leq L^4, \quad |W_i| \geq A_i := \frac{gL^{1/6}}{10K}. \quad (6.7)$$

Here P is a sufficiently large fixed envelope power as in Theorem 5.1. Multiplying the fields by L is essential: the square envelopes in the following estimates belong to W_i .

We claim

$$\int S_i^2 d\nu \leq C m_1 L^3, \quad \int S_1^2 S_2^2 d\nu \leq C_K m_1 L^3. \quad (6.8)$$

A width- L tube over the given time range has mass $O(m_1 L^2)$ by (6.2). Two such tubes with velocities separated by at least δ intersect only in a time interval of length $O_K(L)$; their intersection therefore has mass $O_K(m_1 L)$. These statements also control the tails in (6.6). More explicitly, a width- aL tube, with $a \geq 1$, has mass at most $C m_1 L^2(1 + a)$, and the intersection of width- aL and width- bL tubes at the separated velocities has mass at most $C_K m_1 L(1 + a + b)^2$. To see the latter, subtract the two tube equations to obtain a time interval of length $O_K((a + b)L)$, then apply (6.2) to the narrower tube. If this time length or width exceeds L^2 , use the original time restriction and, if necessary, the total mass bound in (6.4); the same polynomial bound results. Dyadic enlargement and the decay 2^{-Pj} absorb these polynomial costs. Multiplying the individual tube bound by $L \sum |c_\ell|^2 \leq L$, and the intersection bound by $L^2(\sum_{\xi_\ell \in I_{j_1}} |c_\ell|^2)(\sum_{\xi_\ell \in I_{j_2}} |c_\ell|^2) \leq L^2$, proves (6.8).

Take $q = 9/4$. For $g \geq 1$, discard the points where $S_i > g^{q/2} = g^{9/8}$ for either i . The first bound in (6.8) charges these points at most

$$C m_1 L^3 g^{-q}. \quad (6.9)$$

On what remains, use dyadic classes $b_i \leq S_i < 2b_i$, with $b_i \leq g^{9/8}$. At a witness in such a class the frame upper thresholds can be taken as $B_i = 2b_i$. By (6.5),

$$\frac{A_i}{B_i} \geq c_K L^{1/6} g^{-1/8} \geq c'_K L^{1/16}. \quad (6.10)$$

Thus, for L exceeding a fixed bound depending on K , both the large-ratio hypothesis in Theorem 5.1 and its condition $\delta \geq L^{-1/2}$ hold. All smaller scales belong to the induction base. An envelope cannot vanish at a witness with $|W_i| > 0$, so these positive dyadic classes cover the retained set.

Apply Theorem 5.1 to count the unit squares having a witness in a fixed class. Their masses are bounded by Cm_1 , even when the witness is not a lattice point. Equations (6.7)–(6.10) give the bound

$$C_K m_1 \frac{b_1^2 b_2^2}{A_1^2 A_2^2} \left(\frac{L^4}{A_1^2} + \frac{L^4}{A_2^2} \right) \leq C_K m_1 L^3 (b_1 b_2)^2 g^{-6}.$$

On the other hand, the second bound in (6.8) gives $C_K m_1 L^3 (b_1 b_2)^{-2}$ for the mass of the class. Consequently its mass is at most

$$C_K m_1 L^3 \min\{(b_1 b_2)^{-2}, (b_1 b_2)^2 g^{-6}\}. \quad (6.11)$$

The minimum is bounded by g^{-3} . There are $O(\log^2(2+g))$ classes with both $b_i \geq g^{-10}$. For the other classes use the second term in (6.11) and sum geometrically: for example,

$$g^{-6} \sum_{b_1 < g^{-10}} b_1^2 \sum_{b_2 \leq g^{9/8}} b_2^2 \leq C g^{-6-20+9/4} \leq C g^{-3}.$$

The analogous sum handles $b_2 < g^{-10}$. Including (6.9) and the choice of the pair proves

$$\nu(\text{broad points}) \leq C_K m_1 L^3 (g^{-9/4} + g^{-3} \log^2(2+g)) \leq C_K m_1 L^3 g^{-p_0}, \quad p_0 = 21/10. \quad (6.12)$$

The narrow alternative and the exact child hypotheses. At a point which is not broad, call I_j significant if $|V_j| \geq A/(10K)$. All significant indices are within two steps of each other. Choose three consecutive whole partition intervals containing every significant bin, and let J be the closed hull of those three intervals. Let V_J be the sum of exactly their three bin fields, with the original endpoint assignments. The list of all such clusters lies inside I , and each label belongs to at most three clusters. Near an endpoint use the first or last three intervals. Every bin outside the selected cluster is insignificant; hence

$$|V_J| \geq |V| - \sum_{I_j \not\subset J} |V_j| \geq \frac{9}{10} A. \quad (6.13)$$

In particular, the lower fraction in this narrow alternative is independent of K . Using whole partition intervals avoids dropping only part of any field at a cluster boundary.

Put $u = 3/K$ and choose v_J so that $J = v_J + uI$. The change of variables and packet parameters are

$$x' = u(x - 2v_J t), \quad t' = u^2 t, \quad L' = uL, \quad b'_\ell = ub_\ell, \quad \xi'_\ell = \frac{\xi_\ell - v_J}{u} \in I. \quad (6.14)$$

Up to a common factor of modulus one,

$$V_J(x, t) = u^{1/2} V'_J(x', t'), \quad (6.15)$$

where V'_J has exactly the form (6.1) at scale L' , with unchanged profiles and coefficients. Indeed, $b'_\ell/L' = b_\ell/L$ and $L'\xi'_\ell = L\xi_\ell - Lv_J$, so phase-space ball counts are unchanged. This observation also explains why real centers cause no difficulty in the induction.

Let ν' be the pushforward of ν under (6.14). An arbitrary child test, with parameters $t'_0, b' \in \mathbb{R}$, $v' \in I$, and $1 \leq \rho' \leq r' \leq (L')^2$, pulls back to an old test with

$$t_0 = \frac{t'_0}{u^2}, \quad b = \frac{b'}{u}, \quad v = v_J + uv' \in J \subset I, \quad r = \frac{r'}{u^2}, \quad \rho = \frac{\rho'}{u}.$$

Here $1 \leq \rho \leq r \leq L^2$ and $\rho/L = \rho'/L'$. Thus the full hypothesis (6.2), including its lower scale cutoff and its range of velocities, survives with

$$m'_1 = u^{-2}m_1. \quad (6.16)$$

There is no density Jacobian to insert: ν' is a pushforward measure. The time restriction becomes $|t'| \leq (L')^2$.

The transformed spatial domain can be longer than a single interval of length $(L')^2$. Partition it into such intervals. In each one retain only child packets whose compact support meets that interval during $|t'| \leq (L')^2$. The field agrees with this subarray on the restricted measure. Because the child velocities are in the fixed interval I and $L' \geq 1$, each compact child packet meets only a fixed number of these spatial intervals. If $e_{J,Q}$ is the coefficient norm of the subarray for cluster J and spatial interval Q , bounded spatial multiplicity and the threefold frequency overlap give

$$\sum_{J,Q} e_{J,Q}^2 \leq C \|c\|_2^2 = C, \quad \sum_{J,Q} e_{J,Q}^{p_0} \leq \left(\sum_{J,Q} e_{J,Q}^2 \right)^{p_0/2} \leq C. \quad (6.17)$$

These constants are independent of K , even though the number of spatial intervals can grow with K .

By (6.13) and (6.15), the child level parameter is

$$g' = \frac{9}{10}gu^{1/3}, \quad \frac{(9/10)gL^{-5/6}}{u^{1/2}} = g'(L')^{-5/6}.$$

Suppose the homogeneous estimate is known at the smaller scale L' with induction constant C_{ind} . Sum it over the clusters and spatial intervals, using (6.16) and (6.17). The narrow contribution is at most

$$C_{\text{ind}} \sum_{J,Q} m'_1(L')^3 (g')^{-p_0} e_{J,Q}^{p_0} \leq C_0 C_{\text{ind}} u^{1-p_0/3} m_1 L^3 g^{-p_0}. \quad (6.18)$$

The constant C_0 does not depend on K . Since $1 - p_0/3 = 3/10 > 0$, fix K sufficiently large that $C_0(3/K)^{3/10} \leq 1/2$. Next fix a base scale $L_*(K)$ large enough for (6.10), the frame separation condition, and $uL \geq 1$ whenever $L > L_*(K)$. Choose C_{ind} to cover the bounded-scale estimate and twice the constant in (6.12). Induction over successive scale ranges with ratio u^{-1} now closes, proving (6.3) for all real $L \geq 1$. $\square$

6.2 Projection of a thin frequency strip

Proposition 6.2 (Thin-strip estimate). *Let U be a finite two-dimensional packet array of the form (2.2), with the fixed profile and phase-space bounds stated there. Suppose all frequency labels lie within distance C/L of an affine line in frequency space. Let μ be supported in a spatial square of side $O(L^2)$ and in $|t| \leq L^2$, and suppose $\mu(B_r) \leq mr^2$ for every ball with arbitrary center and every $r \geq 1$. Then, with the exponent p_0 of Proposition 6.1,*

$$\mu\{|U| \geq gL^{-4/3}\} \leq C mL^4 g^{-p_0} \|c\|_2^{p_0}, \quad g > 0. \quad (6.19)$$

The constant depends only on the fixed bounds, including the strip width constant. No sparsity hypothesis is required.

Proof. Rotate the spatial and frequency coordinates so that the frequency line has constant normal coordinate. Subtract this central normal frequency by the corresponding Galilean

shear and modulation. These transformations preserve the field modulus, up to composing with the change of variables, and change the ball bound by a fixed factor: the spatial rotation is orthogonal and the space-time shear and its inverse have bounded norms. They also preserve uniform profile and phase-space bounds. In the new coordinates write the spatial variables as (x, y) and the frequency labels as $(\xi_\alpha, \eta_\alpha)$, where

$$|L\eta_\alpha| \leq C. \quad (6.20)$$

The packet fields, including their coefficients, are now

$$L^{-1}c_\alpha e^{i(x\xi_\alpha + y\eta_\alpha - t(\xi_\alpha^2 + \eta_\alpha^2))} H_\alpha \left(\frac{x - b_{\alpha,x} - 2\xi_\alpha t}{L}, \frac{y - b_{\alpha,y} - 2\eta_\alpha t}{L}, \frac{t}{L^2} \right).$$

We keep m for the ball parameter after its fixed enlargement.

Partition the normal coordinate into half-open slabs of width L , and let y_s denote the center of slab s . By (6.20) and compact profile support, only labels in a set $\mathcal{A}_s$ with

$$|b_{\alpha,y} - y_s| \leq CL \quad (6.21)$$

can contribute in this slab during $|t| \leq L^2$. Every label belongs to only boundedly many such sets. Moreover, the projected pairs $(b_{\alpha,x}/L, L\xi_\alpha)$ have bounded counts in unit balls for $\alpha \in \mathcal{A}_s$. Indeed, over any such ball, the remaining coordinates $(b_{\alpha,y}/L, L\eta_\alpha)$ lie in a bounded box by (6.20) and (6.21). Its center may depend on s , but it is covered by a fixed number of unit balls. The original four-dimensional phase-space bound therefore bounds the number of projected labels uniformly. Projection does not cost a power of L .

Let ν_s be the pushforward to (x, t) of the restriction of the transformed measure to slab s . For any bounded tangential velocity v and $1 \leq \rho \leq r \leq L^2$, the inverse image of a test in (6.2) has dimensions comparable to r , ρ , and L , after a shear of bounded norm. It can be covered by

$$C \frac{r}{\rho} \left(1 + \frac{L}{\rho} \right)$$

balls of radius $C\rho$. These radii are at least one. Applying the ball bound gives

$$\nu_s\{|t - t_0| \leq r, |x - b - 2vt| \leq \rho\} \leq Cmr(L + \rho) = CmLr \left(1 + \frac{\rho}{L} \right). \quad (6.22)$$

Thus Proposition 6.1 applies to ν_s with $m_1 = CmL$ and any fixed tangential frequency interval containing the labels.

It remains to handle the dependence of the field on the normal coordinate. A pointwise choice of a normal slice would not suffice for an arbitrary measure, so we use a single Fourier expansion valid throughout the slab. Put

$$z = \frac{x - b_{\alpha,x} - 2\xi_\alpha t}{L}, \quad \tau = \frac{t}{L^2}, \quad w = \frac{y - y_s}{L}, \quad \zeta_{\alpha,s} = \frac{y_s - b_{\alpha,y}}{L}, \quad a_\alpha = L\eta_\alpha.$$

Choose a fixed smooth cutoff $\chi(w)$ equal to one for $|w| \leq 1/2$ and supported strictly inside one period of fixed length T . Expand

$$\begin{aligned} F_{\alpha,s}(z, \tau, w) &:= \chi(w) e^{ia_\alpha w - ia_\alpha^2 \tau} H_\alpha(z, w + \zeta_{\alpha,s} - 2a_\alpha \tau, \tau) \\ &= \sum_{n \in \mathbb{Z}} e^{2\pi i n w / T} F_{\alpha,s,n}(z, \tau). \end{aligned} \quad (6.23)$$

The constants a_α and $\zeta_{\alpha,s}$ are bounded on $\mathcal{A}_s$. Consequently, if D is the finite derivative order needed in Proposition 6.1, integration by parts in w gives, for any sufficiently large fixed N for which the original profiles have $D + N$ derivatives,

$$\max_{j+k \leq D} \|\partial_z^j \partial_\tau^k F_{\alpha,s,n}\|_\infty \leq C_N (1 + |n|)^{-N}. \quad (6.24)$$

All these profiles have support in a common bounded (z, τ) box. The residual normal phase has bounded derivatives in these scaled variables; this is the reason for first removing the central normal frequency. The remaining constant phase $e^{iy_s \eta_\alpha}$ can be included in the coefficient.

Absorb $C_N(1 + |n|)^{-N}$ from (6.24) into the coefficients and normalize each remaining profile. This produces one-dimensional fields $V_{s,n}$ with uniform admissible profiles and the already verified projected phase-space counts, such that in slab s

$$U(x, y, t) = L^{-1/2} \sum_{n \in \mathbb{Z}} e^{2\pi i n(y - y_s)/(TL)} V_{s,n}(x, t), \quad \|c_{s,n}\|_2 \leq C_N(1 + |n|)^{-N} e_s, \quad (6.25)$$

where $e_s^2 = \sum_{\alpha \in \mathcal{A}_s} |c_\alpha|^2$. The series converges absolutely, and all its external phase factors have modulus one at every witness. The factor $L^{-1/2}$ outside the series is the difference between the two-dimensional packet normalization L^{-1} and the one-dimensional normalization $L^{-1/2}$.

Choose positive numbers $a_n = c_*(1 + |n|)^{-2}$ with $\sum_n a_n = 1/2$. If $|U(x, y, t)| \geq gL^{-4/3}$ in slab s , then (6.25) forces, for at least one n ,

$$|V_{s,n}(x, t)| \geq a_n g L^{-5/6}.$$

Taking the projected measure of this union and applying Proposition 6.1 gives

$$\begin{aligned} \mu(\{|U| \geq gL^{-4/3}\} \cap \text{slab } s) &\leq \sum_{n \in \mathbb{Z}} \nu_s\{|V_{s,n}| \geq a_n g L^{-5/6}\} \\ &\leq C m L^4 g^{-p_0} e_s^{p_0} \sum_{n \in \mathbb{Z}} a_n^{-p_0} (1 + |n|)^{-N p_0} \\ &\leq C m L^4 g^{-p_0} e_s^{p_0}, \end{aligned} \quad (6.26)$$

provided $(N - 2)p_0 > 1$. Only finitely many derivative bounds are needed to choose such an N . Finally,

$$\sum_s e_s^2 \leq C \|c\|_2^2, \quad \sum_s e_s^{p_0} \leq \left(\sum_s e_s^2 \right)^{p_0/2} \leq C \|c\|_2^{p_0},$$

by bounded slab overlap and $p_0 > 2$. Summing (6.26) proves (6.19). $\square$

Remark 6.3 (Fixed enlargements of the time domain). The preceding conclusions also hold on $|t| \leq C_t L^2$ for any fixed C_t , with constants depending on C_t . Partition this range into boundedly many intervals of radius at most L^2 , and translate the center of each interval to zero. A time translation by t_* replaces the packet intercept b by $b + 2\omega t_*$ and shifts the profile time argument by t_*/L^2 ; fixed support and derivative bounds remain uniform. On phase space this is the bounded shear $(b/L, L\omega) \mapsto (b/L + 2(t_*/L^2)L\omega, L\omega)$, so real-center ball counts remain bounded. The one-dimensional tube hypothesis is invariant under this translation because its time centers and intercepts are arbitrary; the two-dimensional ball hypothesis is translation invariant as well. Restriction to each time interval and a bounded sum therefore give the asserted extension.

7 Rich tubes lie in a short list of plates

We next isolate a geometric consequence of two simultaneous properties: every tube sees substantial mass, and the tubes available at each point have a spread of directions. Both properties must concern the same measure. The conclusion is a short list of planes near which

all the tube trajectories lie. The first part of the proof finds a rich plane through each target line; the second part packs these planes, keeping track of their offsets as well as their normals.

Divide all three physical coordinates by L , put $H = L$, and divide the original measure by mL^2 . In these coordinates the ball hypothesis becomes $\nu(B(q, r)) \lesssim r^2$ for $r \geq 1$, the relevant domain has diameter $O(H)$, and the tubes have unit width. This section uses these coordinates throughout.

Proposition 7.1 (Rich-tube geometry). *Fix bounded velocity, domain, ball, and neighborhood constants, a richness constant $c_r > 0$, an angular constant $C_a > 0$, exponents $A_r, A_a \geq 0$, and $\eta > 0$. There are constants A_*, C_*, C' depending only on these fixed data such that the following holds whenever $X \geq 2$ and $H \geq C_* X^{A_*}$.*

Let ν be a nonnegative measure supported in a ball of radius $O(H)$, with

$$\nu(B(q, r)) \leq C_b r^2 \quad (r \geq 1).$$

Let $\mathcal{T}$ be a finite family of lines

$$\ell_T = \{(b_T + 2\omega_T t, t) : t \in \mathbb{R}\}, \quad |\omega_T| \leq C_v,$$

viewed as unit-width tubes over a fixed time interval of length $O(H)$ containing the time projection of $\text{supp } \nu$. Write $d_T(x, t) = |x - b_T - 2\omega_T t|$ and $e_T = (2\omega_T, 1)/|(2\omega_T, 1)|$. Assume:

(i) *Every tube is rich against ν :*

$$\int (1 + d_T(q))^{-M} d\nu(q) \geq c_r H X^{-A_r}, \quad M = 100. \quad (7.1)$$

(ii) *At ν -almost every q there is a measurably chosen probability $(p_T(q))_{T \in \mathcal{T}}$, supported on tubes with $d_T(q) \leq C_n$, such that every spherical direction ball satisfies*

$$\sum_{e_T \in B_{\mathbb{S}^2}(e, r)} p_T(q) \leq C_a X^{A_a} r^\eta \quad (r \geq H^{-1}). \quad (7.2)$$

Then each tube can be assigned to one of at most $X^{C'}$ plates (planes thickened by $X^{C'}$). Its assigned plate contains its central line segment over the relevant time interval. If n is a unit normal to that plate, then

$$|n \cdot (2\omega_T, 1)| \leq X^{C'} / H. \quad (7.3)$$

The constants are independent of the number of tubes. In particular, the richness in (7.1) and the pointwise law in (7.2) use exactly the same measure ν .

Fixed factors may be absorbed into fixed powers of X , since $X \geq 2$. We write $X^{O(1)}$ only for powers depending on the data in the proposition; the order in which all additional exponents are chosen is specified at the end of the proof. No estimate below uses a ball bound at radii smaller than one. Bounded velocities also give $\text{dist}(q, \ell_T) \leq d_T(q) \leq C \text{dist}(q, \ell_T)$, so transverse deviation and orthogonal distance define comparable neighborhoods. For a finite family, measurability of the neighboring law adds no restriction: the set $\{T : d_T(q) \leq C_n\}$ takes only finitely many values, and one may choose an admissible probability once for each occurring incidence pattern. These patterns are Borel, since each d_T is continuous, so the resulting law is measurable.

Lemma 7.2 (A truncated parameter energy). *Let $H \geq 1$ and let σ be a finite measure on $\mathbb{R}^2$ of mass m , satisfying*

$$\sigma(B(z, r)) \leq A r^{1+\eta} \quad (z \in \mathbb{R}^2, r \geq H^{-1}), \quad A > 0.$$

Set $\gamma = \eta/(1 + \eta)$. Then

$$\iint \min\{H, |z - z'|^{-1}\} d\sigma(z') d\sigma(z) \leq C_\eta A^{1/(1+\eta)} m^{1+\gamma}, \quad (7.4)$$

where the kernel on the diagonal is H .

Proof. The assertion is immediate when $m = 0$. Otherwise put $R = (m/A)^{1/(1+\eta)}$. If $R < H^{-1}$, the inner integral is at most

$$Hm \leq m/R = A^{1/(1+\eta)} m^\gamma.$$

If $R \geq H^{-1}$, the contribution of $|z - z'| \leq H^{-1}$ is at most $AH^{-\eta}$. The dyadic annuli between H^{-1} and R contribute at most

$$C \sum_{H^{-1} \leq 2^j H^{-1} \lesssim R} A(2^j H^{-1})^\eta \leq C_\eta AR^\eta.$$

The remaining region contributes at most m/R . Thus the inner integral is bounded by $C_\eta A^{1/(1+\eta)} m^\gamma$ in both cases. Integrating in z proves (7.4). The cutoff at H is what permits the first case without a hypothesis below H^{-1} . $\square$

Proof of Proposition 7.1. 1. Localize richness and choose a longitudinal point. For $1 \leq w \leq H$, a width- w neighborhood of any relevant trajectory can be covered by $O(H/w)$ balls of radius $O(w)$. Consequently

$$\nu\{d_T \leq w\} \lesssim Hw. \quad (7.5)$$

For $w \geq H$ the same upper bound follows from $\nu(\mathbb{R}^3) \lesssim H^2$. Therefore the part of the smoothed integral outside width $W \geq 1$ is at most

$$C \sum_{j \geq 0} (2^j W)^{-M} H(2^{j+1} W) \lesssim HW^{1-M}.$$

Choose $W = X^{c_1}$, with $(M-1)c_1 > A_r$ by a sufficiently large fixed margin. It follows from (7.1) that, for every T ,

$$\nu\{d_T \leq W\} \geq c_H X^{-A_r}. \quad (7.6)$$

We always take H large enough that $W \ll H$.

Fix a target tube T_0 , and abbreviate its line and unit direction by ℓ and e . Translate the origin to a point of ℓ at distance $O(W)$ from the original domain. Such a point exists by (7.6). The measure domain, and all relevant trajectory segments, remain inside a ball of radius $O(H)$: the latter assertion follows for every tube from its own localized richness, bounded velocity, and the common $O(H)$ time range.

Draw q_0 according to normalized ν restricted to $\{d_{T_0} \leq W\}$, and let $s = H^{-1}e \cdot q_0$. Thus Ws is the longitudinal coordinate of the orthogonal projection of q_0 onto ℓ . A longitudinal window of relative length $r \geq H^{-1}$ inside this width- $O(W)$ tube is covered by $O(1 + Hr/W)$ balls of radius $O(W)$. Before normalization its mass is at most $C(W^2 + HrW)$; hence

$$\mathbb{P}\{s \in I\} \lesssim X^{A_r}(W^2/H + rW) \leq X^{O(1)}r, \quad |I| \leq r, \quad r \geq H^{-1}. \quad (7.7)$$

In the last inequality we used precisely $H^{-1} \leq r$.

2. *Sample a transverse neighbor and keep bin measures unnormalized.* Choose $\delta = X^{-c_0}$, with c_0 large relative to A_a/η . At every sampled point discard directions with $|e \times e_T| < \delta$. They lie in the union of two spherical balls about e and $-e$ of radius $O(\delta)$, so (7.2) makes their total probability less than $1/2$. Here $H^{-1} \leq \delta$ is included in the scale hierarchy. Renormalize

the remaining probabilities separately at each q_0 . This costs at most two in the angular bound and does not change the marginal law of q_0 or (7.7). Draw a tube T from this remaining law. It is a physical neighbor of q_0 , and is rich against the original common measure ν .

Partition azimuth about the target axis e into half-open intervals of length comparable to H^{-1} . For a bin π , let u_π be its central radial unit vector in $e^\perp$, and let P_π be the exact plane through ℓ spanned by e, u_π . For a sampled unit direction e' in this bin define

$$a = \frac{e \cdot e'}{u_\pi \cdot e'}.$$

The denominator is at least $c\delta$, so $|a| \lesssim \delta^{-1}$. Let σ_π be the pushforward to (s, a) of the sampling probability *restricted* to this azimuth bin, and write $m_\pi = \|\sigma_\pi\|$. We have

$$\sum_{\pi} m_\pi = 1. \quad (7.8)$$

In particular, we have not divided σ_π by m_π .

For completeness, the angular estimate remains effective in these coordinates without conditioning on the bin. If n_π completes e, u_π to an orthonormal basis and θ is azimuth relative to u_π , then

$$e' = \frac{ae + u_\pi + \tan \theta n_\pi}{(1 + a^2 + \tan^2 \theta)^{1/2}}, \quad |\theta| \lesssim H^{-1}.$$

An interval of length $O(r)$ in a , within this bin, is therefore contained in a direction ball of radius $O(r)$ when $r \geq H^{-1}$; allowing $X^{O(1)}r$ would also suffice. For a ball in the (s, a) -plane of radius r , integrate the conditional angular bound over the corresponding longitudinal event in (7.7). This gives

$$\sigma_\pi(B(z, r)) \leq X^{A_F} r^{1+\eta} \quad (r \geq H^{-1}) \quad (7.9)$$

for any $A_F > A_r + 2c_1 + A_a$ with a sufficiently large fixed margin. This is a bound for the unnormalized measure. No assertion about angular nonconcentration conditional on landing in a small bin is needed.

3. *Strips in the bin plane.* Write $x = e \cdot q$ and $y = u_\pi \cdot q$. Choose $w = X^{c_2}$ with $c_2 > c_0 + c_1$ by a sufficiently large fixed margin, and define the plate and its strip multiplicity by

$$Q_\pi = \{q \in \text{supp } \nu : \text{dist}(q, P_\pi) \leq w\},$$

$$F_\pi(q) = \int \mathbf{1}_{\{|x - Hs - ay| \leq w\}} d\sigma_\pi(s, a). \quad (7.10)$$

For each sampled pair (q_0, T) in bin π , the whole width- W rich portion of T lies in Q_π and in its associated strip. Indeed, write $q = q_0 + \tau e_T + \varepsilon$ with $|\varepsilon| \lesssim W$ and $|\tau| \lesssim H$. Azimuth binning gives $|n_\pi \cdot e_T| \lesssim H^{-1}$, while $e \cdot e_T - au_\pi \cdot e_T = 0$. Since the displacement of q_0 from its projection Hse is $O(W)$, the plate error is $O(W + 1)$ and the strip error is $O((1 + |a|)W)$, both bounded by w .

Consider two strips with parameters $z = (s, a)$ and $z' = (s', a')$. At an intersection point,

$$|H(s - s') + (a - a')y| \leq 2w, \quad |y| \lesssim H. \quad (7.11)$$

In particular,

$$|s - s'| \leq C|a - a'| + 2w/H.$$

Put $d = |z - z'|$. If $d \leq Cw/H$, covering the full strip by unit balls costs $X^{O(1)}H$, which is at most $X^{O(1)} \min\{H, d^{-1}\}$ after increasing the fixed power. If $d > Cw/H$, with C chosen from the domain bound, the existence of an intersection in (7.11) forces $|a - a'| \gtrsim d$. The allowed y -interval then has length $O(w/d)$. Its cross-section, the plate thickness, and the factor $1 + |a|$

cost only fixed powers of X in a unit-ball covering. The extra constant in the count of unit intervals is also absorbed, because $d \leq X^{O(1)}$. If the offset difference dominates so strongly that (7.11) cannot hold, the intersection is empty. Thus in all cases the intersection has a covering by

$$X^{A_I} \min\{H, |z - z'|^{-1}\} \quad \text{unit balls} \quad (7.12)$$

For example, the covering argument gives $Cw^3(1 + \delta^{-1}) \min\{H, d^{-1}\}$, so any $A_I > 3c_2 + c_0$ with a sufficiently large fixed margin suffices in (7.12).

Use the ball bound at radius one, Fubini's theorem, and Lemma 7.2 with (7.9). For $\gamma = \eta/(1 + \eta)$ this yields

$$\int_{Q_\pi} F_\pi^2 d\nu \leq X^{A_E} m_\pi^{1+\gamma}. \quad (7.13)$$

One can take any sufficiently enlarged exponent $A_E \geq A_I + A_F/(1 + \eta)$. There is no remaining factor of H in this estimate.

4. *Richness away from the target and overlap of the far plates.* Put $R_f = H/X^{c_3}$, choosing c_3 after c_0, c_1, c_2 , and let

$$Q_\pi^{\text{far}} = Q_\pi \cap \{q : \text{dist}(q, \ell) \geq R_f\}.$$

The geometric meaning of the strip and this exclusion is shown in Figure 2. Because a sampled tube makes sine-angle at least δ with ℓ , the length of its width- W portion within distance R_f of ℓ is at most $C\delta^{-1}(W + R_f)$. Covering it by radius- $O(W)$ balls gives the mass bound

$$C\delta^{-1}W(W + R_f) \leq CX^{c_0+2c_1} + CHX^{c_0+c_1-c_3}. \quad (7.14)$$

Choose $c_3 > A_r + c_0 + c_1$ by a large fixed margin, and then require $H \gg X^{A_r+c_0+2c_1}$. The bound in (7.14) is less than half the lower bound in (7.6), uniformly for every sampled tube. Integrating their retained richness against the sampling law therefore gives

$$\int_{Q_\pi^{\text{far}}} F_\pi d\nu \geq cHX^{-A_r} m_\pi. \quad (7.15)$$

This step uses the richness of every neighboring tube against ν , including mass far from the point at which it was sampled.

Cauchy–Schwarz, (7.13), and (7.15) imply

$$\nu(Q_\pi^{\text{far}}) \geq H^2 X^{-K_0} m_\pi^{1-\gamma} \quad (7.16)$$

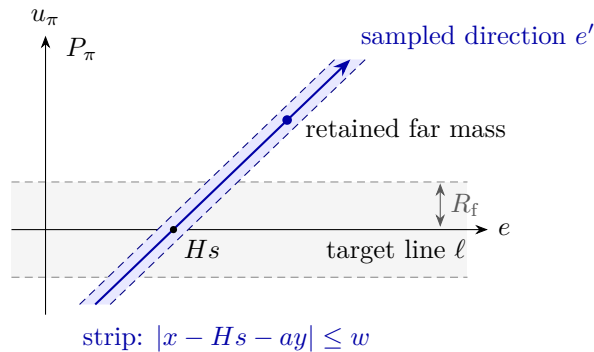


Figure 2: Geometry in the bin plane P_π (schematic projection). The shaded horizontal band is the near region removed in (7.14); the thin sloping strip contains the sampled tube after enlargement. Since its direction is separated from the target, only a controlled length lies in the near band. The true plate also has thickness w perpendicular to this picture.

for some $K_0 \geq 2A_r + A_E$, enlarged to absorb fixed constants.

All planes P_π pass through the same line ℓ . At a point q at distance $\rho \geq R_f$ from that line, the condition $\text{dist}(q, P_\pi) \leq w$ restricts u_π to two azimuth arcs of length $O(w/\rho)$. The two arcs account for opposite radial directions. Since bins have size comparable to H^{-1} , their overlap multiplicity is at most

$$C(1 + Hw/R_f) \leq X^{A_0}, \quad A_0 > c_2 + c_3$$

after a fixed enlargement. Take $H \gg X^{c_2+c_3}$ so that $w \ll R_f$. Summing (7.16), using this overlap and $\nu(\mathbb{R}^3) \lesssim H^2$, gives

$$\sum_{\pi} m_{\pi}^{1-\gamma} \leq X^{K_1} \quad (7.17)$$

for a fixed $K_1 > K_0 + A_0$. But

$$1 = \sum_{\pi} m_{\pi} \leq (\max_{\pi} m_{\pi})^{\gamma} \sum_{\pi} m_{\pi}^{1-\gamma}.$$

Hence some bin has $m_{\pi} \geq X^{-K_1/\gamma}$. Its plane contains the exact target line, and its width- w plate has mass at least

$$H^2 X^{-K}, \quad K \geq K_0 + (1 - \gamma)K_1/\gamma. \quad (7.18)$$

We have produced such a rich plane for every target tube.

5. *Pack rich planes using both normal and offset separation.* Return to a common origin at the center of the original domain. Write each plane just obtained as

$$P_j = \{q : n_j \cdot q = b_j\}, \quad |n_j| = 1,$$

with (n_j, b_j) identified with $(-n_j, -b_j)$. The width- w plates, restricted to $\text{supp } \nu$, all have mass at least λH^2 , where $\lambda = X^{-K}$. Let C_t be a fixed bound for $\nu(\mathbb{R}^3)/H^2$.

Choose an angular threshold $\vartheta = X^{c_4}/H$, with $c_4 > 2c_2 + 2K$ by a sufficiently large fixed margin. If the projective angle between two normals is at least ϑ , the intersection of their width- w plates inside the domain is contained in a box with dimensions

$$O(H) \times O(w) \times O(w/|\sin \angle(n_i, n_j)|).$$

Here projective angles lie in $[0, \pi/2]$, and $\vartheta \ll 1$. Unit-ball covering thus gives

$$\nu(Q_i \cap Q_j) \leq CH^2 X^{2c_2-c_4} \leq \varepsilon H^2, \quad \varepsilon \leq \lambda^2/(2C_t). \quad (7.19)$$

For a pair of normals at projective angle below ϑ , align their signs. Changing the normal then changes its defining linear function by at most $CH\vartheta$ on the domain. Choose an offset threshold

$$\Delta = X^{c_5} > 2w + CH\vartheta, \quad c_5 > \max\{c_2, c_4\} \quad (7.20)$$

with sufficiently large fixed margins. Such a close-normal pair with $|b_i - b_j| \geq \Delta$ has disjoint plates on the domain.

Select a maximal sublist in which every pair is separated either by projective normal angle at least ϑ , or, after aligning closer normals, by offset at least Δ . Finiteness makes this selection immediate. If the list has length N , set $G = \sum_{j=1}^N \mathbf{1}_{Q_j}$. Its first and second moments obey

$$\int G d\nu \geq N\lambda H^2, \quad \int G^2 d\nu \leq C_t N H^2 + \varepsilon N^2 H^2.$$

Cauchy–Schwarz and the total mass bound give

$$N^2 \lambda^2 \leq C_t^2 N + C_t \varepsilon N^2.$$

By the choice in (7.19), the last term is absorbed, and

$$N \leq 2C_t^2 \lambda^{-2} \leq X^{O(1)}. \quad (7.21)$$

Maximality says that each original plane is close to a selected one in both its projective normal and its aligned offset. On the entire bounded range of relevant trajectory segments, this means that its central line lies within $\Delta + CH\vartheta = X^{O(1)}$ of the selected plane. Enlarge the selected plates accordingly. If n_0 is the original normal of the plane through the target line and n is the aligned selected normal, then

$$n_0 \cdot (2\omega_T, 1) = 0, \quad |n - n_0| \lesssim \vartheta.$$

Bounded velocity now gives $|n \cdot (2\omega_T, 1)| \lesssim X^{c_4}/H$, proving (7.3). Offset control is essential for containment of the segment; angular control supplies the asserted directional bound.

Constant hierarchy. Here is one complete order of choices. First fix the data in the proposition and $M = 100$. Choose c_0, c_1 so that

$$c_0\eta > A_a, \quad (M-1)c_1 > A_r,$$

with fixed margins that give the two required half-mass losses. Next choose $c_2 > c_0 + c_1$, followed by

$$A_F > A_r + 2c_1 + A_a, \quad A_I > 3c_2 + c_0, \quad A_E > A_I + A_F/(1+\eta),$$

with sufficient fixed margins for (7.9)–(7.13). Choose

$$c_3 > \max\{c_2, A_r + c_0 + c_1\},$$

and then choose A_O, K_0, K_1, K as in (7.16)–(7.18). Choose $c_4 > 2c_2 + 2K$ to ensure (7.19), and then $c_5 > \max\{c_2, c_4\}$ to ensure (7.20). Choose C' larger than $c_5, c_4, 2K$, with a fixed margin absorbing the final widths, counts, and direction constants. Finally take

$$A_* > \max\{A_r + c_0 + 2c_1, c_2 + c_3, c_0, c_4, C'\},$$

and C_* sufficiently large. Every strict inequality here includes an additional fixed margin if necessary for the fixed constants. All choices precede X, H ; in particular no exponent depends on the number of tubes or on an azimuth-bin mass. $\square$

8 The dyadic distribution estimate

The sparse estimate supplies a gain when every packet meets little measure. For a general array, we first remove sparse packets and narrow angular contributions. The remaining packets lie in a controlled collection of plates. The main analytic step then replaces all intermediate packet scales by a single family at a shorter scale, to which the bilinear frame estimate applies.

Theorem 8.1 (Dyadic distribution bound). *There are fixed $p > 2$ and $\sigma > 0$ with the following property. Let U be a finite sum of the atoms in (2.2), of packet size $L \geq 1$, with $\sum_\alpha |c_\alpha|^2 \leq 1$. Suppose that μ is supported in a spatial box of side $O(L^2)$ and in $|t| \leq L^2$, that $\mu(B_r) \leq mr^2$ for every ball of radius $r \geq 1$, and that (2.4) holds for every label, with $1 \leq D \leq L$. Then, for every $g > 0$,*

$$\mu(A) \leq C mL^4 D^{-\sigma} g^{-p}, \quad A = \{z : |U(z)| \geq \lambda\}, \quad \lambda = gL^{-4/3}. \quad (8.1)$$

The constant is uniform in L, D, g, m and the size of the array. It may depend on the fixed packet class, including sufficiently many finite profile derivative bounds. For an arbitrary coefficient array, the right side is multiplied by $\|c\|_{\ell_2}^p$.

We fix the packet class throughout the proof. In particular, its frequency square, phase-space counting constant, profile support, and required derivative bounds are fixed. Recursive cap scalings will be performed in the original, unrotated coordinates, using squares and retaining only the specified subarray. This preserves the class, including its derivative bounds. Rotations used later are only in nonrecursive estimates. The regularity order can be chosen after all small exponents: the saving c_s in Proposition 4.1, the geometric powers in Proposition 7.1, and the separation power in Theorem 5.1 do not increase when additional profile regularity is required.

8.1 Parameters and recursive estimates

Write c_s for the saving in (4.1), and $p_0 > 2$ for the exponent in (6.3) and (6.19). Choose $2 < p_{\max} < \min(p_0, 3)$ and then $\eta > 0$ so small that

$$\gamma_0 = 1 - p_{\max}/3 - p_{\max}\eta > 0. \quad (8.2)$$

We introduce the parameters used in the construction and work conditionally on the inequalities below. Their compatible order of choice will be checked after the estimates, when the losses determining them have been identified. For fixed constants B, C_a and a small exponent $u > 0$, set

$$X = BD, \quad D_* = BD^2, \quad \delta_0 \asymp X^{-u}, \quad a \asymp X^{C_a}, \quad h = L/a. \quad (8.3)$$

Round δ_0 and h to convenient dyadic scales, changing only fixed constants. The parameter D_* will be used in the sparse-packet deletion; δ_0 is the upper scale for cap recursion. The shorter packet scale h will be used after the geometric reduction.

We split the proof at $D = L^\kappa$, with $\kappa > 0$ sufficiently small. In the range $D < L^\kappa$, we require h to dominate the fixed powers of X occurring below, after enlarging a fixed base scale L_0 . We also require $2 < p \leq p_{\max}$ and $0 < \sigma \leq 1$ with

$$\sigma + \frac{4(p-2)}{3\kappa} < c_s, \quad C_f u + C_g(p-2) + \sigma < c_s, \quad \sigma < u\gamma_0. \quad (8.4)$$

Here C_g is the fixed exponent separating the large levels $g \geq X^{C_g}$ from the remaining levels, and C_f is a fixed loss exponent, independent of u . The frame application below will determine a sufficient value of C_f for (8.35). All three inequalities have strict margins. We take B sufficiently large for the recursive terms to contract, and choose the overall induction constant only after the nonrecursive estimates have been obtained.

Two elementary estimates start the induction. At a point, compact support allows only $O(L^2)$ labels, each atom is $O(L^{-1})$, and hence

$$\|U\|_\infty \lesssim \|c\|_{\ell^2}. \quad (8.5)$$

Also $\mu(\text{supp } \mu) \lesssim mL^4$. Thus all bounded scales are covered by a constant depending on L_0 . When $D \geq L^\kappa$, unit-cube mass is $O(m)$, so Proposition 4.1 gives

$$\int |U|^2 d\mu \leq Cm^{1/3} \sum_q \mu(q)^{2/3} \sup_q |U|^2 \lesssim_\kappa mL^{4/3} D^{-c_s}.$$

Chebyshev therefore gives $CmL^4 D^{-c_s} g^{-2}$. A nonempty level set has $g \lesssim L^{4/3}$ by (8.5); the first inequality in (8.4) implies $D^\sigma g^{p-2} \lesssim D^{c_s}$. This proves (8.1) in the sparse range independently of induction.

For $D < L^\kappa$, use outer induction on dyadic ranges of L and inner induction on the number of iterations of $d \mapsto Bd^2$ needed to reach L^κ . Thus the result is available at a fixed smaller

packet scale, and at the present scale with parameter D_* . For sufficiently large L in this range, $D_* \leq L$; it suffices to have $BL^{2\kappa} \leq L$. Denote the inductive constant by C_{ind} . We will bound recursive contributions by a fixed fraction of $C_{\text{ind}}mL^4D^{-\sigma}g^{-p}$ and all other contributions by a constant independent of C_{ind} .

Lemma 8.2 (Cap gain). *For any subarray in a frequency square of side comparable to r , where $L^{-1} \lesssim r \leq \delta_0$, outer induction gives*

$$\mu\{|U_{\mathcal{C}}| \geq r^\eta \lambda\} \leq C_{\text{cap}}C_{\text{ind}}mL^4g^{-p}r^{1-p/3-p\eta}\left(\sum_{\alpha \in \mathcal{C}}|c_\alpha|^2\right)^{p/2}. \quad (8.6)$$

The estimate holds for any submeasure of μ , with the same upper ball parameter. At the smallest scales choose comparable squares so that the child packet size is at least 1.

Proof. For a square centered at v , use $(x', t') = (r(x - 2vt), r^2t)$, up to fixed normalization conventions. The child packet size is $L' = rL$, the field factor is r , and the pushed measure has ball parameter $m' \leq Cmr^{-3}$. The new threshold parameter is comparable to $gr^{1/3+\eta}$, since

$$r^{-1}r^\eta gL^{-4/3} = gr^{1/3+\eta}(rL)^{-4/3}.$$

After a fixed inflation of m' , covering tubes and their transverse annuli supplies (2.4) with $D = 1$; equivalently $m'(L')^3 \asymp mL^3$. The time interval is still $|t'| \leq (L')^2$. Partition the transformed spatial range into boxes of side comparable to $(L')^2$. A compact child packet meets only boundedly many such boxes during this time interval. If their relevant coefficient energies are e_j , then $\sum_j e_j \lesssim \sum_{\mathcal{C}} |c_\alpha|^2$ and $\sum_j e_j^{p/2} \leq (\sum_j e_j)^{p/2}$. Consequently outer induction costs no number of boxes. Its dimensional factor is

$$(mr^{-3})(rL)^4(gr^{1/3+\eta})^{-p} = mL^4g^{-p}r^{1-p/3-p\eta},$$

as asserted. These parabolic transformations translate the pairs $(b_\alpha/L, L\omega_\alpha)$ and leave the normalized profiles unchanged. Enclosing a specified cap in a comparable square does not add any other labels. $\square$

We need the same estimates when the output point determines a prefix of the array. The next elementary device avoids a loss depending on the number of labels. It adapts the dyadic input-energy decomposition of Christ and Kiselev [8, Theorem 1.1 and its proof in Section 2] to the weak-type block estimates on restricted submeasures used here. We give the proof for these hypotheses, rather than applying their maximal theorem as a black box.

Lemma 8.3 (Energy truncation). *Let a finite coefficient array be ordered, with total energy E , and let each output point select a prefix ending between whole coefficients. Suppose the distribution estimate for every block, on the submeasure where that block is selected, has right side $Kt^{-p}\|c_{\text{block}}\|_{\ell^2}^p$, where $p > 2$. The field formed from the selected prefix then satisfies the same estimate with right side $C_p Kt^{-p}E^{p/2}$.*

Proof. Zero coefficients may be omitted, and $E = 0$ is trivial. Place successive intervals of lengths $|c_\alpha|^2/E$ in $[0, 1]$. For a dyadic interval $I \subset [0, 1]$, apportion c_α by the fraction of its interval lying in I . If that fraction is $f_{\alpha, I}$, its apportioned energy obeys

$$\sum_{\alpha} |f_{\alpha, I} c_\alpha|^2 \leq \sum_{\alpha} f_{\alpha, I} |c_\alpha|^2 = E|I|.$$

These apportioned blocks satisfy the assumed distribution estimate, up to a constant depending on p . Each is either a single apportioned coefficient or the sum of a whole interior block

and at most two fractional endpoint coefficients. Homogeneity gives the estimate for the fractional singletons. Split the threshold equally among these at most three pieces and use $\sum_i \|c_i\|_2^p \leq (\sum_i \|c_i\|_2^2)^{p/2}$; the pieces have disjoint coefficient supports. The resulting bound uses exactly the apportioned energy above.

Every prefix interval $[0, s)$ is a disjoint union of at most one dyadic interval at each depth $k \geq 0$. Every coefficient contributing to a used interval belongs to the prefix, because s is between whole coefficient intervals. The sum of these block fields equals the prefix field, also when the binary expansion is infinite; the original array is finite and the coefficient fractions sum exactly.

Choose $0 < \epsilon < (p/2 - 1)/p$ and positive thresholds $t_k = c_\epsilon t 2^{-\epsilon k}$ with $\sum_k t_k = t$. If a prefix field exceeds t , some used block exceeds t_k . There are at most 2^k blocks at depth k , each with energy at most $E 2^{-k}$. Restricting the measure separately to points using a block, and summing the distribution bounds, gives

$$3^p K t^{-p} E^{p/2} c_\epsilon^{-p} \sum_{k \geq 0} 2^{k(1-p/2+p\epsilon)} \leq C_p K t^{-p} E^{p/2}.$$

This argument also applies inside each fixed frequency cap, with its own total energy E . $\square$

8.2 Peeling to a common rich measure

Starting with A , repeatedly perform the following two deletions. First remove all labels whose smoothed mass against the measure on the current set is less than mL^3/D_* . Next remove every point at which the remaining physically neighboring labels do not support a probability law with direction-ball bounds

$$\mathbb{P}\{\omega \in B_r(v)\} \leq C_{\text{ang}} \delta_0^{-\eta} r^\eta, \quad L^{-1} \leq r \lesssim 1. \quad (8.7)$$

Here a physical neighbor means that the point is within a fixed multiple of L of the packet trajectory; the multiple contains the compact profile support. Directions and frequency centers are interchangeable up to fixed constants on the bounded frequency square.

The angular test can use a finite dyadic label tree, ending at scale comparable to L^{-1} , so it depends measurably on one of finitely many incidence patterns. If a round deletes points but no subsequent label, the angular test is unchanged and no further point can be deleted. Since there are finitely many labels, the process terminates. Write A_∞ for its stabilized set and $\mathcal{R}$ for its surviving labels. At termination all labels in $\mathcal{R}$ satisfy

$$\int_{A_\infty} (1 + |x - b_\alpha - 2\omega_\alpha t|/L)^{-M} d\mu \geq mL^3/D_*; \quad (8.8)$$

every point of this same set satisfies (8.7) using physically neighboring labels in $\mathcal{R}$.

To estimate what was deleted, order removed labels by their removal round, breaking ties arbitrarily. Each original point selects the prefix removed before it exits, including all labels removed in its exit round; a point of A_∞ selects all removed labels. Let $P(z)$ be this prefix field. For any binary block in Lemma 8.3, restrict the measure to points whose prefix uses the block. Such points were still present when each contributing label was removed. By monotonicity of the smoothed integral, every label in that block satisfies (2.4) with D_* . Inner induction and Lemma 8.3 therefore give, for any fixed small $c > 0$,

$$\mu\{|P| \geq c\lambda\} \leq C_p C_{\text{ind}} mL^4 D_*^{-\sigma} g^{-p}. \quad (8.9)$$

This restriction of measure is essential: a removed label need not be sparse on all of the original set.

For both the original field and this prefix field, also remove the exceptional sets where a dyadic cap contribution at any scale $L^{-1} \lesssim r \leq \delta_0$ exceeds $r^\eta \lambda$. Lemma 8.2 and, for prefixes, Lemma 8.3, imply a total cost

$$C' C_{\text{ind}} m L^4 g^{-p} \delta_0^{\gamma_0}. \quad (8.10)$$

Indeed cap energies at each scale partition the original energy, so their $p/2$ powers sum to at most 1, and $1 - p/3 - p\eta \geq \gamma_0$ makes the dyadic scale sum geometric.

Lemma 8.4 (The angular test cannot fail elsewhere). *Outside the exceptions in (8.9) and (8.10), an original point cannot exit in an angular deletion. On a surviving such point, $|U_{\mathcal{R}}| \geq c\lambda$ for an absolute $c > 0$.*

Proof. At a proposed exit, subtract the selected prefix from U . The remaining field has modulus at least a fixed fraction of λ , and every fine cap sum has modulus at most $2r^\eta \lambda$. Only physically neighboring remaining labels can contribute at the point.

Mark the terminal cells of the finite direction tree that contain such a neighboring label. Give unmarked terminal cells capacity zero, fine cells of side $r \leq \delta_0$ capacity r^η , and coarser cells capacity 1. A cut covering the marked leaves either contains a coarse cell, and hence has capacity at least 1, or consists of fine cells. In the latter case its cell fields sum to the remaining field, so the triangle inequality gives total capacity at least an absolute positive constant. Nonneighboring labels in these cells contribute zero at the point.

For completeness, the maximal mass that can pass through a node is the minimum of its capacity and the sum of the maximal masses of its children. Induction up the finite tree shows that this is the minimum cut capacity. Thus a mass bounded below can be sent from the root to marked leaves. Choose a neighboring label in each such leaf and normalize the resulting flow to a probability. A fine cell of side r then has probability $O(r^\eta)$. A ball of radius $r \geq L^{-1}$ is covered by boundedly many cells of comparable size. At coarser scales $r \geq \delta_0$, the desired bound follows from $\delta_0^{-\eta} r^\eta \geq 1$. Taking C_{ang} large enough gives (8.7), a contradiction to the exit test. The assertion about the final field follows by subtracting its prefix. $\square$

Relative to the target with constant C_{ind} , the recursive costs so far are bounded by fixed multiples of

$$(D_*/D)^{-\sigma} = X^{-\sigma}, \quad D^\sigma \delta_0^{\gamma_0} \lesssim D^\sigma X^{-u\gamma_0}. \quad (8.11)$$

The later single-cap exceptions have the second form as well. The last inequality in (8.4) and a sufficiently large B make the sum of all these costs less than, say, half the target. Their constants include the fixed prefix constant C_p , but do not include any of the final nonrecursive constants.

We apply geometry to $\mu|_{A_\infty}$ itself, before subtracting any of the proof-only exceptional sets. Equations (8.7) and (8.8) therefore hold for the same measure. Subsequent upper bounds may use the original μ : its ball and sparsity upper bounds still hold for every surviving label.

8.3 Plates and coarse frequency intervals

Divide physical coordinates by L , and divide $\mu|_{A_\infty}$ by mL^2 . In the notation of Proposition 7.1, $H = L$, the richness is at least $L/D_* \geq L/X^2$, and the angular constant in (8.7) is at most a fixed power of X . The required comparisons $X^C \ll H$ hold by the choice of κ . Proposition 7.1 assigns every surviving tube to one of at most X^{C_1} plates of width X^{C_1} in these coordinates, with

$$|n \cdot (2\omega, 1)| \leq X^{C_1}/L.$$

Enlarge the plates by a fixed polynomial factor to contain the physical packet supports in the domain. All assignments are disjoint at the level of coefficient labels.

The spatial component n_x of such a unit normal is bounded below. Indeed $|\omega| \leq C$, so if $|n_x|$ were sufficiently small, then $|n_t + 2n_x \cdot \omega|$ would be bounded below by an absolute constant, contrary to the displayed estimate for large L . Thus the assigned frequencies lie in a strip of width $X^{O(1)}/L$, which can be partitioned into polynomially many strips of width $O(L^{-1})$. If the total number of resulting arrays is $N \leq X^{C_2}$, the triangle inequality and Proposition 6.2 give

$$\mu\{|U_{\mathcal{R}}| \geq c\lambda\} \lesssim mL^4 N^{p_0} g^{-p_0} \sum_{j=1}^N \|c_j\|_{\ell^2}^{p_0} \leq X^{C_3} mL^4 g^{-p_0}. \quad (8.12)$$

Choosing $C_g(p_0 - p_{\max}) > C_3 + 1$ makes this at most a fixed multiple of the target whenever $g \geq X^{C_g}$. For $g < D^{-1}$, total mass suffices, since $D^{-\sigma} g^{-p} \geq D^{p-\sigma} \geq 1$ after decreasing σ if necessary. We henceforth work in

$$D^{-1} \leq g \leq X^{C_g}. \quad (8.13)$$

Discard intersections of enlarged plates whose projective normal directions differ by at least a constant multiple of a/L . For two plates of normalized width $w \leq X^{C_4}$ and normal separation β , their intersection is in a box of length $O(L)$ and transverse sides $O(w)$ and $O(w/\beta)$. Its normalized mass is at most $X^{O(1)}L/\beta$ by unit-ball covering. Summing over the polynomial plate list, the discarded mass in the original coordinates is at most

$$mL^4 X^{C_5}/a. \quad (8.14)$$

Choose $C_a > C_5 + 1 + C_g p_{\max}$, with room for fixed constants. Then (8.14) is bounded by the target throughout (8.13).

Partition projective normal directions into cells of diameter comparable to a/L , with bounded overlap on balls of that radius. At a retained point all active plate normals have pairwise distance $O(a/L)$. Consequently only boundedly many normal cells contribute, and at least one cluster field has modulus at least $c\lambda$. Relative to a representative normal in each nonempty cell, its label frequencies lie in a strip of width $O(a/L) = O(h^{-1})$. The representative still has spatial component bounded below, and this follows by perturbing the previous direction inequality by $O(a/L)$. The clusters partition the surviving coefficient energy.

For each cluster separately rotate the spatial coordinates and remove its central normal frequency by a bounded shear and modulation. Write the resulting coordinates as (x, y, t) , with x tangential and y normal. Partition tangential label frequencies into coarse intervals τ of length comparable to δ_0 , then into fine intervals θ of length comparable to h^{-1} , using nested grids and consistent endpoint assignments. Each coarse array fits in a comparable two-dimensional cap, because $h^{-1} \ll \delta_0$. In applying Lemma 8.2, enclose that array in an unrotated square in the original coordinates, keeping only its labels. At each scale bounded overlap of the enclosing squares, together with the partitioned coefficient energy across clusters, gives a total exceptional cost

$$C'' C_{\text{ind}} mL^4 g^{-p} \delta_0^{\gamma_0}$$

for the sets on which a coarse cluster field exceeds $\delta_0^\eta \lambda$. This is the further recursive cost already allowed in (8.11).

At any remaining point where a cluster is high, at least $c\delta_0^{-\eta}$ distinct coarse intervals satisfy

$$|U_\tau| \geq c' \delta_0 \lambda. \quad (8.15)$$

To see this, there are $O(\delta_0^{-1})$ coarse intervals, so the sum of those below the displayed small threshold has modulus less than half the cluster threshold if c' is sufficiently small. Each of the others has modulus at most $\delta_0^\eta \lambda$, forcing the stated number. Full partition intervals are used throughout; the complement estimate never discards only part of an interval field.

8.4 Flattening all intermediate scales

The cluster frequencies now lie in a strip of width $O(h^{-1})$, whereas Proposition 6.2 at packet size L requires width $O(L^{-1})$. Splitting into thinner strips produced the large-level estimate, but paid a fixed power of X . To handle the remaining levels, we instead make one smooth packet out of each entire fine frequency array on each short physical cell. We will apply Theorem 5.1 directly to these packets and sum their energies. The construction uses only the general compact profiles in (2.2); no equation or exact Fourier support is imposed on those profiles.

For a fine interval θ centered at v_θ , put

$$z'_\theta = F_\theta(x, y, t) = ((x - 2v_\theta t)/h, y/h, t/h^2). \quad (8.16)$$

The central normal frequency has already been removed. The array is contained in a cap of size $O(h^{-1})$ and therefore has the form

$$U_\theta(x, y, t) = h^{-1} e^{i(xv_\theta - tv_\theta^2)} U'_\theta(z'_\theta), \quad (8.17)$$

up to a common harmless modulation. Here U'_θ is an array of packet size $a = L/h$, with bounded frequencies and the original coefficient norm.

Choose a fixed smooth partition of unity $\{\psi_T\}$ on the unit lattice in $\mathbb{R}^3$, with bounded overlap and fixed enlarged supports. Take a finite derivative order N exceeding all later Sobolev and frame requirements, and let

$$s_{\theta, T} = h^{-1} \max_{|\beta| \leq N} \sup_{z' \in T^*} |\partial^\beta U'_\theta(z')|,$$

where T^* is a fixed sufficiently large enlargement of T . Enlarging the fixed constant in this definition would give the same estimates. Fix also a large polynomial decay power P , larger than $M + 3$ and all tail moments that will occur, and define

$$d(T) = \frac{1}{mh^3} \int (1 + \text{dist}(F_\theta(z), T))^{-P} d\mu(z). \quad (8.18)$$

We abbreviate this to $d(T)$ when the cluster and θ are specified. All coordinate changes push forward the measure without changing its mass. Their ball constants are uniformly bounded after the displayed factor h^3 . In particular $d(T) \lesssim 1$ by annular covering.

Only cells whose partition supports meet the testing domain need be retained. The associated summand

$$e^{i(xv_\theta - tv_\theta^2)} h^{-1} \psi_T(z'_\theta) U'_\theta(z'_\theta) \quad (8.19)$$

is a single composite packet: it has tangential width $O(h)$, time length $O(h^2)$, localized dependence on y/h , and all required normalized derivatives bounded by $Cs_{\theta, T}$.

There are two energy bounds to retain. On a time interval of length h^2 , spatial almost orthogonality controls the total squared amplitudes of the composite packets. Over the full time range, the sparse estimate controls the same amplitudes weighted by $d(T)^{2/3}$. The first bound will supply the large threshold-to-envelope ratios required by the frame theorem; the second will retain the factor D^{-c_s} when the frame estimates are summed.

Lemma 8.5 (Flattened energy estimates). *For a time interval J of length h^2 , let $\mathcal{T}(J)$ denote the boundedly many native time-cell layers that can contribute on J . Summing over the disjoint fine arrays, and also over clusters when needed, one has*

$$\begin{aligned} \sum_{\theta, T \in \mathcal{T}(J)} s_{\theta, T}^2 &\lesssim h^{-2}, \\ \sum_{\theta, T} d(T)^{2/3} s_{\theta, T}^2 &\lesssim_B h^{-2} a^{4/3} D^{-c_s}. \end{aligned} \quad (8.20)$$

The right sides are multiplied by the total original coefficient energy if it is not normalized to be at most 1.

Proof. We first justify spatial almost orthogonality for arbitrary smooth packet profiles. At native time $t' = O(a^2)$, a child atom and all its fixed derivatives have spatial support within $O(a)$ of its transported intercept $b'_\alpha + 2\omega'_\alpha t'$. Spatial integration by parts in the difference of the carrier frequencies gives Gram entries bounded by

$$C_N(1 + a|\omega'_\alpha - \omega'_\beta|)^{-N}, \quad (8.21)$$

and the entries vanish unless the transported intercepts differ by $O(a)$. The joint centers

$$(b'_\alpha/a, a\omega'_\alpha) \mapsto (b'_\alpha/a + 2(t'/a^2)a\omega'_\alpha, a\omega'_\alpha)$$

undergo a uniformly bounded shear. Thus their phase-space counts remain bounded. Summing (8.21) over these centers gives uniformly summable Gram rows when N is sufficiently large. It follows that, for every required derivative,

$$\|\partial^\beta U'_\theta(\cdot, t')\|_{L^2(\mathbb{R}^2)}^2 \lesssim \sum_{\alpha \in \theta} |c_\alpha|^2.$$

Time derivatives are included: bounded child frequencies and factors a^{-1} or a^{-2} from differentiating profiles preserve the same packet bounds. Unit-cell Sobolev estimates, bounded overlap of the enlarged cells, and integration over the $O(1)$ native time layers in J now give the first assertion of (8.20). Summing the coefficient energies across fine arrays and clusters costs at most 1.

For the second assertion, let $\mu_\theta = F_{\theta\#}\mu$ in the coordinates of the relevant cluster. Covering the inverse image of a ball shows that its ball parameter is at most Cmh^3 for radii at least 1. For every child label, the transverse ratio in (2.4) is unchanged by (8.16), and

$$(mh^3)a^3/D = mL^3/D. \quad (8.22)$$

Thus μ_θ satisfies the same sparsity condition with D after a fixed inflation of its ball parameter.

Convolve μ_θ with $K_P(z') = (1 + |z'|)^{-P}$ and restrict the resulting measure to $|t'| \leq C'a^2$, with a fixed enlargement sufficient for the cells and derivatives in use. Call the resulting measure $\tilde{\mu}_\theta$. Its ball parameter remains Cmh^3 . Its smoothed tube moments are still bounded as in (8.22): translating a child tube kernel by w changes it by at most a factor $C(1 + |w|)^M$, and

$$\int_{\mathbb{R}^3} (1 + |w|)^{M-P} dw < \infty.$$

Every relevant unit cell q in a fixed enlargement of T has

$$\tilde{\mu}_\theta(q) \gtrsim mh^3 d(T), \quad (8.23)$$

after using the time enlargement; this follows by comparing the kernel at all points of q with its distance to T .

Apply Proposition 4.1, including its derivative version, at packet size a to U'_θ and $\tilde{\mu}_\theta$. Unit cells in T^* , of which there are boundedly many, control the suprema defining $s_{\theta,T}$. Using (8.23) and dividing out $(mh^3)^{2/3}$ yields

$$\sum_T d(T)^{2/3} s_{\theta,T}^2 \lesssim h^{-2} a^{4/3} D^{-c_s} \sum_{\alpha \in \theta} |c_\alpha|^2.$$

The transformed spatial domain may exceed size a^2 . To use the local form of the sparse estimate, divide it into boxes of side comparable to a^2 and retain only intersecting compact

child packets. Each such packet meets boundedly many boxes on $|t'| \leq C'a^2$; summing the squared estimates therefore uses only bounded total coefficient energy.

This application requires a fixed positive-power lower bound on D in terms of a . By $a \asymp (BD)^{C_a}$ and $a \geq D$, we have $a^{1/(2C_a+2)} \leq D \leq a$ once D exceeds a constant depending on B . Use this fixed exponent in Proposition 4.1. For the remaining bounded values of D , also a is bounded in terms of B . The first assertion summed over $O(a^2)$ time layers, and $d(T) \lesssim 1$, gives the second assertion with a different allowed constant. Finally sum over the partitioned coefficient arrays. $\square$

We sort the cells into dyadic density bands $d(T) \asymp d_\ell$, where $d_\ell = d_{\max} 2^{-\ell}$, $\ell \geq 0$, and $d_{\max}$ is a fixed upper bound. Shifting this index by a fixed amount has no effect. Only $O(\log h)$ bands need occur. Indeed if the total measure of the domain is already less than $mL^4 D^{-\sigma} g^{-p}$ there is nothing to prove. Otherwise, for a relevant cell the transformed domain diameter is $O(L^2/h)$, and the strictly positive weight in (8.18) gives

$$d(T) \gtrsim (mh^3)^{-1} \mu(\text{domain}) (1 + CL^2/h)^{-P} \gtrsim a^{4-2P} h^{1-P} D^{-\sigma} g^{-p}. \quad (8.24)$$

Here $L = ah$, and fixed powers of a, D, g are bounded by fixed powers of X , which in turn are bounded by fixed powers of h . Thus $d(T) \geq c_B h^{-C}$ with a fixed C . This band count will only be used with a power gain in h ; it will not remain in the final bound.

8.5 Two bands and the bilinear frame estimate

For every coarse interval satisfying (8.15), summably split its threshold over the density bands. At least one band field then has modulus at least

$$A_\ell = c\delta_0(1 + \ell)^{-2}\lambda. \quad (8.25)$$

The constant $c > 0$ is fixed sufficiently small that the sum of these thresholds is less than the significant-interval threshold. Choose one such band for each significant interval. Fix a sufficiently large exponent C_d , to be chosen in the tiny-band estimate below. If one selected band has $d < X^{-C_d}$, pair its interval with a separated significant interval. Otherwise there are only $O(C_d \log X)$ selected bands. By increasing B , the number $c\delta_0^{-\eta}$ in (8.15) exceeds a sufficiently large multiple of this band count. Pigeonholing gives two separated coarse intervals with the same band. Here “separated” means distance at least a fixed multiple of δ_0 ; excluding a bounded number of neighboring intervals suffices. The list of significant intervals is also large enough to supply the partner in the tiny-band case.

Partition time into intervals J of length h^2 . For each cluster, J , separated pair (τ_1, τ_2) , and selected bands (ℓ_1, ℓ_2) , form the measurable subset where both band fields exceed the corresponding levels in (8.25). Restrict to the high points under consideration if desired. Denote its mass by M_0 , and put

$$E_i = \sum_{\substack{\theta \subset \tau_i, T \in \mathcal{T}(J) \\ d(T) \asymp d_i}} s_{\theta, T}^2, \quad d_i = d_{\ell_i}, \quad i = 1, 2. \quad (8.26)$$

The witness subsets need not be disjoint; we use an upper bound on their sum of masses. There are at most $X^{C_6}(\log h)^2$ relevant tuples: the cluster list and interval pairs have polynomial size in X , and the time partition contributes $O(a^2)$ intervals.

Tuples with

$$M_0 \leq mh^{4-1/12} \quad (8.27)$$

have total mass at most $X^{C_6}(\log h)^2 mh^{4-1/12}$. Relative to the target this is at most

$$CX^{C_6+\sigma+C_g p_{\max}} a^{-4} h^{-1/12} (\log h)^2,$$

which is bounded, and can be made small at large scales, by the choice of κ . This discard counts tuples before normal spatial slabs are introduced, so it incurs no number of slabs.

Fix a remaining tuple. Define its two full square envelopes by

$$S_i(z)^2 = \sum_{\substack{\theta \subset \tau_i, T \in \mathcal{T}(J) \\ d(T) \asymp d_i}} s_{\theta, T}^2 (1 + \text{dist}(F_\theta(z), T))^{-P}.$$

By (8.18), $\int S_i^2 d\mu \lesssim m h^3 d_i E_i$. Markov's inequality retains at least half the witness mass with both bounds

$$S_i \leq B_i, \quad B_i = C \left(\frac{m h^3 d_i E_i}{M_0} \right)^{1/2}. \quad (8.28)$$

The constant C here is large and absolute. The first estimate in (8.20), $d_i \lesssim 1$, and the reverse of (8.27) give

$$B_i \lesssim h^{-3/2+1/24}, \quad \frac{A_{\ell_i}}{B_i} \gtrsim h^{1/8} \delta_0 g a^{-4/3} (1 + \ell_i)^{-2}. \quad (8.29)$$

Since $g \geq D^{-1}$ and $\ell_i = O(\log h)$, the right side of the ratio estimate is bounded below by $h^{1/8}$ divided by a fixed power of X and by $C_B(\log h)^2$. Our scale hierarchy makes it larger than the fixed threshold-ratio constant in Theorem 5.1. The same hierarchy ensures $\delta_0 \geq h^{-1/2}$. Zero class energy would give an identically zero class field, so it cannot occur in a tuple with positive witness mass.

Lemma 8.6 (Frame estimate for one tuple). *For every tuple remaining after (8.27),*

$$M_0 \lesssim \delta_0^{-C} m h \frac{B_1^2 B_2^2}{A_{\ell_1}^2 A_{\ell_2}^2} \left(\frac{h^3 E_1}{A_{\ell_1}^2} + \frac{h^3 E_2}{A_{\ell_2}^2} \right). \quad (8.30)$$

In particular,

$$\lambda^6 M_0^3 \lesssim \delta_0^{-C} (1 + \ell_1 + \ell_2)^C m^3 h^{10} d_1 d_2 E_1 E_2 (E_1 + E_2). \quad (8.31)$$

Proof. Partition the normal coordinate into slabs of width h . In a slab centered at y_0 , only boundedly many normal cell indices T_y can contribute, and on J only boundedly many time indices T_t can contribute. Write $t = t_0 + s$ with t_0 an endpoint of J . For a composite cell in (8.19), the tangential carrier has frequency ν_θ , and its intercept in the time coordinate s is

$$b_{\theta, T} = h T_x + 2\nu_\theta t_0. \quad (8.32)$$

The fine centers are spaced at scale h^{-1} . At each such center, T_x runs over an integer lattice translated by $2\nu_\theta t_0/h$; the bounded number of normal and time indices thus gives the required one-dimensional phase-space counts uniformly in t_0 . The packet profiles are compact and smooth on the h, h^2 scales, by the definition of $s_{\theta, T}$.

Expand these profiles, with a fixed smooth cutoff on an enlargement of the slab, in the variable $(y - y_0)/h$. The mode packets can be written with scalar amplitudes at most

$$C s_{\theta, T} \langle n \rangle^{-N_1}, \quad \langle n \rangle = (1 + n^2)^{1/2},$$

and uniformly bounded required tangential profile derivatives, for a fixed large N_1 . This follows by integrating the Fourier coefficients by parts, using enough derivatives in the definition of s . The external Fourier phases have modulus one at every witness.

For the cells belonging to this slab and time bin, the omitted normal and native time displacements in $F_\theta(z) - T$ are bounded at a witness. Hence the one-dimensional envelope

weight in Theorem 5.1, after (8.32), is at most a fixed multiple of the full weight defining S_i there. On the set retained in (8.28), the n th mode envelope is consequently bounded by $CB_i \langle n \rangle^{-N_1}$. Its energy parameter H_i^* in (5.1) is at most $Ch^3 E_{i,\text{slab}} \langle n \rangle^{-2N_1}$, where $E_{i,\text{slab}}$ is the part of (8.26) near this slab.

Whenever a band field has modulus at least A_{ℓ_i} , some Fourier mode has modulus at least $cA_{\ell_i} \langle n \rangle^{-2}$. For each pair of selected modes apply Theorem 5.1 at separation δ_0 . The threshold-to-envelope ratios only improve with $|n|$ when $N_1 > 2$, so (8.29) verifies its ratio hypothesis uniformly in the modes. The factors in the frame estimate are summable over the two mode indices: for example its first energy term has powers $\langle n_1 \rangle^{-4N_1+8} \langle n_2 \rangle^{-2N_1+4}$. Take $N_1 > 3$, with sufficient additional regularity for all derivatives.

The projection to (x, t) of the measure restricted to a slab has mass at most Cmh on any unit square: its inverse image has dimensions $O(1), O(h), O(1)$ and is covered by $O(h)$ unit balls. This is an upper bound on the full projected measure, regardless of which normal coordinate supplied a witness. Multiplying the unit-square count from (5.1) by Cmh , and summing the modes, bounds the retained witness mass in this slab by

$$C\delta_0^{-C}mh \frac{B_1^2 B_2^2}{A_{\ell_1}^2 A_{\ell_2}^2} \left(\frac{h^3 E_{1,\text{slab}}}{A_{\ell_1}^2} + \frac{h^3 E_{2,\text{slab}}}{A_{\ell_2}^2} \right).$$

Each composite cell belongs to boundedly many enlarged slabs, and therefore $\sum_{\text{slab}} E_{i,\text{slab}} \lesssim E_i$. Summing costs no number of slabs and proves (8.30), since at least half of M_0 was retained by Markov.

Finally substitute (8.28) and (8.25). The product of the two squared envelope bounds supplies $m^2 h^6 d_1 d_2 E_1 E_2 / M_0^2$; the projected square mass and frame energy supply $mh \cdot h^3$. Thus the total factor is $m^3 h^{10} / M_0^2$. The six threshold powers contribute λ^{-6} times a fixed power of δ_0^{-1} and of $1 + \ell_1 + \ell_2$. Rearranging proves (8.31). $\square$

8.6 Summing the bands and closing the induction

First suppose at least one selected band is tiny. The unweighted energy bound $E_i \lesssim h^{-2}$ in (8.20) and the cube root of (8.31) give

$$\lambda^2 M_0 \lesssim \delta_0^{-C} (1 + \ell_1 + \ell_2)^C m h^{4/3} (d_1 d_2)^{1/3}. \quad (8.33)$$

For dyadic bands, geometric decay absorbs the polynomial index factors:

$$\sum_{\min(d_1, d_2) < X^{-C_d}} (1 + \ell_1 + \ell_2)^C (d_1 d_2)^{1/3} \lesssim X^{-C_d/6}.$$

The constants here depend on the fixed exponent C . After summing over clusters, time bins and coarse interval pairs, all other costs in (8.33) are at most a fixed power of X , uniformly for $u \leq 1$. Choose C_d large enough to absorb this power, together with $D^\sigma g^{p-2}$ for (8.13); using $h^{4/3} \leq L^{4/3}$, this gives

$$\lambda^2 \sum_{\text{tiny}} M_0 \lesssim m L^{4/3} D^{-\sigma} g^{-(p-2)}.$$

This is precisely the target multiplied by λ^2 , with a constant independent of induction.

For the other tuples the two density bands coincide, say $d_1 \asymp d_2 \asymp d \geq X^{-C_d}$. Then $\ell_i = O(C_d \log X)$, and

$$(E_1 E_2 (E_1 + E_2))^{1/3} \leq C(E_1 + E_2).$$

The cube root of (8.31) gives

$$\lambda^2 M_0 \lesssim \delta_0^{-C} (\log X)^C m h^{10/3} d^{2/3} (E_1 + E_2). \quad (8.34)$$

We now sum energies rather than the number of tuples. A composite cell appears only in its own cluster, in boundedly many J , in its specified density band and coarse interval, and for at most $O(\delta_0^{-1})$ choices of the other coarse interval at that band. There is consequently only an additional fixed power of δ_0^{-1} in the multiplicity. In particular, no factor a^2 from time bins, no number of clusters, and no number of density bands is paid in this sum. The weighted estimate in (8.20) therefore gives

$$\begin{aligned} \lambda^2 \sum_{\text{common}} M_0 &\lesssim_B X^{C_f u} (\log X)^{C_f} m h^{10/3} (h^{-2} a^{4/3} D^{-c_s}) \\ &= C_B X^{C_f u} (\log X)^{C_f} m L^{4/3} D^{-c_s}. \end{aligned} \quad (8.35)$$

The fixed exponent C_f includes the just described partner multiplicity; it is the exponent used in choosing u . The transforms vary with θ and with the cluster, but their coefficient arrays are disjoint, exactly as required in the proof of (8.20).

The target after multiplication by λ^2 is $m L^{4/3} D^{-\sigma} g^{-(p-2)}$. Since $g \leq X^{C_g}$, the ratio of (8.35) to this quantity is at most

$$C_B X^{C_f u + C_g(p-2)} (\log X)^{C_f} D^{\sigma - c_s}.$$

It is bounded by a fixed constant depending on B by the strict second inequality in (8.4). This proves the required estimate for the common-band tuples with no growing loss in L when D and g are fixed.

Compatible order of parameter choices. We now verify that the conditions used in the proof can be imposed together. The values of $p_{\max}$ and η were fixed in (8.2). Temporarily restricting $0 < u \leq 1$ and $0 < \sigma \leq 1$ bounds all geometric losses by fixed powers of X . Choose C_g large enough for (8.12), using only these powers and the positive gap $p_0 - p_{\max}$. Choose C_a larger still for (8.14), and then choose C_d for the tiny-band sum. These exponents are independent of how small u will be.

The equal-band estimate (8.35) has loss $X^{C_f u} (\log X)^{C_f}$, where the fixed C_f accounts for the frame separation loss and the coarse-interval multiplicity. Choose $u > 0$ so small that $C_f u < c_s/4$. Next choose $\kappa > 0$ small in terms of all the fixed powers just chosen. For any fixed required K , the relations $X \leq B L^\kappa$ and $h \asymp L/X^{C_a}$ give $h \geq X^K$ at sufficiently large L if $\kappa(C_a + K) < 1$. This supplies the geometric scale conditions, $\delta_0 \geq h^{-1/2}$, the frame threshold ratios, and the small-mass discard.

Choose $p > 2$ sufficiently close to 2, with $p \leq p_{\max}$, and then choose $\sigma > 0$ sufficiently small to satisfy all three strict inequalities in (8.4). Finally choose B large enough to contract the recursive terms and to supply the angular pigeonhole argument. Constants depending on B are allowed in the nonrecursive estimates. Enlarge the fixed base scale L_0 to cover the remaining comparisons involving B . Every fixed constant and exponent is chosen before the induction constant and the variable scales L, D .

Completion of the proof of Theorem 8.1. The sparse range and bounded scales were handled directly. In the remaining range, the prefix exceptions, all dyadic-cap exceptions, and the coarse cluster exceptions cost less than half the desired bound with constant C_{ind} , by (8.11). Every other original high point either is handled by the large-level strip estimate or total mass, lies in a discarded plate intersection, belongs to a small-mass tuple, or is covered by the tiny-band or common-band bounds. All these latter costs are bounded by a fixed multiple of $m L^4 D^{-\sigma} g^{-p}$ independently of C_{ind} . Choose C_{ind} once to be twice their total constant and large enough for the initial cases. The two inductions close. Homogeneity supplies the factor $\|c\|_{\ell^2}^p$ for arbitrary subarrays and completes the theorem. $\square$

9 Endpoint square summation and the order of limits

The exponent $p > 2$ in Theorem 8.1 has a final, essential use. It supplies a positive convexity deficit on a common dyadic time tree. Combined with spatial localization, this gives a square sum over frequency scales and hence the Sobolev norm at the endpoint. We then construct a representative continuous in time and verify that the prescribed Gaussian limits recover that representative at every real time on one full-measure set.

Throughout this section, $S(t)f = e^{it\Delta}f$ denotes the L^2 evolution, whose Fourier multiplier is $e^{-it|\xi|^2}$. When $\hat{f}$ has compact support, we also use $S(t)f(x)$ for its absolutely convergent Fourier integral. This is continuous in (x, t) , since then $\hat{f} \in L^1$. Constants may depend on fixed annular supports and smooth cutoff functions, but are independent of frequency, the center of a spatial ball, and every measurable choice of time made below.

9.1 From packet distributions to a short-time maximal estimate

For $p > 0$ and measurable h on $A \subset \mathbb{R}^2$, we use the weak L^p quasinorm

$$\|h\|_{L^{p,\infty}(A)} = \sup_{\lambda>0} \lambda |\{x \in A : |h(x)| > \lambda\}|^{1/p}.$$

Proposition 9.1 (Short-time maximal estimate). *Let $p > 2$ be the exponent in Theorem 8.1. Fix $0 < c_0 < C_0 < \infty$. If $N \geq 1$ is dyadic and*

$$\text{supp } \hat{f}_N \subset \{\xi \in \mathbb{R}^2 : c_0 N \leq |\xi| \leq C_0 N\},$$

then, on every unit ball $B \subset \mathbb{R}^2$,

$$\left\| \sup_{0 < t < 1/N} |S(t)f_N| \right\|_{L^{p,\infty}(B)} \leq CN^{1/3} \|f_N\|_2. \quad (9.1)$$

The same estimate holds on every translated time interval of length at most $1/N$, with either endpoint included.

Proof. We first apply Theorem 8.1 at $D = 1$ to an arbitrary measurable spatial graph in unit-frequency variables. Let $\mathcal{B}$ be a spatial ball of radius $O(L^2)$, let $T : \mathcal{B} \rightarrow [0, L^2]$ be measurable, and define

$$\mu(F) = \int_{\mathcal{B}} \mathbf{1}_F(X, T(X)) dX. \quad (9.2)$$

This is the pushforward of spatial area, regardless of the regularity of T . A space-time ball of radius r projects into a spatial disk of the same radius, so $\mu(B_r) \leq \pi r^2$. A tube of width L and time length L^2 , with one of the bounded velocities in the packet class, is covered by $O(L)$ space-time balls of radius CL . Its μ -mass is therefore $O(L^3)$. A transverse enlargement by 2^ℓ can be covered by $O(L2^{2\ell})$ such balls. Summing these bounds against the decay $2^{-M\ell}$, where M is the fixed large exponent in (2.4), proves that condition with $D = 1$ after a fixed enlargement of m . All constants here are uniform over T , the packet intercepts, and the center of $\mathcal{B}$.

We explain why the compact packet estimate applies to an exact unit-frequency extension

$$Ev(X, T) = \int e^{i(X \cdot \eta - T|\eta|^2)} v(\eta) d\eta, \quad \text{supp } v \subset \{c_0 \leq |\eta| \leq C_0\}.$$

Apply Lemma 2.2 at packet width L . It gives $v = \sum_{\omega, n} c_{\omega, n} v_{\omega, n}$, with $\omega \in L^{-1}\mathbb{Z}^2$, $n \in \mathbb{Z}^2$, and

$$\sum_{\omega, n} |c_{\omega, n}|^2 \leq C \|v\|_2^2.$$

Its exact profile formula (2.11) reads

$$Ev_{\omega,n}(X, T) = L^{-1} e^{i(X \cdot \omega - T|\omega|^2)} H\left(\frac{X - Ln - 2\omega T}{L}, \frac{T}{L^2}\right),$$

$$H(z, s) = \frac{1}{2\pi} \int \psi(\zeta) e^{i(z \cdot \zeta - s|\zeta|^2)} d\zeta,$$

where ψ is the fixed smooth frequency cutoff in that lemma. The phase-space counts are bounded, and, on every fixed bounded s -interval, all required derivatives of H decrease faster than any fixed power of $|z|$. The remaining issue is its noncompact spatial support; we resolve it by summing compact layers.

To obtain the compact profiles of (2.2), choose a smooth time cutoff equal to one on $[0, 1]$, and a spatial partition of unity $\sum_{q \in \mathbb{Z}^2} \rho(z - q) = 1$ with ρ compactly supported. Fix the finite profile derivative order required by Theorem 8.1. In the q th term replace the intercept Ln by $Ln + Lq$. The new profile is the old profile evaluated at $z + q$, multiplied by $\rho(z)$ and the time cutoff. For any prescribed large A , its required derivative bounds are $O((1 + |q|)^{-A})$. Put this factor into the coefficients. For each fixed q , all phase-space intercepts undergo the same translation, so the counting bounds are preserved, and the new coefficient norm is at most

$$C_A(1 + |q|)^{-A} \|v\|_2.$$

Thus every layer is an admissible array for Theorem 8.1.

The exponent A is chosen once, after the required derivative order; the fixed profile bound may depend on both, but never on L or q . For completeness, let $b_q > 0$ satisfy $\sum_q b_q = 1$ and $b_q \asymp (1 + |q|)^{-3}$. Apply (8.1) to the q th layer at threshold $b_q \lambda$, using homogeneity in its coefficients. Choosing A sufficiently large makes $\sum_q b_q^{-p} (1 + |q|)^{-Ap}$ finite and yields

$$\mu\{|Ev| > \lambda\} \leq CL^4 (\lambda L^{4/3})^{-p} \|v\|_2^p. \quad (9.3)$$

These operations can first be made on finite packet sums. At a point, each frequency label has only $O(1)$ intercepts contributing to a fixed compact layer, and there are $O(L^2)$ frequency labels. Cauchy–Schwarz therefore bounds that layer by $C(1 + |q|)^{-A} \|v\|_2$, uniformly in L ; the factor L from the square root of the label count cancels the packet amplitude L^{-1} . The same bound holds with the coefficient norm of any tail in place of $\|v\|_2$. Taking $A > 2$ makes the layer sum absolutely and uniformly convergent, and square-summability makes finite coefficient truncations converge uniformly as well. Fourier-series partial sums also converge in L^2 on a fixed bounded frequency set, hence their exact extensions converge uniformly by Cauchy–Schwarz in frequency. These observations identify the compact layer sum with the exact extension. Passing to the limit in the distribution bound, first with a slightly smaller threshold, proves (9.3) for arbitrary $v \in L^2$ with the indicated support. This passage uses exact extensions and therefore introduces no time-dependent choices of representatives.

Now take $L = \sqrt{N}$ and $v(\eta) = N \hat{f}_N(N\eta)$. Then $\|v\|_2 = \|\hat{f}_N\|_2$, and

$$S(t)f_N(x) = (2\pi)^{-2} N Ev(Nx, N^2 t). \quad (9.4)$$

A unit spatial ball becomes a ball of radius $N = L^2$, and a time interval of length $1/N$ becomes one of length $N = L^2$. For a measurable physical graph $t = t(x)$, its lifted spatial area in the new coordinates is N^2 times the original area. This cancels L^4 in (9.3). The threshold scale in the physical field is

$$NL^{-4/3} = N^{1/3}.$$

Consequently, for every such graph and every $\lambda > 0$,

$$|\{x \in B : |S(t(x))f_N(x)| > \lambda\}| \leq C \left(\frac{N^{1/3}\|f_N\|_2}{\lambda} \right)^p.$$

For finitely many times choose, measurably, the first time attaining the largest modulus. Apply this graph bound, then increase the finite list to a countable dense set. Continuity of the bounded-frequency evolution identifies the resulting supremum with the supremum over all times and allows the endpoints. Finally, replacing f_N by $S(t_0)f_N$ translates the interval without changing either its Fourier support or its L^2 norm. $\square$

9.2 Two dual estimates on a short time interval

Fix a unit spatial ball Ω , a measurable function $t : \Omega \rightarrow [0, 1)$, and a measurable complex function w with $|w| \leq 1$. These choices remain the same at every frequency. For $N = 2^j$, $j \geq 1$, let $\chi_N(\xi) = \chi(\xi/N)$ be a smooth cutoff supported on a fixed annulus away from zero. The same argument permits finitely many fixed choices of χ . For a half-open time interval $I \subset [0, 1)$, put

$$\begin{aligned} E_I &= \{x \in \Omega : t(x) \in I\}, & m_I &= |E_I|, \\ G_N(I)(\xi) &= \chi_N(\xi) \int_{E_I} w(x) e^{i(x \cdot \xi - t(x)|\xi|^2)} dx, & e_j(I) &= N^{-2/3} \|G_N(I)\|_2^2. \end{aligned} \quad (9.5)$$

The quantities m_I are the masses of a single measure on time, the pushforward of $\mathbf{1}_\Omega dx$ under t . This measure can have atoms; all interval partitions below are half-open so membership is unique.

We will use the elementary weak-space integration bound

$$\int_F |h| \leq \frac{p}{p-1} \|h\|_{L^{p,\infty}} |F|^{1-1/p}. \quad (9.6)$$

Indeed the layer-cake formula bounds its left side by $\int_0^\infty \min\{|F|, (\|h\|_{L^{p,\infty}}/\lambda)^p\} d\lambda$; splitting where the two terms agree gives the stated expression.

The kernel for pairings of the dual functions is

$$K_N(z, \tau) = \int |\chi_N(\xi)|^2 e^{i(z \cdot \xi - \tau|\xi|^2)} d\xi = N^2 \int |\chi(\eta)|^2 e^{i(Nz \cdot \eta - N^2\tau|\eta|^2)} d\eta. \quad (9.7)$$

In particular,

$$\langle G_N(I), G_N(J) \rangle = \int_{E_I} \int_{E_J} w(x) \overline{w(y)} K_N(x - y, t(x) - t(y)) dy dx.$$

Only the smooth frequency variable will be differentiated in estimates for this kernel. No regularity of the graph $t(x)$ is needed.

Lemma 9.2 (Separation in time). *Let $\mathcal{D}_j$ be the dyadic partition of $[0, 1)$ into intervals of length $2^{-j} = 1/N$. Then*

$$e_j([0, 1)) \leq C \sum_{I \in \mathcal{D}_j} e_j(I) + CN^{-2}. \quad (9.8)$$

Proof. For two intervals whose indices differ by at most a sufficiently large fixed integer C_1 , Hilbert-space Cauchy–Schwarz bounds their pairing by the geometric mean of their squared norms. Each interval has only $O(C_1)$ such neighbors, so their total contribution is at most $C \sum_I \|G_N(I)\|_2^2$.

For all other pairs, $|t(x) - t(y)| \geq (C_1 - 1)/N$, while $|x - y| \leq 2$. Write $z = x - y$ and $\tau = t(x) - t(y)$. On the support of χ , the gradient

$$\nabla_\eta(Nz \cdot \eta - N^2\tau|\eta|^2) = Nz - 2N^2\tau\eta$$

has magnitude at least $cN^2|\tau|$ if C_1 is large enough, since the annulus has a positive inner radius. Dividing the phase by $N^2|\tau|$ gives a smooth phase with uniformly bounded derivatives and a uniformly nonvanishing gradient. Repeated integration by parts gives, for every fixed integer M ,

$$|K_N(z, \tau)| \leq C_M N^2 (N^2|\tau|)^{-M} \leq C_M N^{2-M}.$$

Integrate this bound once over the full set of far pairs, whose spatial measure is at most $|\Omega|^2$. There is no additional factor for the number of interval pairs. After normalization by $N^{-2/3}$, taking $M = 4$ gives an error $O(N^{-8/3})$, which is bounded by $O(N^{-2})$. $\square$

Lemma 9.3 (Mass and spatial localization). *If $I \in \mathcal{D}_j$ and $J \subset I$ is a dyadic descendant at relative depth k , where $0 \leq k \leq j$, set $r = 2^{-k}$. Then*

$$e_j(J) \leq C m_J^{2-2/p}, \quad (9.9)$$

$$e_j(J) \leq C r^{2/3} m_J. \quad (9.10)$$

Consequently, with

$$\alpha = \frac{3}{2} - \frac{1}{p} = 1 + \frac{1-2/p}{2} > 1,$$

we have

$$e_j(J) \leq C 2^{-k/3} m_J^\alpha. \quad (9.11)$$

Proof. For the first estimate, dualize the frequency L^2 norm in (9.5). Against a frequency test function of norm one, the spatial integrand is an annular Schrödinger extension with input norm bounded by a fixed constant. Its values at $t(x)$ are bounded by its short-time maximal function on the interval J , whose length is at most $1/N$. Proposition 9.1 and (9.6) therefore give

$$\|G_N(J)\|_2 \leq C N^{1/3} m_J^{1-1/p}.$$

Squaring and normalizing proves (9.9).

For the second estimate, partition space into half-open squares Q of side r , and write $G_N(J, Q)$ for the same dual integral over $E_J \cap Q$. Set $m_{J,Q} = |E_J \cap Q|$. Declare two squares neighbors if their grid indices differ by at most a sufficiently large fixed constant C_2 . Neighbor interactions are bounded by $C \sum_Q \|G_N(J, Q)\|_2^2$, using Cauchy–Schwarz and the bounded number of neighbors of each square.

For non-neighboring squares, $|x - y| \geq cC_2 r$, whereas $|t(x) - t(y)| \leq |J| = r/N$. Choose C_2 using the outer radius of the annular support. The spatial part now dominates the gradient of the phase in (9.7), and integration by parts gives

$$|K_N(x - y, t(x) - t(y))| \leq C N^2 (1 + N|x - y|)^{-4}.$$

The right side has a uniformly bounded spatial integral. Since the graph measure is projected spatial area and $|w| \leq 1$, the total normalized contribution of these far pairs is at most

$$C N^{-2/3} \int_{E_J} \int_{\mathbb{R}^2} N^2 (1 + N|x - y|)^{-4} dy dx \leq C N^{-2/3} m_J \leq C r^{2/3} m_J. \quad (9.12)$$

The last inequality is exactly where $k \leq j$, equivalently $Nr \geq 1$, is used.

It remains to estimate a single square contribution. Let x_Q be the center of Q and t_J the left endpoint of J . Make the change

$$x = x_Q + rX, \quad t = t_J + r^2s, \quad \xi = \eta/r.$$

The new graph lies in a unit spatial ball, has spatial area $r^{-2}m_{J,Q}$, and has times in an interval of length

$$\frac{|J|}{r^2} = \frac{1}{Nr}.$$

The new annular frequency is $Nr \geq 1$. The dual function acquires the factor r^2 from dx and a unimodular factor from x_Q, t_J ; changing frequency measure gives

$$\|G_N(J, Q)\|_2^2 = r^2 \|\tilde{G}_{Nr}\|_2^2.$$

Apply Proposition 9.1 and (9.6) directly at the normalized scale, including its lowest possible value $Nr = 1$. This gives

$$\begin{aligned} N^{-2/3} \|G_N(J, Q)\|_2^2 &\leq C N^{-2/3} r^2 (Nr)^{2/3} (r^{-2} m_{J,Q})^{2-2/p} \\ &= C r^{-4/3+4/p} m_{J,Q}^{2-2/p} \leq C r^{2/3} m_{J,Q}, \end{aligned}$$

where the last step uses $m_{J,Q} \leq r^2$ and $2 - 2/p > 1$. Sum this estimate over Q , include the bounded neighbor interactions and (9.12), and obtain (9.10). Finally, the geometric mean of (9.9) and (9.10) is (9.11). $\square$

9.3 The common time tree and the Sobolev square norm

The next step uses the same mass assignment $I \mapsto m_I$ for every j . Choosing a different graph separately at each frequency would destroy the telescoping argument.

Lemma 9.4 (Summation on the common time tree). *For every graph and phase choice in (9.5),*

$$\sum_{j \geq 1} 2^{-2j/3} \|G_{2^j}([0, 1])\|_2^2 \leq C. \quad (9.13)$$

The constant is independent of t , w , and the center of Ω .

Proof. Fix $j \geq 1$ and $I \in \mathcal{D}_j$. Starting from $H_{I,0} = I$, select at each step a child with larger mass, breaking ties consistently by choosing the left child. Call the selected child $H_{I,k}$ and the other child $P_{I,k}$, for $1 \leq k \leq j$. Thus

$$m_{P_{I,k}} \leq m_{H_{I,k}}, \quad I = \left(\bigcup_{k=1}^j P_{I,k} \right) \dot{\cup} H_{I,j}.$$

The second identity is an exact partition of time, including any atoms of the time measure. Figure 3 records this partition and the two different depths used in the summation.

By additivity of G_N and the triangle inequality in frequency L^2 , followed by (9.11),

$$e_j(I)^{1/2} \leq C \sum_{k=1}^j 2^{-k/6} m_{P_{I,k}}^{\alpha/2} + C 2^{-j/6} m_{H_{I,j}}^{\alpha/2}.$$

Use Cauchy–Schwarz with weights $2^{-k/6}$, including one additional terminal weight $2^{-j/6}$. Their sum is bounded independently of j . Since $m_{H_{I,j}} \leq m_I$, we obtain

$$e_j(I) \leq C \sum_{k=1}^j 2^{-k/3} m_{P_{I,k}}^{\alpha} + C 2^{-j/3} m_I^{\alpha}. \quad (9.14)$$

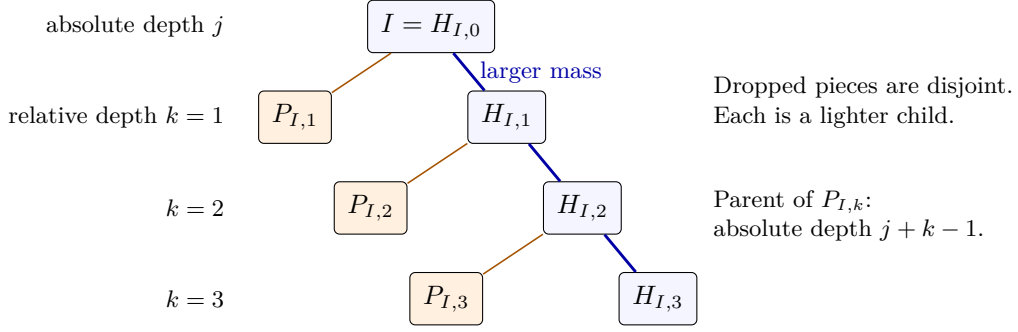


Figure 3: Three steps of the heavier-child path. At each step the lighter sibling is set aside; the chosen child is split again. The dropped siblings together with the terminal child partition I . For fixed relative depth k , changing the frequency index j changes the absolute depth of the parent. This makes the parents distinct in the sum controlled by the convexity deficit.

We give the convexity calculation behind the sum of the first terms. If a parent has child masses $a \geq b \geq 0$, then

$$(a + b)^\alpha - a^\alpha - b^\alpha \geq (2^\alpha - 2)b^\alpha. \quad (9.15)$$

For $b > 0$, divide by b^α and put $u = a/b \geq 1$. The function $(u + 1)^\alpha - u^\alpha - 1$ is increasing for $u \geq 1$, so its minimum is $2^\alpha - 2$; the case $b = 0$ is immediate. Summing parent-minus-children deficits through depth $d - 1$ telescopes to

$$m_{[0,1]}^\alpha - \sum_{Q \in \mathcal{D}_d} m_Q^\alpha \leq m_{[0,1]}^\alpha.$$

The deficits are nonnegative. Letting $d \rightarrow \infty$ shows that the sum of the lighter-child α -powers over any collection of distinct dyadic parents is at most $m_{[0,1]}^\alpha / (2^\alpha - 2)$.

Now fix $k \geq 1$ and range over all $j \geq k$ and all $I \in \mathcal{D}_j$. The parent of $P_{I,k}$ is $H_{I,k-1}$, at absolute depth $j + k - 1$. Different j give different absolute depths; for the same j , different I have disjoint descendants. These parents are therefore distinct. Consequently,

$$\sum_{j \geq k} \sum_{I \in \mathcal{D}_j} m_{P_{I,k}}^\alpha \leq \frac{m_{[0,1]}^\alpha}{2^\alpha - 2}.$$

For the terminal terms, superadditivity of the α th power gives $\sum_{I \in \mathcal{D}_j} m_I^\alpha \leq m_{[0,1]}^\alpha$. Thus summing (9.14), first at fixed k and then in k , gives

$$\sum_{j \geq 1} \sum_{I \in \mathcal{D}_j} e_j(I) \leq C m_{[0,1]}^\alpha \left(\sum_{k \geq 1} 2^{-k/6} + \sum_{j \geq 1} 2^{-j/6} \right) \leq C.$$

All sums are of nonnegative terms, so these rearrangements also follow from finite truncations. Finally apply Lemma 9.2 and sum its $O(2^{-2j})$ errors. $\square$

Remark 9.5. The strict inequality $p > 2$ is used quantitatively here: $\alpha = 3/2 - 1/p > 1$ and $2^\alpha - 2 > 0$. At $p = 2$ the convexity deficit in (9.15) vanishes. The gain in the time tree therefore cannot be recovered merely from a weak L^2 version of (9.1).

Theorem 9.6 (Local strong maximal estimate). *There is a constant C such that, for every $f \in H^{1/3}(\mathbb{R}^2)$ whose Fourier transform has compact support and every $x_0 \in \mathbb{R}^2$,*

$$\int_{B(x_0,1)} \sup_{0 \leq t < 1} |S(t)f(x)| dx \leq C \|f\|_{H^{1/3}}. \quad (9.16)$$

For arbitrary $f \in H^{1/3}(\mathbb{R}^2)$ there is a jointly measurable representative $v_f(x, t)$, $0 \leq t < 1$, continuous in t for almost every x , equal to $S(t)f$ as an L^2 function for each fixed t , and satisfying the same bound with $S(t)f(x)$ replaced by $v_f(x, t)$.

Proof. First suppose $\widehat{f}$ has compact support. Take a smooth frequency partition

$$\psi_0 + \sum_{j \geq 1} \psi_j = 1,$$

where ψ_0 is supported in a fixed ball and ψ_j is supported on an annulus $|\xi| \asymp 2^j$, with uniformly bounded overlap. Put $\widehat{f}_j = \psi_j \widehat{f}$. Choose enlarged smooth annular cutoffs χ_{2^j} equal to one on $\text{supp } \psi_j$. These are the cutoffs used in Lemma 9.4.

For any common graph $t(x)$ and phase $w(x)$, Fourier inversion and the definition of G_N give

$$\int_{\Omega} w(x) \sum_{j \geq 1} S(t(x)) f_j(x) dx = (2\pi)^{-2} \sum_{j \geq 1} \int \widehat{f}_j(\xi) G_{2^j}([0, 1])(\xi) d\xi.$$

Only finitely many summands are nonzero. Cauchy–Schwarz in frequency and in j bounds the modulus by

$$C \left(\sum_{j \geq 1} 2^{2j/3} \|\widehat{f}_j\|_2^2 \right)^{1/2} \left(\sum_{j \geq 1} 2^{-2j/3} \|G_{2^j}([0, 1])\|_2^2 \right)^{1/2} \leq C \|f\|_{H^{1/3}}. \quad (9.17)$$

The first factor is bounded by the Sobolev square norm because on the support of ψ_j one has $2^{2j/3} \asymp (1 + |\xi|^2)^{1/3}$, and the supports overlap boundedly. The second factor is bounded by (9.13). The low-frequency term satisfies $\sup_{x,t} |S(t)f_0(x)| \leq C \|f_0\|_2$ by Cauchy–Schwarz on its fixed frequency ball, so its integral over Ω has the same bound.

To recover the maximal function, take any finite list of times in $[0, 1)$ and let $t(x)$ be the first time attaining the maximal modulus of $S(t)f(x)$ on that list. Set $w(x) = \overline{S(t(x))f(x)} / |S(t(x))f(x)|$ where the denominator is nonzero, and set $w(x) = 0$ otherwise. These are measurable choices, and the linearized integral is exactly the integral of the finite maximum. Increase the list to a countable dense subset of $[0, 1)$ containing zero. Continuity in time and monotone convergence prove (9.16). The estimates above used only fixed ball radii and spatial differences, so the constant is uniform in x_0 .

For general f , choose f_n with $\widehat{f}_n \in C_c^\infty(\mathbb{R}^2)$ and $\|f_n - f\|_{H^{1/3}} \leq 2^{-n}$. In particular,

$$\sum_{n \geq 1} \|f_{n+1} - f_n\|_{H^{1/3}} < \infty.$$

Define the measurable functions

$$d_n(x) = \sup_{0 \leq t < 1} |S(t)f_{n+1}(x) - S(t)f_n(x)|.$$

The supremum can be taken over rational times, together with zero. By (9.16),

$$\sup_{x_0 \in \mathbb{R}^2} \int_{B(x_0, 1)} \sum_{n \geq 1} d_n(x) dx \leq C \sum_{n \geq 1} \|f_{n+1} - f_n\|_{H^{1/3}} < \infty.$$

A countable covering of $\mathbb{R}^2$ by unit balls therefore gives a single full-measure set E_0 on which $\sum_n d_n(x) < \infty$. For each $x \in E_0$ the sequence $S(t)f_n(x)$ converges uniformly over $0 \leq t < 1$. Denote its limit by $v_f(x, t)$, and define $v_f(x, t) = 0$ for $x \notin E_0$. It is jointly measurable, and its time dependence is continuous at every point of E_0 , as a uniform limit of continuous functions.

For each fixed t , unitarity gives $S(t)f_n \rightarrow S(t)f$ in L^2 . The pointwise limit on E_0 must therefore agree almost everywhere with that L^2 limit. This identifies the L^2 class of $v_f(\cdot, t)$ for every fixed t ; it does not assert a pointwise intersection over uncountably many times. Finally, Fatou's lemma in the maximal estimate gives

$$\int_{B(x_0,1)} \sup_{0 \leq t < 1} |v_f(x, t)| dx \leq C \lim_{n \rightarrow \infty} \|f_n\|_{H^{1/3}} = C \|f\|_{H^{1/3}}.$$

□

9.4 Recovering the Gaussian-first, all-real-time limit

Fix $f \in H^{1/3}(\mathbb{R}^2)$ and the approximants just constructed, and write $v = v_f$. The same argument gives error envelopes with both pointwise decay and spatial control at infinity. Namely, let

$$M_n(x) = \sup_{0 \leq t < 1} |v(x, t) - S(t)f_n(x)|. \quad (9.18)$$

These functions are measurable by the rational-time supremum, since the chosen $v(x, \cdot)$ is continuous even on the complement where it was defined to be zero. On E_0 the convergence is uniform, so $M_n(x) \rightarrow 0$. Applying (9.16) to $f_\ell - f_n$ and then using Fatou yields

$$\sup_{x_0 \in \mathbb{R}^2} \int_{B(x_0,1)} M_n(x) dx \leq C \|f - f_n\|_{H^{1/3}}. \quad (9.19)$$

In particular, every M_n is locally integrable with uniformly bounded integrals over all translated unit balls. This uniformity controls Gaussian convolution far from its evaluation point.

Lemma 9.7 (Gaussian approximation with uniform local mass). *Let $h \geq 0$ be locally integrable on $\mathbb{R}^2$ and suppose*

$$A = \sup_{z \in \mathbb{R}^2} \int_{B(z,1)} h(y) dy < \infty.$$

For $a > 0$ let

$$K_a(z) = \frac{1}{4\pi a} e^{-|z|^2/(4a)}.$$

Then $(K_a * h)(x)$ is finite for every $a > 0$ and every x . At every Lebesgue point x of h ,

$$\lim_{a \downarrow 0} (K_a * h)(x) = h(x). \quad (9.20)$$

Proof. For a fixed $\rho > 0$, cover each annulus $2^\ell \rho \leq |z| < 2^{\ell+1} \rho$ by at most $C(1 + 2^\ell \rho)^2$ unit balls. Positivity gives

$$\begin{aligned} \int_{|z| \geq \rho} K_a(z) h(x - z) dz &\leq \frac{CA}{a} \sum_{\ell \geq 0} (1 + 2^\ell \rho)^2 e^{-4^\ell \rho^2/(4a)} \\ &\leq C_\rho A a^{-1} e^{-\rho^2/(8a)}, \quad 0 < a \leq \min\{1, \rho^2\}. \end{aligned} \quad (9.21)$$

The series in the first line is finite for every fixed $a > 0$. Together with local integrability this proves finiteness of the convolution, and the second line shows that its distant contribution tends to zero.

At a Lebesgue point x , put $H_x(r) = \int_{|z|<r} |h(x-z) - h(x)| dz = o(r^2)$. Given $\varepsilon > 0$, choose $\rho > 0$ so that $H_x(r) \leq \varepsilon r^2$ for $0 < r \leq \rho$. With $k_a(r) = (4\pi a)^{-1} e^{-r^2/(4a)}$, radial integration by parts gives

$$\begin{aligned} \int_{|z|<\rho} K_a(z) |h(x-z) - h(x)| dz &= k_a(\rho) H_x(\rho) + \int_0^\rho (-k'_a(r)) H_x(r) dr \\ &\leq \varepsilon \left(k_a(\rho) \rho^2 + \int_0^\rho (-k'_a(r)) r^2 dr \right) \leq C\varepsilon. \end{aligned}$$

Outside this neighborhood, (9.21) controls the h term, and the constant $h(x)$ is controlled by the vanishing Gaussian mass outside $B(0, \rho)$. Let $a \downarrow 0$ and then $\varepsilon \downarrow 0$. $\square$

Proof of Theorem 1.1. At time zero, the fixed-time L^2 identification in Theorem 9.6 gives $v(\cdot, 0) = f$ in L^2 . Since f^* represents the same L^2 function on the set of Lebesgue points of f , intersect E_0 with the full-measure set where $v(x, 0) = f^*(x)$ and with that Lebesgue-point set. Next intersect with the Lebesgue-point sets of the countably many functions M_n . Call the resulting full-measure set E . For every $x \in E$ we now have, simultaneously,

$$\begin{aligned} v(x, \cdot) \text{ is continuous on } [0, 1), \quad v(x, 0) &= f^*(x), \\ M_n(x) \longrightarrow 0, \quad (K_a * M_n)(x) &\longrightarrow M_n(x) \quad \text{as } a \downarrow 0 \text{ for every } n. \end{aligned} \quad (9.22)$$

The last assertion follows from (9.19) and Lemma 9.7. Only countably many exceptional sets were removed.

For each fixed $t \in [0, 1)$, the function $v(\cdot, t)$ represents $S(t)f$ in L^2 . Since $K_a(x - \cdot) \in L^2$, convolution at a given x is an L^2 pairing and is independent of the representative of this L^2 class. Plancherel's identity therefore gives

$$(K_a * v(\cdot, t))(x) = \frac{1}{(2\pi)^2} \int e^{ix \cdot \xi - it|\xi|^2 - a|\xi|^2} \widehat{f}(\xi) d\xi = U_a f(x, t) \quad (9.23)$$

for every $x \in \mathbb{R}^2$, every $a > 0$, and every $0 < t < 1$. The integral is absolutely convergent by Cauchy–Schwarz. The quantifiers in this step are significant: for each arbitrary t , an identity of L^2 classes gives the convolution identity at every evaluation point. There is no intersection of time-dependent pointwise exceptional sets in (9.23).

For the approximants, absolute Fourier integrability yields

$$\begin{aligned} R_n(a) &:= \sup_{x \in \mathbb{R}^2, 0 < t < 1} |U_a f_n(x, t) - S(t) f_n(x)| \\ &\leq \frac{1}{(2\pi)^2} \int |1 - e^{-a|\xi|^2}| |\widehat{f}_n(\xi)| d\xi \longrightarrow 0 \quad (a \downarrow 0). \end{aligned} \quad (9.24)$$

Positivity of K_a , the error envelope, and (9.23) give, for each $x \in E$,

$$\sup_{0 < t < 1} |U_a f(x, t) - v(x, t)| \leq (K_a * M_n)(x) + M_n(x) + R_n(a). \quad (9.25)$$

Indeed the difference between the two convolutions is bounded by $K_a * M_n$, because $|v(y, t) - S(t)f_n(y)| \leq M_n(y)$ for almost every y , simultaneously for all t ; the remaining two differences are bounded by $R_n(a)$ and $M_n(x)$. All these convolutions are finite by the preceding lemma or by the L^2 pairing.

Fix n and take $\limsup_{a \downarrow 0}$ in (9.25). By (9.22) and (9.24),

$$\limsup_{a \downarrow 0} \sup_{0 < t < 1} |U_a f(x, t) - v(x, t)| \leq 2M_n(x).$$

Let $n \rightarrow \infty$. Thus for every $x \in E$,

$$\lim_{a \downarrow 0} \sup_{0 < t < 1} |U_a f(x, t) - v(x, t)| = 0. \quad (9.26)$$

In particular, at each such x the finite limit defining $u_f(x, t)$ exists for every real $0 < t < 1$, with t fixed before a tends to zero, and it equals $v(x, t)$. Continuity at zero now gives

$$\lim_{\substack{t \downarrow 0 \\ t \in \mathbb{R}}} u_f(x, t) = \lim_{t \downarrow 0} v(x, t) = v(x, 0) = f^*(x).$$

This proves Theorem 1.1. □

References

- [1] J. Bennett, A. Carbery, and T. Tao, *On the multilinear restriction and Kakeya conjectures*, Acta Math. **196** (2006), no. 2, 261–302. doi:10.1007/s11511-006-0006-4. arXiv:math/0509262v1.
- [2] J. Bourgain, *Some new estimates on oscillatory integrals*, in *Essays on Fourier Analysis in Honor of Elias M. Stein* (C. Fefferman, R. Fefferman, and S. Wainger, eds.), Princeton Math. Ser., vol. 42, Princeton Univ. Press, 1995, pp. 83–112. Author preprint.
- [3] J. Bourgain, *On the Schrödinger maximal function in higher dimension*, Proc. Steklov Inst. Math. **280** (2013), 46–60. doi:10.1134/S0081543813010045. arXiv:1201.3342.
- [4] J. Bourgain, *A note on the Schrödinger maximal function*, J. Anal. Math. **130** (2016), 393–396. doi:10.1007/s11854-016-0042-8. arXiv:1609.05744.
- [5] J. Bourgain and C. Demeter, *The proof of the l^2 Decoupling Conjecture*, Ann. of Math. (2) **182** (2015), no. 1, 351–389. doi:10.4007/annals.2015.182.1.9. arXiv:1403.5335v3.
- [6] J. Bourgain and L. Guth, *Bounds on oscillatory integral operators based on multilinear estimates*, Geom. Funct. Anal. **21** (2011), no. 6, 1239–1295. doi:10.1007/s00039-011-0140-9. arXiv:1012.3760.
- [7] L. Carleson, *Some analytic problems related to statistical mechanics*, in *Euclidean Harmonic Analysis* (J. J. Benedetto, ed.), Lecture Notes in Math., vol. 779, Springer, Berlin, 1980, pp. 5–45. doi:10.1007/BFb0087666.
- [8] M. Christ and A. Kiselev, *Maximal functions associated to filtrations*, J. Funct. Anal. **179** (2001), no. 2, 409–425. doi:10.1006/jfan.2000.3687.
- [9] B. E. J. Dahlberg and C. E. Kenig, *A note on the almost everywhere behavior of solutions to the Schrödinger equation*, in *Harmonic Analysis* (F. Ricci and G. Weiss, eds.), Lecture Notes in Math., vol. 908, Springer, Berlin, 1982, pp. 205–209. doi:10.1007/BFb0093289.
- [10] Y. Deng, C. Fan, S. Guo, Z. Guo, and Y. Luo, *Sharp Strichartz estimates on the one-dimensional torus*, arXiv:2609.18670v1 (2026). arXiv:2609.18670v1.
- [11] X. Du, L. Guth, and X. Li, *A sharp Schrödinger maximal estimate in $\mathbb{R}^2$* , Ann. of Math. (2) **186** (2017), no. 2, 607–640. doi:10.4007/annals.2017.186.2.5.
- [12] X. Du, L. Guth, X. Li, and R. Zhang, *Pointwise convergence of Schrödinger solutions and multilinear refined Strichartz estimates*, Forum Math. Sigma **6** (2018), e14, 18 pp. doi:10.1017/fms.2018.11. arXiv:1803.01720v2.
- [13] X. Du and R. Zhang, *Sharp L^2 estimates of the Schrödinger maximal function in higher dimensions*, Ann. of Math. (2) **189** (2019), no. 3, 837–861. doi:10.4007/annals.2019.189.3.4. arXiv:1805.02775v3.
- [14] S. Guo, Z. K. Li, and P.-L. Yung, *Improved discrete restriction for the parabola*, Math. Res. Lett. **30** (2023), no. 5, 1375–1409. arXiv:2103.09795v2.
- [15] L. Guth and N. H. Katz, *On the Erdős distinct distances problem in the plane*, Ann. of Math. (2) **181** (2015), no. 1, 155–190. doi:10.4007/annals.2015.181.1.2. arXiv:1011.4105v3.

- [16] L. Guth, D. Maldague, and H. Wang, *Improved decoupling for the parabola*, J. Eur. Math. Soc. (JEMS) **26** (2024), no. 3, 875–917. doi:10.4171/JEMS/1295. arXiv:2009.07953.
- [17] W. He, *Orthogonal projections of discretized sets*, J. Fractal Geom. **7** (2020), no. 3, 271–317. doi:10.4171/JFG/92. arXiv:1710.00795v2.
- [18] S. Lee, *On pointwise convergence of the solutions to Schrödinger equations in $\mathbb{R}^2$* , Int. Math. Res. Not. (2006), Art. ID 32597, 21 pp. doi:10.1155/IMRN/2006/32597.
- [19] Z. K. Li and P.-L. Yung, *An improved high-low method for the parabola*, arXiv:2609.33949v1 (2026).
- [20] A. Moyua, A. Vargas, and L. Vega, *Schrödinger maximal function and restriction properties of the Fourier transform*, Int. Math. Res. Not. **1996** (1996), no. 16, 793–815. doi:10.1155/S1073792896000499.
- [21] Y. Pan, W. Sun, and J. Tan, *Pointwise convergence of Schrödinger operators in Bessel potential spaces*, arXiv:2605.25833v2 (2026). arXiv:2605.25833v2.
- [22] P. Sjögren and P. Sjölin, *Convergence properties for the time-dependent Schrödinger equation*, Ann. Acad. Sci. Fenn. Ser. A I Math. **14** (1989), 13–25. Journal PDF.
- [23] P. Sjölin, *Regularity of solutions to the Schrödinger equation*, Duke Math. J. **55** (1987), no. 3, 699–715. doi:10.1215/S0012-7094-87-05535-9.
- [24] T. Tao, *A sharp bilinear restriction estimate for paraboloids*, Geom. Funct. Anal. **13** (2003), no. 6, 1359–1384. doi:10.1007/s00039-003-0449-0. arXiv:math/0210084.
- [25] T. Tao and A. Vargas, *A bilinear approach to cone multipliers. II. Applications*, Geom. Funct. Anal. **10** (2000), no. 1, 216–258. doi:10.1007/s000390050007.
- [26] L. Vega, *Schrödinger equations: pointwise convergence to the initial data*, Proc. Amer. Math. Soc. **102** (1988), no. 4, 874–878. doi:10.1090/S0002-9939-1988-0934859-0.
- [27] R. Wongkew, *Volumes of tubular neighbourhoods of real algebraic varieties*, Pacific J. Math. **159** (1993), no. 1, 177–184. doi:10.2140/pjm.1993.159.177.